

Master's Thesis

Regularisation in Functional Linear Regression: Theory, Methods, and Applications

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Abstract

Scalar-on-function linear regression is an inverse problem in which the predictor covariance operator determines both identification and the stability of slope recovery. This thesis examines how regularisation, numerical representation, and the purpose of an analysis interact. It compares functional principal component regression, raw and Arnoldi implementations of functional partial least squares, and a released-code conjugate-gradient-type procedure based on moment-residual minimisation.

The theoretical discussion distinguishes ordinary slope error, covariance-weighted prediction error, and the residual approximation needed for inference. Monte Carlo experiments with 5,000 replications in each of three controlled designs show that the implemented early-stopping rule can reduce slope variance while leaving predictive signal unresolved. Raw and orthogonalised Krylov fits agree closely when their coordinates are reliable, but rare ill-conditioned raw fits create severe error tails. A separate inference study shows that an estimation-oriented stopping variant can leave a substantial moment residual on the root- n scale. Late iteration makes this algorithmic discrepancy negligible in the studied designs; calibration remains conditional on the reference distribution, discretisation, and finite-sample approximation.

An application uses instrument-1 near-infrared spectra to predict assay for pharmaceutical tablets, with 195 development observations and a designated 460-tablet benchmark. Raw and Arnoldi FPLS achieve benchmark RMSEs of 4.607 and 4.643 in the supplied assay units. Their predictions are very close, while Arnoldi uses a stable five-dimensional representation and the selected raw representation is severely ill-conditioned. The benchmark had been inspected during dataset selection, and the uncertainty intervals condition on the fitted models. The findings support matching regularisation and stopping to the statistical target, monitoring numerical reliability, and distinguishing controlled inferential benchmarks from feasible data analysis.

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Chapter 1

Introduction

Many modern data sets contain observations that are most naturally represented as functions rather than as finite collections of unrelated measurements. Examples include growth trajectories, spectrometric curves, electricity-demand profiles, biomedical signals, and records collected continuously over time or space. Treating such observations as functional data preserves their ordering and smooth structure and makes it possible to study features of the complete trajectory. Functional data analysis provides the statistical framework for this purpose; see, for example, Ramsay and Silverman (2005). One of its central regression models relates a scalar response to a functional predictor. For a square-integrable random function X on a compact interval $\mathcal{I}$, the scalar-on-function linear model is

$$Y = \alpha + \int_{\mathcal{I}} X(s)\beta(s) \, ds + \varepsilon, \quad (1.1)$$

where Y is a scalar response, α is an intercept, β is an unknown slope function, and ε is an error term with conditional mean zero. The function β describes how variation in the predictor at different locations of its domain is associated with the response. This model is conceptually similar to multiple linear regression, but estimation is substantially more difficult because the unknown parameter belongs to an infinite-dimensional function space (Cardot et al. 1999; Hall and Horowitz 2007).

After centring X and Y and, for notational simplicity, denoting the centred variables again by X and Y , the population normal equation corresponding to (1.1) can be written as

$$K\beta = r, \quad (1.2)$$

where K is the covariance operator of X and $r = \mathbb{E}(YX)$ is the cross-covariance element between the response and the predictor. If K has a non-trivial null space, the data identify β only up to additions from $\text{Null}(K)$; the conventional target is the minimum-norm representative $\beta^\dagger = K^\dagger r$ in the identified space $\mathcal{H}_{\text{id}} := \text{Null}(K)^\perp$. Unless the

distinction is material, the notation β below refers to this identified representative. In typical functional settings, K is compact. When it has infinite rank, its positive eigenvalues decrease to zero and its generalised inverse is unbounded: small sampling errors in directions associated with small eigenvalues may be greatly magnified when (1.2) is solved. Estimating the identified slope is therefore an ill-posed inverse problem even when it is unique on the chosen parameter space. A direct solution of the empirical normal equation is generally unstable, including after the functional observations have been represented on a finite basis. Some form of regularisation is indispensable (Engl et al. 1996; Hall and Horowitz 2007).

Regularisation deliberately restricts or modifies the inverse problem so that the estimator does not attempt to recover directions that are weakly identified by the data. In functional principal component regression (FPCR), the covariance operator is replaced by a truncated spectral expansion and the inverse is computed only on the span of selected empirical eigenfunctions. This produces an interpretable and widely used estimator, and the truncation level acts as the regularisation parameter. Its components, however, are ordered according to variation in the predictor alone; a direction that explains relatively little variation in X may still be important for predicting Y . Functional partial least squares (FPLS) instead constructs response-informed directions and has consequently become an important alternative to purely spectral regularisation (Preda and Saporta 2005; Delaigle and Hall 2012; Febrero-Bande et al. 2017).

An especially useful view of FPLS is obtained through Krylov subspaces. At iteration m , the estimator is sought in

$$\mathcal{K}_m(K, r) = \text{span}\{r, Kr, K^2r, \dots, K^{m-1}r\}. \quad (1.3)$$

This formulation shows that the method progressively explores directions determined jointly by the covariance structure of X and its relationship with Y . It also connects FPLS with the conjugate-gradient (CG) algorithm for linear systems and inverse problems (Hestenes and Stiefel 1952; Phatak and Hoog 2002; Babii et al. 2026). The connection is statistically important because the number of Krylov iterations is itself a regularisation parameter. Early iterations recover well-identified signal directions, whereas later iterations increasingly expose the solution to sampling noise in small-eigenvalue directions. Early stopping can therefore regularise the functional inverse problem without explicitly truncating an eigendecomposition (Blanchard and Krämer 2016; Babii et al. 2026).

The mathematical equivalence of alternative Krylov formulations does not imply identical numerical behaviour. A direct implementation of (1.3) uses successive powers of the empirical covariance operator. These vectors can become nearly linearly dependent,

causing the associated Gram matrix to be severely ill-conditioned. In exact arithmetic, this construction spans the intended space; in finite-precision arithmetic, however, loss of independence can produce unstable coefficients and occasionally very large estimation errors. Orthogonal Krylov algorithms such as Arnoldi replace the raw power basis by an orthonormal basis, while CG generates the relevant iterates through short recurrences. The practical performance of regularisation therefore has both a statistical and a numerical dimension: one must decide how far to iterate and how to represent the space explored by those iterations.

This thesis studies that interaction through four estimators: FPCR, raw FPLS based on unorthogonalised Krylov powers, Arnoldi FPLS with two-pass classical Gram–Schmidt orthogonalisation, and a direct CG–FPLS implementation with a data-dependent stopping rule. These estimators represent two broad regularisation principles. FPCR uses a spectral cut-off, while the three FPLS implementations use response-informed Krylov spaces and iterative regularisation. At a common component count and under the required rank conditions, raw and Arnoldi FPLS target the same restricted response fit, so their comparison isolates the effect of numerical stabilisation. CG–FPLS, in turn, permits a direct examination of early stopping and of the computational geometry underlying FPLS.

The study is organised around four questions. First, how do spectral and iterative regularisation methods compare in slope estimation and out-of-sample prediction when the covariance structure and slope smoothness change? Second, when theoretically equivalent FPLS formulations are implemented in finite precision, how strongly do Krylov-basis conditioning and orthogonalisation affect their reliability? Third, how does the CG stopping index affect the validity of statistical inference, and why may an inference-compatible index differ from one chosen for estimation or prediction? Fourth, how do these regularisation and stability trade-offs appear in a real spectroscopic prediction problem?

The third question reflects a fundamental distinction. For estimation and prediction, stopping CG early controls variance and prevents weakly identified directions from dominating the slope estimate. For inference on the moment equation, the remaining algorithmic residual must instead be negligible relative to sampling uncertainty. Following Babii et al. (2026), this thesis considers, for $b \in \mathcal{H}_{\text{id}}$, the global hypothesis

$$H_0 : \beta = b \quad \text{against} \quad H_1 : \beta \neq b, \quad (1.4)$$

using the statistic

$$\mathcal{T}_n(b) = n \left\| \widehat{K}(\widehat{\beta}_m - b) \right\|_{L^2}^2. \quad (1.5)$$

For this statistic, iteration must continue until the moment residual $\widehat{r} - \widehat{K}\widehat{\beta}_m$ is negligible at the root- n sampling scale. Under the conditions developed later, this makes $\mathcal{T}_n(b)$ asymptotically equivalent to a direct-moment statistic and supports testing, power analysis, and confidence sets obtained by inversion. The essential contrast is one of purpose: early stopping regularises estimation, whereas inference requires the algorithmic residual to be asymptotically negligible.

The simulation programme has two connected components. The first is a reproducible Monte Carlo comparison of the four regularised estimators across three functional linear models. Performance is evaluated through integrated squared error for the slope, mean squared prediction error, selected complexity, conditioning, and complete error distributions. Particular attention is paid to rare numerical failures, since averages or medians alone can conceal the practical consequences of an unstable Krylov basis. The second component studies CG-FPLS inference through null calibration, empirical size, power, confidence-set coverage, and convergence of the finite-iteration statistic towards its direct-moment limit. These controlled studies are complemented by a concise spectroscopy application, which asks whether the statistical and numerical trade-offs isolated in simulation remain visible with observed functional predictors. Chapter 8 reports the principal empirical evidence; Appendix D records the complete protocol, diagnostics, and sensitivity analysis.

The contribution is therefore methodological, computational, and expository rather than a new asymptotic theory. The thesis brings spectral regularisation, raw and orthogonalised Krylov formulations, direct CG iteration, and CG-based inference into a common, reproducible analysis. Its evidence yields conditional trade-offs rather than a universal method ranking: performance depends jointly on the target geometry, tuning rule, numerical representation, and data structure. The later application provides a first assessment of how those mechanisms transfer beyond the controlled designs.

The argument proceeds in four stages. Chapters 2 and 3 develop the functional-analytic foundation and turn it into four fully specified estimators. Chapters 4 and 5 use controlled changes in the slope and covariance structure to study estimation, prediction, and numerical stability. Chapters 6 and 7 then develop and evaluate the late-stopped inference procedure. Chapter 8 applies the four methods to spectroscopic data and relates the empirical findings to the mechanisms isolated in simulation; Appendix D provides the full supporting analysis. Chapter 9 closes by synthesising the results, limitations, and implications for further work.

Chapter 2

Literature and Functional Regression Background

The scalar-on-function linear model is easy to state but cannot be understood fully through finite-dimensional regression intuition alone. Its regressor is a random element of a function space, its covariance is a linear operator rather than a matrix, and estimation of the slope requires the inversion of that operator. This chapter introduces the mathematical objects needed to formulate the problem precisely and places the regularisation methods studied in this thesis within the functional regression literature. The presentation draws on the functional-analysis material developed in the MIT 18.102 lecture notes (Lin 2021), the operator-theoretic treatment of Hsing and Eubank (2015), and the statistical accounts of Ramsay and Silverman (2005), Horváth and Kokoszka (2012), and Kokoszka and Reimherr (2017).

Definitions and proofs of the foundational results used below are collected in Appendix A. Keeping the arguments there makes the assumptions and derivations available for reference while allowing this chapter to retain a continuous statistical narrative.

2.1 Functional observations from discrete measurements

In an ordinary multivariate data set, the observational unit is a vector with a fixed number of coordinates. In functional data analysis, the observational unit is a curve or another structured function. The available data are still recorded at finitely many points, but those measurements are regarded as observations of an underlying object $X(t)$ defined over a continuum $t \in \mathcal{I}$. This viewpoint is appropriate when the order and

proximity of the measurement locations have substantive meaning and when values at neighbouring locations are related. Daily demand profiles, growth trajectories, spectra, and biomedical signals are typical examples (Ramsay and Silverman 2005; Kokoszka and Reimherr 2017).

Suppose that the i th curve is observed at grid points $t_1, \dots, t_T$. A functional representation may be obtained directly from a dense grid, by interpolation or smoothing, or through a finite basis expansion

$$X_i(t) \approx \sum_{j=1}^J x_{ij} \phi_j(t), \quad (2.1)$$

where $\phi_1, \dots, \phi_J$ may be Fourier, spline, wavelet, or other basis functions. The coefficient vector in (2.1) is a computational representation of the curve, not a change in the conceptual model: the statistical object remains a function. Increasing the grid size or the number of basis functions allows progressively richer variation and therefore exposes the infinite-dimensional character of the problem.

Functional observations may be densely and regularly sampled, sparsely sampled, or contaminated by measurement error. These regimes require different preliminary reconstruction methods. The simulations in this thesis use densely evaluated curves generated from a known basis, so reconstruction uncertainty is deliberately separated from regularisation of the regression slope. This controlled setting allows the comparison to focus on the inverse problem and on the numerical behaviour of the four estimators. The spectroscopy application in Chapter 8 will state the preprocessing choices needed to interpret its principal findings, while Appendix D will document them in full, since smoothing, registration, and scaling can form an additional layer of regularisation with observed curves.

Once a curve has been reconstructed, its statistical analysis requires a geometry in which distances, projections, and linear functionals are well defined. The square-integrable function space introduced next supplies that geometry.

2.2 Hilbert-space foundations

2.2.1 The space of square-integrable functions

Let $\mathcal{I} = [0, 1]$ and consider real-valued functions on this interval. The interval can be replaced by any compact interval through an affine transformation. The natural state

space for the present model is

$$\mathcal{H} = L^2(\mathcal{I}) = \left\{ f : \mathcal{I} \rightarrow \mathbb{R} : \int_{\mathcal{I}} f(t)^2 dt < \infty \right\}, \quad (2.2)$$

where functions that differ only on a set of Lebesgue measure zero are identified. The space is equipped with the inner product

$$\langle f, g \rangle = \int_{\mathcal{I}} f(t)g(t) dt \quad (2.3)$$

and induced norm $\|f\| = \langle f, f \rangle^{1/2}$. The Cauchy–Schwarz inequality,

$$|\langle f, g \rangle| \leq \|f\| \|g\|, \quad (2.4)$$

ensures, among other things, that the regression functional $f \mapsto \langle f, \beta \rangle$ is continuous whenever $\beta \in L^2(\mathcal{I})$.

Definition 2.1 (Hilbert space). An inner-product space is a *Hilbert space* if it is complete under the norm induced by its inner product. Completeness means that every Cauchy sequence has a limit in the space.

The space $L^2(\mathcal{I})$ is a separable Hilbert space. Completeness makes limits of approximating sequences mathematically well defined, while separability guarantees the existence of a finite or countable orthonormal basis. Both properties are essential in functional statistics. They permit functions to be approximated by finite expansions without reducing the population model itself to a fixed finite dimension (Lin 2021; Hsing and Eubank 2015).

The use of L^2 is also matched to the regression model. The quantity $\langle X, \beta \rangle$ is the integrated contribution of the predictor curve to the response, and $\|\hat{\beta} - \beta\|^2$ is the integrated squared error used later to evaluate slope recovery. Other function spaces may encode smoothness or pointwise behaviour more directly, but L^2 supplies the geometry in which covariance operators, principal components, and the CG–FPLS moment equation have particularly convenient forms.

2.2.2 Orthogonality, bases, and projections

A sequence $\{e_j\}_{j \geq 1} \subset \mathcal{H}$ is orthonormal if $\langle e_j, e_k \rangle = 0$ for $j \neq k$ and $\|e_j\| = 1$. It is an orthonormal basis if its closed linear span equals $\mathcal{H}$. For such a basis, every $f \in \mathcal{H}$ has the expansion

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j, \quad (2.5)$$

where convergence is in the L^2 norm. Parseval's identity gives

$$\|f\|^2 = \sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2. \quad (2.6)$$

Thus, the coefficient sequence contains the same norm information as the function itself.

For the finite-dimensional subspace $V_m = \text{span}\{e_1, \dots, e_m\}$, the orthogonal projection is

$$P_m f = \sum_{j=1}^m \langle f, e_j \rangle e_j. \quad (2.7)$$

It satisfies the best-approximation property

$$\|f - P_m f\| = \inf_{v \in V_m} \|f - v\|. \quad (2.8)$$

This basic geometric result appears repeatedly in functional regression. A predetermined basis approximation restricts the slope to a chosen subspace; FPCR projects onto a subspace estimated from the covariance operator; and Arnoldi FPLS constructs an orthonormal basis for a response-informed Krylov subspace. The distinction between these methods therefore lies less in whether projection occurs than in how the target subspace is selected.

The Riesz representation theorem provides a second connection between Hilbert-space geometry and regression. Every continuous linear functional $\ell : \mathcal{H} \rightarrow \mathbb{R}$ can be written uniquely as $\ell(f) = \langle f, b \rangle$ for some $b \in \mathcal{H}$. Consequently, assuming a linear and continuous effect of a functional predictor naturally leads to an inner-product model with slope β . This representation is the infinite-dimensional analogue of expressing a linear functional on $\mathbb{R}^p$ as multiplication by a coefficient vector (Lin 2021; Hsing and Eubank 2015).

Figure 2.1 places this Hilbert-space geometry within the broader family of ℓ^p norms. In the two-dimensional coordinate space $\mathbb{R}^2$, write

$$\|x\|_p = (|x_1|^p + |x_2|^p)^{1/p}, \quad 1 \leq p < \infty, \quad \|x\|_\infty = \max\{|x_1|, |x_2|\}.$$

The corresponding metric is $d_p(x, z) = \|x - z\|_p$, so a unit-ball boundary is also a unit-distance contour about the origin; translating and rescaling it gives the contour about any other centre. The diamond, circle, and square in the left panel therefore show both different unit balls and different ways of measuring the same displacement. These are finite-dimensional sections used for visualisation. Although all norms on $\mathbb{R}^2$ are equivalent, the infinite-dimensional sequence spaces ℓ^p are genuinely different.

More formally, if $1 \leq p < \infty$ and q satisfies $p^{-1} + q^{-1} = 1$, then $(\ell^p)^*$ is isometrically ℓ^q and

$$\|y\|_q = \sup_{\|x\|_p \leq 1} \left| \sum_{j \geq 1} x_j y_j \right|.$$

The case $p = q = 2$ is the self-dual member of this conjugate family. For an orthonormal basis, Parseval's identity identifies L^2 functions with their ℓ^2 coefficient sequences without changing distances, while Riesz identifies continuous linear effects with elements of the same space. This common geometry is particularly well matched to orthogonal projections, covariance operators, and the conjugate-gradient constructions used later in the thesis. It does not assert that L^2 is universally preferable to every other function space: in particular, the prediction criterion later uses a covariance-weighted seminorm. The full sequence-space statement and its finite-dimensional polar interpretation are proved in Theorem A.13; Remark A.14 records the exceptional endpoint behaviour of $(\ell^\infty)^*$ (Lin 2021).

2.3 Random functions and covariance operators

2.3.1 Random elements in L^2

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A functional observation is represented by a random element $X : \Omega \rightarrow \mathcal{H}$. Assume throughout that

$$\mathbb{E}\|X\|^2 < \infty. \tag{2.9}$$

Its mean $\mu_X \in \mathcal{H}$ is defined by the relation

$$\langle \mu_X, f \rangle = \mathbb{E}\langle X, f \rangle, \quad f \in \mathcal{H}. \tag{2.10}$$

This definition does not require pointwise expectations to exist at every t ; it characterises the mean as an element of the Hilbert space. Write $X^c = X - \mu_X$ for the centred predictor.

For $u, v \in \mathcal{H}$, define the rank-one operator $u \otimes v$ by

$$(u \otimes v)f = \langle v, f \rangle u. \tag{2.11}$$

The covariance operator of X is

$$K = \mathbb{E}(X^c \otimes X^c), \quad Kf = \mathbb{E}[\langle X^c, f \rangle X^c]. \tag{2.12}$$

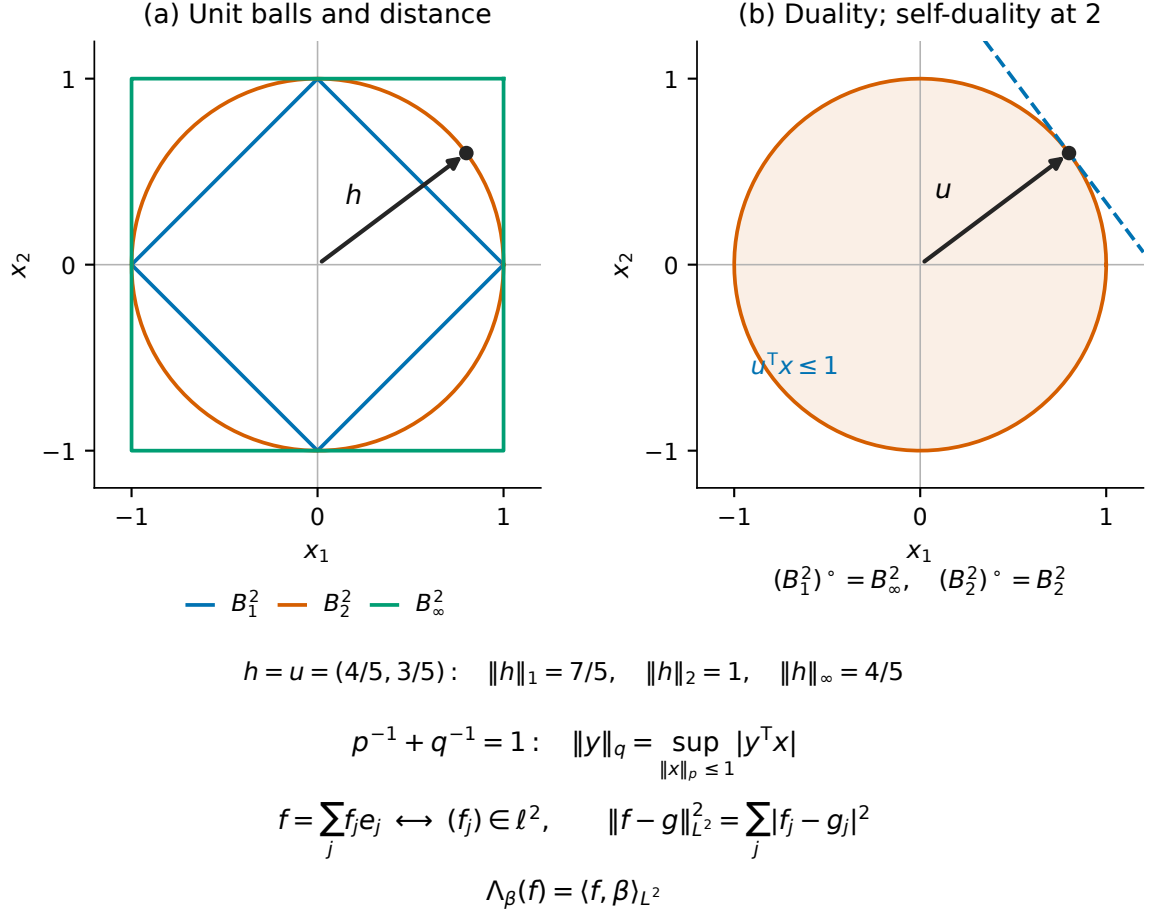


Figure 2.1: Unit-ball geometry, norm-induced distance, and duality. The left panel plots $B_p^2 = \{x \in \mathbb{R}^2 : \|x\|_p \leq 1\}$ for $p = 1, 2, \infty$ and assigns three different lengths to the same displacement $h = (4/5, 3/5)$. In the right panel, q is the Hölder-conjugate exponent and $(B_p^2)^\circ = B_q^2$ under the standard finite-dimensional pairing. The dashed line $u^\top x = 1$ supports the Euclidean ball at $u = (4/5, 3/5)$; the same u is both the boundary point and its norming vector, illustrating $(B_2^2)^\circ = B_2^2$. The lower strip records how Parseval's identity and Riesz representation transfer this self-dual coefficient geometry to $L^2(\mathcal{I})$. The plotted balls are two-dimensional analogues, not depictions of infinite-dimensional sets.

If a jointly measurable representative of X^c has pointwise second moments, define

$$c(s, t) = \text{Cov}\{X(s), X(t)\}.$$

Provided expectation and integration may be interchanged—for example, if, for the function f under consideration,

$$\int_{\mathcal{I}} \mathbb{E}|X^c(s)X^c(t)| |f(t)| dt < \infty \quad \text{for almost every } s,$$

the same operator can be written as

$$(Kf)(s) = \int_{\mathcal{I}} c(s, t) f(t) dt. \quad (2.13)$$

The qualification is important because an element of L^2 is an equivalence class rather than a pointwise-defined curve. The operator formulation is therefore preferable: it remains meaningful without choosing pointwise values or imposing sample-path smoothness.

2.3.2 Properties and spectral representation

The covariance operator is bounded, self-adjoint, and non-negative:

$$\langle Kf, g \rangle = \langle f, Kg \rangle, \quad \langle Kf, f \rangle = \mathbb{E}\langle X^c, f \rangle^2 \geq 0. \quad (2.14)$$

Under (2.9), it is also trace class and hence compact, with

$$\text{tr}(K) = \mathbb{E}\|X^c\|^2 < \infty. \quad (2.15)$$

These facts place K within the compact self-adjoint operator theory covered by the spectral theorem (Lin 2021; Hsing and Eubank 2015; Horváth and Kokoszka 2012).

Proposition 2.2 (Spectral representation of the covariance operator). *Let $\mathcal{J}_K$ be either a possibly empty finite index set $\{1, \dots, q\}$, with $q \geq 0$, or $\mathbb{N}$. There exist positive eigenvalues $\{\lambda_j : j \in \mathcal{J}_K\}$ and orthonormal eigenfunctions $\{v_j : j \in \mathcal{J}_K\}$ spanning $\overline{\text{Range}(K)}$ such that*

$$Kf = \sum_{j \in \mathcal{J}_K} \lambda_j \langle f, v_j \rangle v_j, \quad f \in \mathcal{H}, \quad (2.16)$$

where the eigenvalues are ordered non-increasingly. If $\mathcal{J}_K = \mathbb{N}$, then $\lambda_j \rightarrow 0$; on $\text{Null}(K)$ the operator is zero.

The eigenfunctions define the functional principal component directions. With scores

$\xi_j = \langle X^c, v_j \rangle$, the Karhunen–Loève representation is

$$X = \mu_X + \sum_{j \in \mathcal{J}_K} \xi_j v_j, \quad (2.17)$$

with convergence in mean square. The scores are uncorrelated, $\mathbb{E}\xi_j = 0$, $\mathbb{E}(\xi_j \xi_k) = 0$ for $j \neq k$, and $\mathbb{E}\xi_j^2 = \lambda_j$. When at least m positive eigenvalues exist, the first m eigenfunctions span the m -dimensional subspace that minimises the expected squared reconstruction error of X among all m -dimensional subspaces. This optimality concerns reconstruction of the predictor; it does not by itself guarantee that the same directions are the most informative for a response variable.

Given independent observations $X_1, \dots, X_n$, the empirical mean and covariance operator are

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad (2.18)$$

$$\hat{K}f = \frac{1}{n} \sum_{i=1}^n \langle X_i - \bar{X}, f \rangle (X_i - \bar{X}). \quad (2.19)$$

Even though $\hat{K}$ acts on an infinite-dimensional space, its rank is at most $n - 1$. It is therefore non-invertible on $\mathcal{H}$. A basis or grid representation replaces it by a finite matrix, but this does not remove the underlying difficulty: the resulting matrix is singular or increasingly ill-conditioned as more directions are admitted.

2.4 The scalar-on-function linear model

Let (Y, X) be a scalar–function pair with $\mathbb{E}(Y^2) < \infty$ and $X \in \mathcal{H}$. The scalar-on-function linear model is

$$Y = \alpha + \langle X, \beta \rangle + \varepsilon, \quad \mathbb{E}(\varepsilon \mid X) = 0, \quad (2.20)$$

where $\beta \in \mathcal{H}$ is the slope function. If $\mu_Y = \mathbb{E}Y$, then

$$\alpha = \mu_Y - \langle \mu_X, \beta \rangle. \quad (2.21)$$

Consequently, centring gives

$$Y^c = Y - \mu_Y = \langle X^c, \beta \rangle + \varepsilon. \quad (2.22)$$

All subsequent derivations can therefore be carried out for centred variables without an intercept.

Define the cross-covariance element

$$r = \mathbb{E}(Y^c X^c) \in \mathcal{H}. \quad (2.23)$$

For any $f \in \mathcal{H}$, the conditional mean restriction implies

$$\langle r, f \rangle = \mathbb{E}[Y^c \langle X^c, f \rangle] \quad (2.24)$$

$$= \mathbb{E}[\langle X^c, \beta \rangle \langle X^c, f \rangle] = \langle K\beta, f \rangle. \quad (2.25)$$

Since this holds for every f , the population normal equation is

$$K\beta = r. \quad (2.26)$$

This is the operator counterpart of the finite-dimensional normal equations: the covariance matrix and cross-covariance vector are replaced by K and r , respectively.

The same equation follows from least squares. For a candidate $b \in \mathcal{H}$, let

$$R(b) = \mathbb{E} (Y^c - \langle X^c, b \rangle)^2. \quad (2.27)$$

The error orthogonality condition gives the risk identity

$$R(b) - R(\beta) = \mathbb{E} \langle X^c, b - \beta \rangle^2 = \langle K(b - \beta), b - \beta \rangle = \|K^{1/2}(b - \beta)\|^2. \quad (2.28)$$

Thus, prediction error and slope estimation error use different geometries. Integrated squared error is $\|b - \beta\|^2$, whereas excess prediction risk is the weaker covariance-weighted seminorm $\|K^{1/2}(b - \beta)\|^2$ on $\mathcal{H}$. It becomes a norm after the parameter space is restricted to $\text{Null}(K)^\perp$. Directions with small eigenvalues contribute little to prediction but may contribute substantially to integrated slope error. The distinction is central to the theoretical results of Cai and Hall (2006) and motivates reporting both criteria in the simulation study.

The empirical counterparts of r and the normal equation are

$$\hat{r} = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X}), \quad (2.29)$$

$$\hat{K}\hat{\beta} = \hat{r}. \quad (2.30)$$

Equation (2.30) is the common starting point for all four estimators. Before regularising

it, however, one must separate which part of the slope is identified from whether that identified part can be recovered stably.

2.5 Identification and ill-posedness

2.5.1 Identification of the slope

For two slopes β and $\tilde{\beta}$, put $h = \beta - \tilde{\beta}$. They produce the same conditional mean if and only if

$$h \in \text{Null}(K). \quad (2.31)$$

Indeed, if $h \in \text{Null}(K)$, then

$$\mathbb{E}\langle X^c, h \rangle^2 = \langle Kh, h \rangle = 0, \quad (2.32)$$

and hence $\langle X^c, h \rangle = 0$ almost surely. Conversely, equality of the conditional means implies this almost-sure equality and therefore $Kh = \mathbb{E}\{\langle X^c, h \rangle X^c\} = 0$. Thus the observationally identified parameter is the equivalence class $[\beta] = \beta + \text{Null}(K)$. Equivalently, one may work on the canonical parameter space

$$\mathcal{H}_{\text{id}} = \text{Null}(K)^\perp = \overline{\text{Range}(K)}, \quad (2.33)$$

where the conditional-mean map is injective. The identified representative is the unique minimum-norm element of $[\beta]$; it is denoted by $\beta^\dagger$. Full identification on all of $\mathcal{H}$ occurs if and only if $\text{Null}(K) = \{0\}$. When the distinction is not material, later chapters write β for the canonical representative $\beta^\dagger$.

Expand $\beta^\dagger$ and r in the positive-eigenvalue system of K :

$$\beta^\dagger = \sum_{j \in \mathcal{J}_K} \beta_j v_j, \quad r = \sum_{j \in \mathcal{J}_K} r_j v_j. \quad (2.34)$$

For positive eigenvalues, (2.26) implies

$$r_j = \lambda_j \beta_j, \quad \beta_j = \frac{r_j}{\lambda_j}. \quad (2.35)$$

Together with $r \perp \text{Null}(K)$, a square-integrable solution exists if and only if the coefficients satisfy the Picard condition

$$\sum_{j \in \mathcal{J}_K} \frac{r_j^2}{\lambda_j^2} < \infty. \quad (2.36)$$

Under the correctly specified model, $r = K\beta$ ensures this population condition for the true square-integrable slope. It does not ensure that replacing r and K by noisy empirical quantities yields a stable estimator.

2.5.2 Why inversion is unstable

Compactness is the source of the inverse problem. If K has infinite rank, its positive eigenvalues converge to zero. The appropriate inverse for the identified parameter is the Moore–Penrose inverse. On its natural domain it is

$$K^\dagger f = \sum_{j \in \mathcal{J}_K} \frac{\langle f, v_j \rangle}{\lambda_j} v_j, \quad \mathcal{D}(K^\dagger) = \left\{ f \in \mathcal{H} : \sum_{j \in \mathcal{J}_K} \frac{|\langle f, v_j \rangle|^2}{\lambda_j^2} < \infty \right\}, \quad (2.37)$$

with $K^\dagger f = 0$ for $f \in \text{Null}(K)$. The Picard condition ensures that $r \in \mathcal{D}(K^\dagger)$ and the canonical slope is $\beta^\dagger = K^\dagger r$. The factors $1/\lambda_j$ are nevertheless unbounded. To see the consequence directly, perturb the right-hand side of the normal equation by $a_j v_j$. Although the perturbation has norm $|a_j|$, its effect after inversion has norm $|a_j|/\lambda_j$. For high-index directions, an arbitrarily small error in r can therefore generate a large error in the recovered slope.

Sampling error affects both $\hat{r}$ and $\hat{K}$. Empirical eigenvalues fluctuate, empirical eigenfunctions are estimated imperfectly, and components of $\hat{r}$ along low-variance directions can be dominated by noise. Simply increasing the basis dimension does not solve these problems. It permits more small-eigenvalue directions to enter the calculation and may increase variance and numerical conditioning problems. Functional linear regression is consequently ill posed in the sense that the solution does not depend continuously on the observed moments (Engl et al. 1996; Hall and Horowitz 2007).

Ill-posedness should be distinguished from non-identification. An injective compact covariance operator can identify every coefficient of β at the population level while still possessing an unbounded inverse. Identification states that two different slopes imply different population conditional means. Stability asks whether a small change in the moments produces only a small change in the recovered slope. Regularisation is needed for the second property even when the first holds.

There is also a finite-dimensional numerical manifestation. After discretisation in a J -dimensional basis, the covariance operator becomes a positive semidefinite matrix. Its condition number is the ratio between its largest and smallest retained eigenvalues. As J grows, the smallest eigenvalues typically become very small and the condition number becomes large. Rounding error and statistical noise can then be amplified by a direct linear solve. This link explains why the theoretical inverse problem and the

numerical conditioning diagnostics in the simulation study describe the same underlying geometry at different levels.

2.6 Regularisation principles in functional regression

Regularisation replaces the unstable inverse by a family of stable approximate inverses. The amount of regularisation controls a bias–variance trade-off. Strong regularisation suppresses poorly identified directions and introduces bias; weak regularisation admits more directions and increases sensitivity to noise. A tuning parameter, truncation level, or stopping index determines the operating point.

For orientation, suppose that $\hat{K}$ has eigenpairs $(\hat{\lambda}_j, \hat{v}_j)$. A broad class of spectral estimators has the form

$$\hat{\beta}_\eta = \sum_{j: \hat{\lambda}_j > 0} g_\eta(\hat{\lambda}_j) \langle \hat{r}, \hat{v}_j \rangle \hat{v}_j, \quad (2.38)$$

where g_η is a bounded approximation to $\lambda \mapsto 1/\lambda$ on the positive spectrum and is taken to satisfy $g_\eta(0) = 0$. Different choices of g_η lead to different regularisation schemes. Krylov estimators need not be computed through the empirical eigendecomposition, but they can still be understood as iteration-dependent polynomial approximations to this inverse.

Table 2.1: Regularisation principles relevant to functional linear regression.

Method family	Restricted or modified object	Main tuning quantity	Statistical interpretation
Basis and roughness penalty	Predetermined basis and a penalty on size or derivatives	Basis dimension and penalty weight	Encodes structural smoothness and stabilises correlated coefficients
FPCR	Leading empirical covariance eigenfunctions	Number of components	Removes directions explaining little predictor variation
Tikhonov regularisation	Shifted operator $\hat{K} + \rho I$	Penalty ρ	Continuously shrinks small-eigenvalue directions
FPLS and CG	Response-informed Krylov subspace	Component count or stopping rule	Uses iteration as regularisation and prioritises directions linked to the response

2.6.1 Basis penalties and Tikhonov regularisation

One approach restricts β to a predetermined finite basis and estimates its coefficients by penalised least squares. When L is an unbounded operator such as differentiation,

the optimisation is understood over $b \in \text{Dom}(L)$, or over a finite differentiable basis contained in that domain. A typical criterion is

$$\frac{1}{n} \sum_{i=1}^n \{Y_i - \bar{Y} - \langle X_i - \bar{X}, b \rangle\}^2 + \rho \|Lb\|^2, \quad (2.39)$$

where L is often a derivative operator and $\rho > 0$ controls smoothness. When $L = I$, the population analogue yields the Tikhonov solution $(K + \rho I)^{-1}r$, whose filter on the positive spectrum is $g_\rho(\lambda) = (\lambda + \rho)^{-1}$. We retain the convention $g_\rho(0) = 0$ in (2.38). This gives the same solution because $r \perp \text{Null}(K)$ under the model; likewise, $\hat{r} \perp \text{Null}(\hat{K})$ empirically. Every positive eigendirection is retained, with stronger shrinkage where the eigenvalue is small. Penalised basis and spline estimators for functional linear regression were developed, among others, by Cardot et al. (2003) and Crambes et al. (2009). Although these estimators are not part of the four-method Monte Carlo comparison, they clarify that regularisation can encode smoothness directly rather than through covariance components.

2.6.2 Spectral cut-off and FPCR

Functional principal component regression uses the truncated inverse

$$\hat{\beta}_m^{\text{FPCR}} = \sum_{j=1}^m \frac{\langle \hat{r}, \hat{v}_j \rangle}{\hat{\lambda}_j} \hat{v}_j. \quad (2.40)$$

The cut-off prevents division by empirical eigenvalues beyond the retained rank. FPCR is attractive because it is transparent, computationally convenient, and tied directly to the Karhunen–Loève representation. Its main limitation is that $\hat{v}_j$ and their order are determined only by the marginal variation of X . A direction with modest predictor variance but a large slope coefficient may be important for the response and nevertheless enter late in the principal component ordering.

Cardot et al. (1999) developed functional principal-component estimation, while their later spline construction combines basis expansion with a roughness penalty (Cardot et al. 2003). Hall and Horowitz (2007) analysed convergence rates for principal-component and ridge-type regularisation under eigenvalue and smoothness conditions. Cai and Hall (2006) showed why estimating the slope and predicting a response can have substantially different difficulty, a distinction already visible in (2.28). FPCR therefore provides both a canonical estimator and a benchmark against which response-informed regularisation can be assessed.

2.6.3 Response-informed Krylov regularisation

Partial least squares selects directions using both X and Y . In functional form, the relevant population Krylov space is

$$\mathcal{K}_m(K, r) = \text{span}\{r, Kr, \dots, K^{m-1}r\}. \quad (2.41)$$

An m -component FPLS solution has the polynomial form

$$\beta_m = p_{m-1}(K)r \quad (2.42)$$

for a polynomial p_{m-1} of degree at most $m - 1$. Its moment residual is

$$r - K\beta_m = \pi_m(K)r, \quad \pi_m(\lambda) = 1 - \lambda p_{m-1}(\lambda). \quad (2.43)$$

The iteration constructs an approximation to the inverse only along directions represented in the cross-covariance r . This explains the supervised nature of FPLS and why it can require fewer components than FPCR in some prediction problems.

Preda and Saporta (2005) formulated PLS regression for stochastic-process predictors. Delaigle and Hall (2012) developed an alternative functional PLS representation using successive powers of the covariance operator and established its statistical properties. In finite-dimensional PLS, the connection with Lanczos and conjugate-gradient algorithms was studied by Phatak and Hoog (2002). More recently, Babii et al. (2026) formulated functional PLS explicitly as a CG regularisation method, developed a data-driven early-stopping rule, and derived estimation and inference theory.

The common organising object is the Krylov space, but not every algorithm defines the same finite-step estimate. Raw FPLS uses the displayed power sequence directly, while Arnoldi replaces it by an orthonormal basis of the same space. Under the usual full-rank conditions, classical score-and-deflation FPLS, often implemented by a NIPALS-type algorithm, yields the same restricted response-space fit at a common component count.

Babii’s released-code CG–FPLS recurrence also explores these nested spaces without forming the complete power basis, but it minimises the empirical moment residual rather than the response sum of squares. It is therefore closely related to, but not generally identical with, the finite-step NIPALS/APLS estimator. Even among formulations with the same exact-arithmetic target, the coordinates matter numerically: repeated covariance powers can approach an active dominant eigenspace and become nearly dependent. Because the objectives and coordinates differ, the rule that selects the iteration or component count is part of the procedure rather than an interchangeable afterthought.

2.6.4 Tuning and early stopping

In every method in Table 2.1, tuning determines how much information from weak directions enters the estimator. For FPCR and component-based FPLS, m is often selected by cross-validation or a generalised cross-validation criterion. For iterative inverse methods, a discrepancy rule can stop when the residual is comparable with an estimate of the sampling noise. The precise rule matters: allowing m to increase without control ultimately moves the estimator towards the unstable empirical inverse.

The statistical role of early stopping parallels spectral shrinkage. Initially, the iterative estimator tends to recover directions associated with larger effective eigenvalues. Later iterations fit increasingly delicate features of the sample moment equation. Stopping before this occurs introduces regularisation bias but limits variance. This is the sense in which CG is not merely a faster linear-system solver in the present thesis; its iteration path is a family of regularised estimators.

2.7 Estimation, prediction, and inference

The literature reviewed above shows that the quality of a regularisation method depends on the target. With β now denoting the canonical representative, slope estimation evaluates $\|\hat{\beta} - \beta\|$, whereas prediction evaluates the covariance-weighted seminorm $\|K^{1/2}(\hat{\beta} - \beta)\|$. Inference introduces a third objective: the algorithmic approximation error must be negligible at the scale of the sampling distribution. These objectives do not generally select the same amount of regularisation.

For the hypothesis $H_0 : \beta = b$, with $b \in \mathcal{H}_{\text{id}}$, Babii et al. (2026) base inference on the covariance-weighted difference appearing in $\mathcal{T}_n(b)$ from (1.5):

$$\hat{K}(\hat{\beta}_m - b) = (\hat{r} - \hat{K}b) - (\hat{r} - \hat{K}\hat{\beta}_m). \quad (2.44)$$

The first term is the sample moment under the null; the second is the residual left by stopping CG after m iterations. Early stopping may minimise an estimation or prediction criterion while leaving the second term too large for valid inference. Under independent and identically distributed sampling, $\mathbb{E}(\varepsilon X^c) = 0$, $\mathbb{E}\|\varepsilon X^c\|^2 < \infty$, and the standard measurability conditions for the Hilbert-space central limit theorem,

$$\sqrt{n}(\hat{r} - \hat{K}b) \rightsquigarrow G \quad \text{under } H_0, \quad (2.45)$$

where G is a centred Gaussian element. Sample centring contributes only a lower-order term under the corresponding second-moment conditions. To ensure that the algorithmic

residual does not alter this limit, the inferential requirement is

$$\sqrt{n} \|\hat{r} - \hat{K}\hat{\beta}_m\| \xrightarrow{\mathbb{P}} 0. \quad (2.46)$$

This generally calls for later stopping than estimation or prediction. Under (2.46), the finite-iteration statistic has the same first-order limit as the direct-moment statistic $n\|\hat{r} - \hat{K}b\|^2$, and the spectral representation of G leads to a weighted chi-square law. Chapter 6 states the full testing and calibration argument; Chapter 7 then evaluates its finite-sample behaviour.

This separation also clarifies why late stopping does not claim to improve the slope estimate. The inferential statistic depends on $\hat{K}\hat{\beta}_m$, for which the weakly identified slope directions are attenuated by $\hat{K}$. Late iteration is used to solve the sample moment equation accurately, not to recover β accurately in the unweighted L^2 norm. These target-dependent requirements have so far been expressed at the operator level. Their implementation introduces approximation and floating-point effects in addition to sampling error, motivating the distinction considered next.

2.8 Discretisation and the population–computation distinction

Theoretical objects in this chapter live in $L^2(\mathcal{I})$, whereas computation uses vectors and matrices. Let $t_1, \dots, t_T$ be a numerical grid with quadrature weights $w_1, \dots, w_T$. Then

$$\langle f, g \rangle \approx \sum_{\ell=1}^T w_\ell f(t_\ell) g(t_\ell). \quad (2.47)$$

Equivalently, a J -term basis representation yields a finite coefficient vector and a basis Gram matrix. Both routes approximate the Hilbert-space geometry; neither changes the population estimand.

Three errors should therefore be kept conceptually separate. Approximation error arises from representing a function on a finite grid or basis. Statistical error arises because K and r are estimated from a finite sample. Numerical error arises from floating-point calculations, particularly when a matrix or basis is ill-conditioned. The simulation design uses a sufficiently rich common basis and grid for all methods, so comparisons are not driven by method-specific discretisations. The raw-versus-Arnoldi comparison then isolates numerical stability within the same theoretical Krylov space, while the comparison across component counts describes statistical regularisation.

This distinction is also important for interpreting a successful matrix computation. A finite matrix may be invertible even though it approximates an operator with an unbounded inverse. Conversely, a stable population regularisation principle may be implemented poorly if the chosen coordinates are nearly dependent. A defensible functional regression analysis must therefore align the population theory, statistical tuning, and numerical linear algebra.

2.9 Position of the thesis in the literature

The functional linear model and its principal-component estimators are well established (Cardot et al. 1999; Hall and Horowitz 2007). Penalised and spline approaches provide alternative ways to encode smoothness (Crambes et al. 2009), while the different statistical difficulty of slope recovery and prediction has been studied explicitly (Cai and Hall 2006). The FPLS literature introduced response-guided dimension reduction and later supplied a Krylov representation and convergence theory (Preda and Saporta 2005; Delaigle and Hall 2012). Comparative evidence indicates that neither FPCR nor FPLS can dominate independently of the covariance spectrum, slope alignment, and tuning design (Febrero-Bande et al. 2017). The question pursued here lies where those statistical principles meet finite-precision computation: a Krylov subspace can remain mathematically well defined even when its coordinates have become numerically unusable.

The study isolates that interaction through controlled contrasts. FPCR changes how the search space is selected; raw and Arnoldi FPLS hold the Krylov space and response objective fixed while changing its coordinates and linear solver; and released-code CG-FPLS retains the nested spaces but changes the finite-step objective and stopping rule. The separate inference study then asks how far the moment-residual Krylov path must be continued when the target changes from regularised estimation to testing. With the operator, identification, and target geometries now in place, Chapter 3 turns these contrasts into four fully specified estimators.

Chapter 3

FPLS, Krylov Methods, and Regularisation

Chapter 2 showed that functional linear regression leads to a compact-operator equation whose direct inverse is unstable. This chapter develops the four regularised estimators used in the Monte Carlo study: FPCR, raw FPLS, Arnoldi FPLS, and the released-code CG-FPLS procedure of Babii et al. (2026). Particular attention is paid to a terminological point that otherwise causes confusion. Classical functional PLS can be computed by a sequential score-and-deflation algorithm of NIPALS type or represented as an alternative PLS (APLS) problem on a Krylov subspace. Raw and Arnoldi FPLS are two numerical implementations of this same response-space APLS estimator. Babii’s CG-FPLS explores the same nested Krylov spaces but uses a different finite-iteration objective. The methods are consequently closely related without being interchangeable in every respect.

To align the theoretical comparison with the computation, the definitions below use the same centring, discretisation, rank safeguards, and tuning conventions as the accompanying Python implementation. Proofs of the Krylov-space identities and variational characterisations are collected in Appendix A.

3.1 The empirical operator problem

Let $X_i^c = X_i - \mu_X$ and $y_i = Y_i - \mu_Y$. In an empirical application the unknown means would normally be replaced by their sample counterparts. The controlled simulation designs impose $\mu_X = \mu_Y = 0$, so the generated curves and responses enter the code directly and no intercept is estimated. Define the empirical sampling operator

$T_n : \mathcal{H} \rightarrow \mathbb{R}^n$ and its adjoint by

$$(T_n b)_i = \langle X_i^c, b \rangle, \quad (3.1)$$

$$T_n^* a = \frac{1}{n} \sum_{i=1}^n a_i X_i^c, \quad a \in \mathbb{R}^n, \quad (3.2)$$

where $\langle a, c \rangle_n = n^{-1} a^\top c$ is used in response space. The empirical covariance operator and cross-covariance element are then

$$\hat{K} = T_n^* T_n, \quad \hat{r} = T_n^* y, \quad (3.3)$$

so unregularised least squares would attempt to solve $\hat{K}b = \hat{r}$. With population-centred curves, $\hat{K}$ is generated by n rank-one operators and hence has rank at most n ; when the curves are centred by their sample mean, the sharper bound is $n - 1$. The simulations use the known zero means, so the former bound applies. In either case, $\hat{K}$ is finite rank and approximates a compact population operator, so this equation cannot be used as a stable unrestricted inverse. Unless K is injective, all slope estimands below are understood as the minimum-norm representatives in $\mathcal{H}_{\text{id}} = \text{Null}(K)^\perp$; components in $\text{Null}(K)$ are not identified by the data.

The simulation evaluates functions on an equally spaced grid $t_1, \dots, t_T$ and uses the discrete inner product

$$\langle f, g \rangle_T = \frac{1}{T} \sum_{\ell=1}^T f(t_\ell) g(t_\ell) = \frac{1}{T} f^\top g. \quad (3.4)$$

If X denotes the centred $n \times T$ matrix of curves, the operators in (3.3) are represented exactly as

$$\hat{K} = \frac{X^\top X}{nT}, \quad \hat{r} = \frac{X^\top y}{n}, \quad T_n b = \frac{Xb}{T}. \quad (3.5)$$

These scaling factors matter: omitting T^{-1} would change the numerical operator and the stopping threshold. All four estimators receive the same sample, grid, zero-mean convention, and inner product.

Two residuals will be used repeatedly. The response residual at b is $y - T_n b$, whereas the moment residual is

$$e(b) = \hat{r} - \hat{K}b = T_n^*(y - T_n b). \quad (3.6)$$

The latter is the response residual after it has been mapped back to function space. Although the two contain related information, minimising their norms is not the same finite-dimensional optimisation problem.

3.2 Functional principal component regression

FPCR is the spectral benchmark. Let $(\hat{\lambda}_j, \hat{v}_j)$ denote the empirical eigenpairs of $\hat{K}$, ordered by decreasing eigenvalue. For a component count m , the estimator is

$$\hat{\beta}_m^{\text{FPCR}} = \sum_{j=1}^m \frac{\hat{v}_j^\top \hat{r}}{\hat{\lambda}_j} \hat{v}_j, \quad (3.7)$$

where the coordinate eigenvectors are Euclidean-normalised. Formula (3.7) is the least-squares solution restricted to the first m empirical principal directions. The component count is therefore a hard spectral cut-off: components beyond m are assigned an inverse weight of zero instead of the potentially large value $1/\hat{\lambda}_j$. In the implementation, an empirical eigenvalue is retained only when it exceeds $\varepsilon_{\text{mach}} T \max_j |\hat{\lambda}_j|$; this is a numerical rank safeguard, not an additional statistical tuning parameter. The reported path is further capped at $\min\{m_{\text{max}}, n - 1, T - 1\}$ before GCV, and at the number of numerically positive eigenvalues.

The reported FPCR estimator chooses m by response-space generalised cross-validation (GCV). If $\hat{y}_m = T_n \hat{\beta}_m^{\text{FPCR}}$, the criterion is

$$\text{GCV}_Y(m) = \frac{n^{-1} \|y - \hat{y}_m\|_2^2}{(1 - m/n)^2}. \quad (3.8)$$

The numerator rewards in-sample fit and the denominator penalises the effective dimension. This is response GCV, not a criterion based on the moment residual. Section 3.7 introduces a separate moment-space FPCR pilot used only to estimate the noise scale for CG stopping.

FPCR orders directions by predictor variance. It is therefore expected to be effective when the slope signal is concentrated in leading covariance eigenfunctions. It may be less parsimonious when a response-relevant direction has modest predictor variance, which motivates supervised PLS directions.

3.3 Classical functional PLS and the NIPALS algorithm

Partial least squares originated as a component method for multivariate regression. NIPALS, conventionally expanded as *non-linear iterative partial least squares*, is one of its classical score-and-deflation algorithms (Wold et al. 1984; Geladi and Kowalski 1986). The present model has one scalar response and therefore corresponds to PLS1. Its functional version replaces predictor vectors by functions and ordinary dot products by the L^2 inner product.

To see the construction, start with residual curves $X_i^{(0)} = X_i^c$ and residual responses $y_i^{(0)} = y_i$. At component $h = 1, \dots, m$, a NIPALS-type FPLS algorithm performs the following steps.

1. Form and normalise the residual cross-covariance direction,

$$c_h = \frac{1}{n} \sum_{i=1}^n y_i^{(h-1)} X_i^{(h-1)}, \quad w_h = \frac{c_h}{\|c_h\|}. \quad (3.9)$$

2. Project the residual predictor curves onto this direction to obtain the score

$$t_{ih} = \langle X_i^{(h-1)}, w_h \rangle. \quad (3.10)$$

3. Regress both residual objects on the score. The functional predictor loading and scalar response loading are

$$p_h = \frac{\sum_i t_{ih} X_i^{(h-1)}}{\sum_i t_{ih}^2}, \quad \ell_h = \frac{\sum_i t_{ih} y_i^{(h-1)}}{\sum_i t_{ih}^2}. \quad (3.11)$$

4. Deflate the predictor and response,

$$X_i^{(h)} = X_i^{(h-1)} - t_{ih} p_h, \quad y_i^{(h)} = y_i^{(h-1)} - t_{ih} \ell_h. \quad (3.12)$$

At the first step, $c_1 = \hat{r}$, so the first PLS weight is the normalised sample cross-covariance. Subsequent weights are computed after deflation and are therefore not simply displayed as raw powers of $\hat{K}$. If $W_m a = \sum_{h=1}^m a_h w_h$ and $P_m a = \sum_{h=1}^m a_h p_h$, the corresponding slope can be reconstructed as

$$\hat{\beta}_m^{\text{NIPALS}} = W_m (P_m^* W_m)^{-1} \ell, \quad \ell = (\ell_1, \dots, \ell_m)^\top, \quad (3.13)$$

provided the small loading matrix is nonsingular.

The deflation formulas provide the classical algorithmic view: each score explains the part of the response not captured by earlier scores. Under the usual full-rank conditions, NIPALS and SIMPLS yield the same fixed- m response fit despite using different intermediate coordinates (Jong 1993). The next section identifies that common fit directly as an optimisation over a Krylov subspace. NIPALS is therefore a reference description, not an additional estimator in the simulation.

3.4 APLS and the Krylov representation

Delaigle and Hall (2012) established an alternative functional PLS representation that exposes the operator geometry hidden by sequential deflation. Define the empirical Krylov subspace

$$\hat{\mathcal{K}}_m = \mathcal{K}_m(\hat{K}, \hat{r}) = \text{span}\{\hat{r}, \hat{K}\hat{r}, \dots, \hat{K}^{m-1}\hat{r}\}. \quad (3.14)$$

Standard response-space FPLS is characterised by

$$\hat{\beta}_m^{\text{APLS}} \in \arg \min_{b \in \hat{\mathcal{K}}_m} \|y - T_n b\|_n^2. \quad (3.15)$$

Its first-order condition is

$$\hat{r} - \hat{K} \hat{\beta}_m^{\text{APLS}} \perp \hat{\mathcal{K}}_m. \quad (3.16)$$

The NIPALS scores span the image $T_n \hat{\mathcal{K}}_m$, so the sequential algorithm and (3.15) produce the same fitted responses at a common component count. When the response risk is strictly convex on the restricted space, their slope functions are equal as well. This is the precise similarity the present thesis uses: classical NIPALS-type FPLS and APLS are algorithmic and operator-theoretic descriptions of the same standard PLS1 fit. A proof is given in Proposition A.37.

Every element of (3.14) is a polynomial in the empirical covariance operator applied to the cross-covariance:

$$b = p_{m-1}(\hat{K})\hat{r}. \quad (3.17)$$

Unlike FPCR, this space is response-informed because $\hat{r} = T_n^* y$ determines its starting direction. The iteration does not retain the m largest-variance directions automatically. It constructs a polynomial approximation to the inverse in directions that are visible in the empirical relationship between X and Y .

Let $g_j = \hat{K}^{j-1}\hat{r}$ for $j = 1, \dots, m$ and write $b = \sum_{j=1}^m a_j g_j$. The normal equations for (3.15) are

$$H_m^Y a = h_m^Y, \quad (3.18)$$

with entries

$$(H_m^Y)_{jk} = \langle \hat{r}, \hat{K}^{j+k-1}\hat{r} \rangle, \quad (h_m^Y)_j = \langle \hat{r}, \hat{K}^{j-1}\hat{r} \rangle. \quad (3.19)$$

The matrix is Hankel because each entry depends only on $j + k$. The same system is obtained by regressing y on the design columns $T_n g_1, \dots, T_n g_m$. This representation reveals both the close relationship with PLS and the numerical weakness of raw covariance powers.

3.5 Raw FPLS

Raw FPLS implements (3.15) literally. It forms the matrix

$$G_m = [\hat{r}, \hat{K}\hat{r}, \dots, \hat{K}^{m-1}\hat{r}] \quad (3.20)$$

without rescaling or orthogonalising its columns, constructs $Z_m = T_n G_m$, and solves

$$(Z_m^\top Z_m)a = Z_m^\top y, \quad \hat{\beta}_m^{\text{raw}} = G_m a. \quad (3.21)$$

In exact arithmetic, this is simply APLS in the power basis. It is included because it is the most direct translation of the theoretical Krylov span and because it exposes a numerical issue that can remain hidden when only final prediction errors are reported.

Repeated multiplication by $\hat{K}$ progressively emphasises the leading empirical eigendirections. To see this, expand $\hat{r} = \sum_j \hat{r}_j \hat{v}_j$. Then

$$\hat{K}^k \hat{r} = \sum_j \hat{\lambda}_j^k \hat{r}_j \hat{v}_j. \quad (3.22)$$

As k increases, ratios $(\hat{\lambda}_j/\hat{\lambda}_1)^k$ contract whenever $\hat{\lambda}_j < \hat{\lambda}_1$. More precisely, let $\hat{\lambda}_\star$ be the largest empirical eigenvalue whose eigenspace has a non-zero projection of $\hat{r}$, and let $P_\star$ denote that projection. Then, in the finite empirical spectrum,

$$\frac{\hat{K}^k \hat{r}}{\|\hat{K}^k \hat{r}\|} \longrightarrow \frac{P_\star \hat{r}}{\|P_\star \hat{r}\|}$$

whenever $\hat{\lambda}_\star > 0$. Thus the normalised raw columns eventually become nearly parallel. The effect can be slow at modest k when several active eigenvalues are close, but the condition number of G_m can nevertheless grow rapidly. Moreover, whenever Z_m has full column rank, $\kappa_2(Z_m^\top Z_m) = \kappa_2(Z_m)^2$ exactly; if it is rank deficient, both sides are infinite under the usual extended convention. Small floating-point perturbations can then create large changes in a , despite the fact that the intended exact subspace remains well defined.

The raw path is deliberately not repaired by a pseudoinverse or hidden orthogonalisation in the simulation code. A component count is marked invalid if its normal-equation solve fails or produces non-finite values. Five-fold cross-validation is evaluated only on component counts valid in the full sample and every fold. This makes numerical failure part of the empirical comparison rather than silently changing the estimator.

3.6 Arnoldi FPLS and reorthogonalisation

Arnoldi FPLS retains the response-space objective (3.15) but changes the coordinates used for the Krylov space. Starting from $q_1 = \hat{r}/\|\hat{r}\|$, Arnoldi forms $z = \hat{K}q_j$ and removes its projections onto the existing basis. The implementation uses two passes of classical Gram–Schmidt (CGS2):

$$z \leftarrow z - Q_j(Q_j^\top z), \quad (3.23)$$

$$z \leftarrow z - Q_j(Q_j^\top z), \quad (3.24)$$

$$q_{j+1} = z/\|z\|, \quad (3.25)$$

where $Q_j = [q_1, \dots, q_j]$. A second pass is inexpensive at the component counts considered here and reduces the loss of orthogonality caused by rounding. The implementation terminates basis expansion when

$$\|z\|_2 \leq \varepsilon_{\text{rank}} \max\{\|\hat{K}\|_\infty, \eta_{\text{tiny}}\}, \quad \varepsilon_{\text{rank}} = 10^{-10}, \quad (3.26)$$

where η_{tiny} is the smallest positive normal floating-point number. Expansion also terminates when the candidate norm is non-finite. The resulting columns span the same $\hat{\mathcal{K}}_m$ as the raw powers in exact arithmetic, as proved in Proposition A.35.

The displayed Arnoldi operations use the Euclidean norm of the grid coordinates, exactly as in the code. On an equally weighted grid this norm differs from the discrete L^2 norm in (3.4) only by the common factor $\sqrt{T}$. The factor changes neither the generated span nor the orthogonality and normalised-condition diagnostics.

For each available m , Arnoldi FPLS solves

$$\hat{a}_m \in \arg \min_{a \in \mathbb{R}^m} \|y - T_n Q_m a\|_n^2, \quad \hat{\beta}_m^{\text{Arnoldi}} = Q_m \hat{a}_m. \quad (3.27)$$

The response-space least-squares problem is solved by QR decomposition, not by forming another Gram matrix. If $Z_m^A = T_n Q_m = Q_{Z,m} R_{Z,m}$ denotes the reduced QR factorisation, the response solve is accepted through component j only while

$$|(R_{Z,m})_{jj}| > \varepsilon_{\text{rank}} \max\{|(R_{Z,m})_{11}|, \eta_{\text{tiny}}\}. \quad (3.28)$$

On failure of this response-rank test, or after a non-finite solve, the path is frozen at its last valid fitted estimate for larger requested component counts. Thus, orthogonalisation protects the functional basis, QR protects the subsequent response solve, and the two stated tests handle numerical rank loss. These are fixed computational safeguards rather than data-driven regularisation choices.

Figure 3.1 illustrates the distinction for the population geometry of Model 1. At $m = 12$, every pair among the normalised raw powers has absolute inner product above 0.997, and their condition number is of order 10^{16} . The CGS2 Arnoldi basis has no off-diagonal inner product exceeding the displayed threshold and an orthogonality defect below 10^{-15} . The precise floating-point values can vary slightly across linear-algebra platforms. The figure does not show different statistical information: both panels represent the same Krylov space. It shows why the choice of basis can determine whether that space is numerically usable.

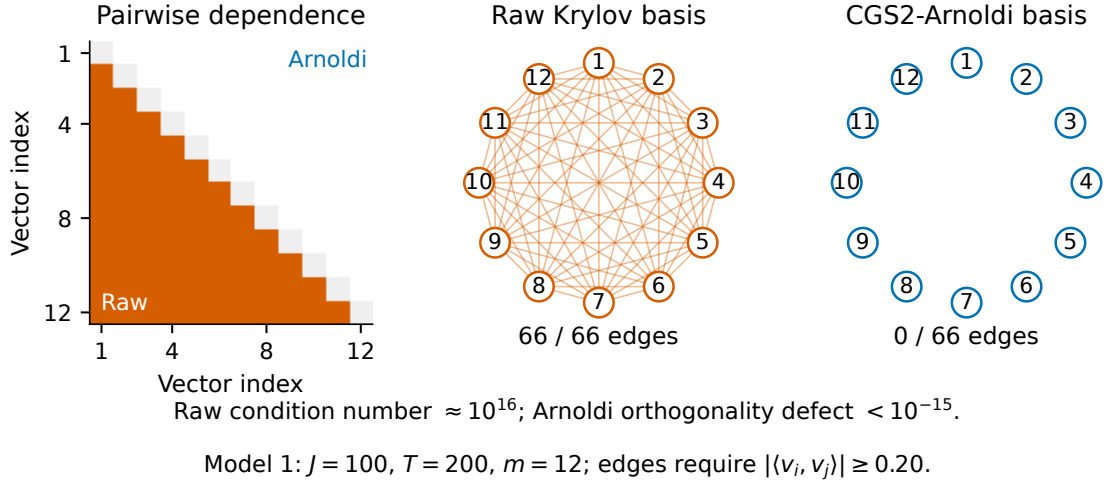


Figure 3.1: Raw and CGS2-Arnoldi representations of the same 12-dimensional population Krylov space in Model 1. An edge marks an absolute pairwise inner product above 0.20. The raw power vectors are almost collinear, whereas reorthogonalisation yields an orthonormal coordinate system. The illustration uses $J = 100$ basis coefficients and a grid of $T = 200$ points.

At a fixed common m , a successful raw solve and a full-rank Arnoldi solve must give the same exact-arithmetic response fit because both minimise the same criterion over the same subspace. Their selected estimates need not be identical in the Monte Carlo study, however. Each method is tuned by its own five-fold cross-validation curve; finite-precision instability can alter the raw curve, invalidate candidate dimensions, and lead to a different selected component count. The comparison therefore separates exact subspace equivalence from the behaviour of complete, tuned numerical procedures.

3.7 Released-code CG–FPLS

The fourth estimator follows the released-code construction of Babii et al. (2026). It searches over the same spaces $\widehat{\mathcal{K}}_m$ but defines the finite- m iterate by

$$\widehat{\beta}_m^{\text{CG}} \in \arg \min_{b \in \widehat{\mathcal{K}}_m} \|\widehat{r} - \widehat{K}b\|^2. \quad (3.29)$$

This is PLS measured in the norm of the empirical normal-equation residual. Its first-order condition is

$$\widehat{r} - \widehat{K}\widehat{\beta}_m^{\text{CG}} \perp \widehat{K}\widehat{\mathcal{K}}_m, \quad (3.30)$$

or, equivalently, $\widehat{K}e_m \perp \widehat{\mathcal{K}}_m$. Compare this with $e_m \perp \widehat{\mathcal{K}}_m$ in (3.16). The subspace is the same, but the projection condition is not.

The difference is also visible in raw coordinates. With $b = \sum_{j=1}^m a_j g_j$, the moment objective produces

$$H_m^M a = h_m^M, \quad (3.31)$$

where

$$(H_m^M)_{jk} = \langle \widehat{r}, \widehat{K}^{j+k} \widehat{r} \rangle, \quad (h_m^M)_j = \langle \widehat{r}, \widehat{K}^j \widehat{r} \rangle. \quad (3.32)$$

Relative to (3.19), every operator power has shifted by one. Hence the NIPALS/APLS and released-code CG estimates generally differ before convergence. Proposition A.38 proves these two characterisations.

The estimator is computed without explicitly constructing either Hankel matrix. Initialise

$$\widehat{\beta}_0 = 0, \quad e_0 = \widehat{r}, \quad d_0 = \widehat{r}. \quad (3.33)$$

For $m = 0, 1, \dots$, the released recurrence is

$$\alpha_m = \frac{\langle e_m, \widehat{K}e_m \rangle}{\|\widehat{K}d_m\|^2}, \quad (3.34)$$

$$\widehat{\beta}_{m+1} = \widehat{\beta}_m + \alpha_m d_m, \quad (3.35)$$

$$e_{m+1} = e_m - \alpha_m \widehat{K}d_m, \quad (3.36)$$

$$\gamma_m = \frac{\langle e_{m+1}, \widehat{K}e_{m+1} \rangle}{\langle e_m, \widehat{K}e_m \rangle}, \quad (3.37)$$

$$d_{m+1} = e_{m+1} + \gamma_m d_m. \quad (3.38)$$

The paper’s and code’s name *CG–FPLS* is retained throughout the thesis. The important reproducibility point is the displayed recurrence, not a generic software routine named `cg`. Its short recurrence avoids the explicit, ill-conditioned power-basis normal equations.

In standard numerical-linear-algebra terminology, the residual-minimising updates in (3.34)–(3.38) are the conjugate-residual recurrence, rather than the textbook energy-norm CG recurrence used in Section 3.8; the nomenclature here follows the paper and released code. The displayed updates are well defined in exact arithmetic while their quadratic-form numerator and squared-norm denominator are positive. The implementation symmetrises $\widehat{K}$ and freezes the remaining path at the last valid iterate if either scalar is non-finite or non-positive, or if

$$\|\widehat{K}d_m\|_2 \leq 10^{-12} \max\{\|\widehat{K}\|_\infty, \eta_{\text{tiny}}\} \max\{\|d_m\|_2, \eta_{\text{tiny}}\}, \quad (3.39)$$

using the same η_{tiny} as above. A non-finite updated iterate or recurrence coefficient raises a floating-point error. These are numerical-breakdown guards, not the statistical stopping rule.

The theoretical discrepancy rule in Babii et al. (2026) uses the population noise scale

$$\nu_n^\circ = \tau \left\{ \frac{2\sigma^2 \mathbb{E}\|X^c\|^2}{\delta n} \right\}^{1/2}, \quad m^\circ = \min\{m \geq 0 : \|e_m\| \leq \nu_n^\circ\}. \quad (3.40)$$

Here $\sigma^2 = \mathbb{E}\varepsilon^2$ in the homoskedastic simulation design. This ideal rule states the theoretical benchmark. Because σ^2 and $\mathbb{E}\|X^c\|^2$ are not supplied to an estimator, the implemented Python workflow instead uses the plug-in scale

$$\widehat{\nu}_n = \tau \left\{ \frac{2\widehat{\sigma}^2 \widehat{\mathbb{E}}\|X^c\|^2}{\delta n} \right\}^{1/2}, \quad \widehat{\mathbb{E}}\|X^c\|^2 = \frac{1}{n} \sum_{i=1}^n \|X_i^c\|^2. \quad (3.41)$$

and selects

$$\widehat{m} = \min \left\{ m \geq 1 : \|\widehat{r} - \widehat{K}\widehat{\beta}_m^{\text{CG}}\| \leq \widehat{\nu}_n \right\}, \quad (3.42)$$

with the maximum permitted component count used if the threshold is not reached. Here $\tau > 1$ is a safety multiplier, $\delta \in (0, 1)$ controls the probability bound underlying the rule, and $\widehat{\sigma}^2$ is the mean squared response residual from an internal FPCR pilot. The pilot dimension is chosen by moment-space GCV solely to estimate the noise level; it does not enter the method comparison as a separate estimator. The code also enforces at least one component, whereas the ideal definition permits $m^\circ = 0$ if the initial residual already lies below the population threshold.

3.8 Why conjugate directions improve on steepest descent

The following comparison supplies geometric intuition for conjugate directions; it does not assert equivalence with the released CG–FPLS recurrence, which uses the conjugate-residual updates specified above. Consider an ordinary symmetric positive-definite quadratic,

$$f(x) = \frac{1}{2}x^\top Ax - b^\top x, \quad A = A^\top > 0. \quad (3.43)$$

Steepest descent moves in the negative-gradient direction $r_k = b - Ax_k$ and, even with exact line search, often zig-zags across a narrow elliptical valley. Its progress deteriorates as the condition number $\kappa_2(A)$ increases. Standard CG instead constructs directions satisfying $p_j^\top Ap_k = 0$ for $j \neq k$. Minimisation along a new direction does not undo minimisation along earlier conjugate directions, and exact-arithmetic CG terminates in at most the dimension of the system (Hestenes and Stiefel 1952).

Figure 3.2 uses the same initial point, quadratic, and convergence tolerance for both methods. The matrix has condition number 50. Steepest descent is given its optimal exact line-search step at every iteration, yet takes 59 iterations; CG reaches the two-dimensional minimiser in two. The contour and surface views depict the identical trajectories from different perspectives.

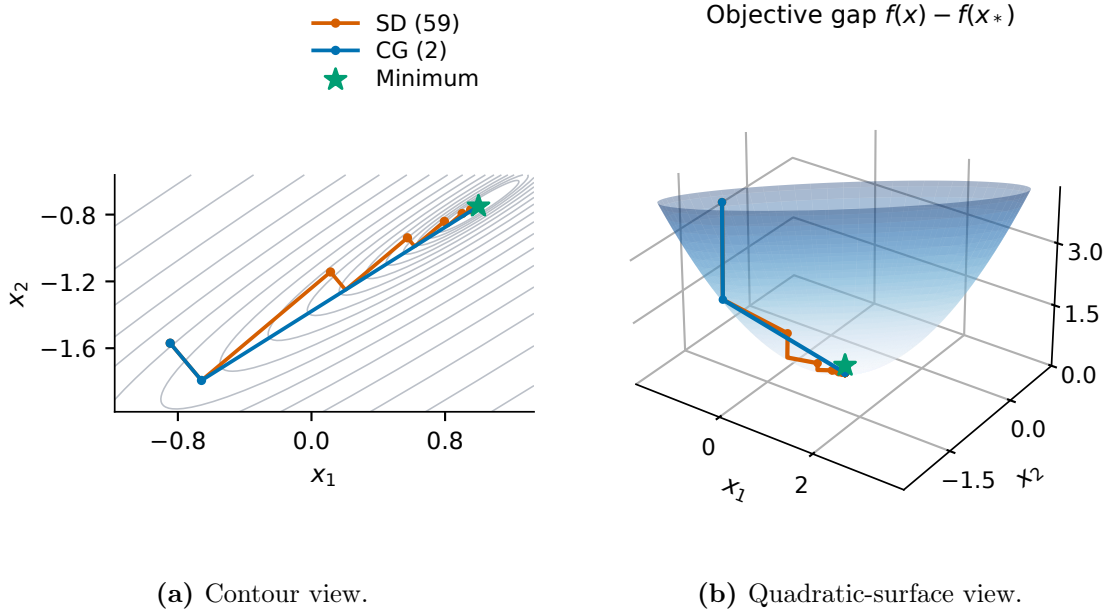


Figure 3.2: Steepest descent and conjugate gradient on the same two-dimensional positive-definite quadratic with $\kappa_2(A) = 50$. Steepest descent uses exact line search. The objective gap is measured relative to the common minimiser.

In this limited role, the illustration motivates conjugate directions without predicting

the statistical ranking of the estimators. That ranking also depends on sampling error, ill-posedness, the restricted objective, and tuning—the distinctions assembled after the stopping rule is considered.

3.9 Early stopping as regularisation

In exact finite-dimensional arithmetic, continued Krylov iteration can eventually solve the estimable part of the empirical equation. That is not the statistical objective. The right-hand side $\hat{r}$ and operator $\hat{K}$ contain sampling noise, and late iterates increasingly fit directions associated with small empirical eigenvalues. The initial iterates are typically smoother and more stable; the sequence may then enter a semi-convergent regime in which the algebraic residual continues to fall while slope error begins to rise. Iteration number is therefore a regularisation parameter rather than only a measure of computational effort (Hanke 1995; Engl et al. 1996).

The discrepancy principle formalises this idea by comparing the moment residual with a sampling-noise scale. If the residual is larger than that scale, the iterate has not yet solved the reliable part of the empirical equation. Once it is comparable with sampling uncertainty, further reduction is more likely to reproduce noise than stable signal. Under the source and moment conditions imposed by Babii et al. (2026), the ideal rule in (3.40) adapts to unknown slope regularity and yields convergence rates for both the unweighted slope norm and covariance-weighted norms. The source condition has the generic form $\beta = K^\mu g$: a larger μ means that the slope is more concentrated in well-identified covariance directions. The practical rule (3.42) replaces the unknown scale by a pilot estimate. Its behaviour is evaluated as part of the complete simulation procedure; the theorem for the ideal threshold should not be read as an automatic guarantee for this particular plug-in pilot without additional consistency and probability-control arguments.

The discrepancy rule above is calibrated for estimation. Inference instead uses the stricter residual requirement derived in Section 2.7, and therefore follows the same path to a later stopping index. For a literal hypothesis $H_0 : \beta = b$, either K must be injective or both β and b must be restricted to $\mathcal{H}_{\text{id}}$; otherwise the data identify only the moment hypothesis $H_0 : K\beta = Kb$.

3.10 Equivalence boundaries and tuning design

The comparison is organised by three decisions: which subspace is searched, which norm is minimised within that space, and how its dimension is chosen. Table 3.1 makes these dependencies explicit; a common component count implies a common estimate only when both the subspace and restricted objective coincide.

Table 3.1: Roles and exact equivalence boundaries of the methods discussed in this chapter.

Procedure	Search space	Finite- m criterion	Relationship and tuning
FPCR	Leading eigenspace of $\hat{K}$	Response least squares	Spectral rather than response-informed; response GCV selects m
NIPALS-type FPLS	Sequential scores spanning $T_n \hat{\mathcal{K}}_m$	Response least squares	Classical PLS1 reference; not an extra reported estimator
Raw FPLS	Raw powers spanning $\hat{\mathcal{K}}_m$	Response least squares	Same fixed- m APLS fit as NIPALS under full rank; five-fold CV
Arnoldi FPLS	CGS2 basis spanning $\hat{\mathcal{K}}_m$	Response least squares solved by QR	Same exact fixed- m fit as raw FPLS; five-fold CV applied separately
Released-code CG-FPLS	Recursive directions spanning $\hat{\mathcal{K}}_m$	Moment-residual least squares	Same nested spaces but generally a different finite- m iterate; discrepancy stopping

Tuning is part of each complete estimator and is therefore retained rather than artificially equalised. Separate tuning means that differences in selected complexity are outcomes to interpret alongside prediction error, slope error, and conditioning. In particular, the raw–Arnoldi contrast asks whether coordinate instability changes an otherwise common APLS target, whereas the Arnoldi–CG contrast changes the finite-step objective itself.

These contrasts determine the Monte Carlo design in Chapter 4, which holds the observations and candidate path common while preserving each procedure’s objective, tuning rule, and numerical representation.

Chapter 4

Monte Carlo Design

This chapter specifies the experiment used to compare the four estimators developed in Chapter 3. Its purpose is to separate the scientific design from the numerical findings. The data-generating process, component-selection rules, diagnostic thresholds, loss functions, and random number streams recorded here define the frozen final experiment; none is redefined using the outcomes reported in the next chapter. The design retains the three functional linear models and principal numerical settings used in the replication study of Babii et al. (2026), while restricting the comparison to the four estimators relevant to this thesis: released-code CG-FPLS, raw FPLS, Arnoldi FPLS, and FPCR.

The three models form controlled comparisons. Model 2 changes only the slope relative to Model 1, whereas Model 3 changes only the score-variance sequence that determines the predictor covariance. Within every Monte Carlo replication, all four estimators receive exactly the same training and test observations. Differences between methods within a model therefore reflect estimation, tuning, and numerical implementation rather than different simulated samples.

4.1 Discretised functional data-generating process

4.1.1 Grid, basis, and inner product

The functional domain is $\mathcal{X} = [0, 1]$. Each curve is observed without additional measurement error on the endpoint-inclusive grid

$$s_t = \frac{t-1}{T-1}, \quad t = 1, \dots, T, \quad T = 200. \quad (4.1)$$

All integrals used by the estimators and evaluation criteria are represented by the equal-weight discrete inner product

$$\langle f, g \rangle_T = \frac{1}{T} \sum_{t=1}^T f(s_t)g(s_t). \quad (4.2)$$

This is the endpoint average used in the code, rather than a trapezoidal rule. Consequently, every displayed finite-sample formula is interpreted under the same numerical geometry as the estimators.

The fixed dictionary contains $J = 100$ functions. Following the indexing in the released replication code, it is

$$\phi_1(s) = 1, \quad \phi_j(s) = \sqrt{2} \cos(j\pi s), \quad j = 2, \dots, J. \quad (4.3)$$

The indexing in (4.3) is intentional. The code first constructs cosine frequencies $1, \dots, J$ and then replaces its first column by the constant function; hence $\sqrt{2} \cos(\pi s)$ is not included. Stating this convention explicitly prevents an apparently harmless re-indexing from creating a different experiment. In continuous $L^2[0, 1]$ the retained functions are mutually orthonormal. On the finite endpoint grid their inner products are evaluated exactly by (4.2) and are not assumed to equal their continuum values.

4.1.2 Predictors and responses

For observation i , draw independent standard normal scores

$$Z_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, J, \quad (4.4)$$

and construct the functional predictor

$$X_i(s_t) = \sum_{j=1}^J \sqrt{\lambda_j} Z_{ij} \phi_j(s_t). \quad (4.5)$$

Here λ_j is exactly the variance multiplier of the independent score Z_{ij} . In the underlying continuous orthonormal system these quantities are covariance eigenvalues. Because the endpoint-grid inner product does not make the retained dictionary exactly orthonormal, however, they should be read as score variances—not as the exact eigenvalues of the finite $T \times T$ discrete covariance matrix. This distinction does not affect any formula below, which is evaluated directly under the discrete geometry. The scalar response is

generated from

$$Y_i = \langle X_i, \beta \rangle_T + \varepsilon_i = \frac{1}{T} \sum_{t=1}^T X_i(s_t) \beta(s_t) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (4.6)$$

where $\sigma = 1$ and the errors are independent of all scores. The population means of X_i and Y_i are therefore zero and the intercept is known to be zero. The simulation passes the generated observations directly to all four estimators; it does not subtract realised sample means. This common convention isolates regularisation and numerical effects from an additional centring step.

For a coefficient vector $b = (b_1, \dots, b_J)^\top$, the slope evaluated on the grid is

$$\beta(s_t) = \sum_{j=1}^J b_j \phi_j(s_t). \quad (4.7)$$

Equations (4.5)–(4.7) define the data directly at the T grid points. There is thus no preliminary smoothing, basis estimation, or interpolation stage in this controlled study.

4.2 The three controlled models

Define baseline slope coefficients and score variances by

$$b_j^{(0)} = \frac{4}{j^{2.7}}, \quad \lambda_j^{(0)} = \frac{2}{j^{1.1}}, \quad j = 1, \dots, J. \quad (4.8)$$

Two targeted modifications are then introduced:

$$b_j^{(1)} = \begin{cases} 4, & j \leq 5, \\ 4j^{-2.7}, & j > 5, \end{cases} \quad (4.9)$$

$$\lambda_j^{(1)} = \begin{cases} 2, & j \leq 5, \\ 2j^{-1.1}, & j > 5. \end{cases} \quad (4.10)$$

The resulting combinations are summarised in Table 4.1.

Figure 4.1 shows both the true slopes and the first twenty coefficient magnitudes. Models 1 and 3 have the same target function. Model 2 distributes substantial slope energy across the first five directions and is therefore more oscillatory. This distinction tests whether a supervised method can exploit response-relevant directions differently from a method ordered only by predictor variance.

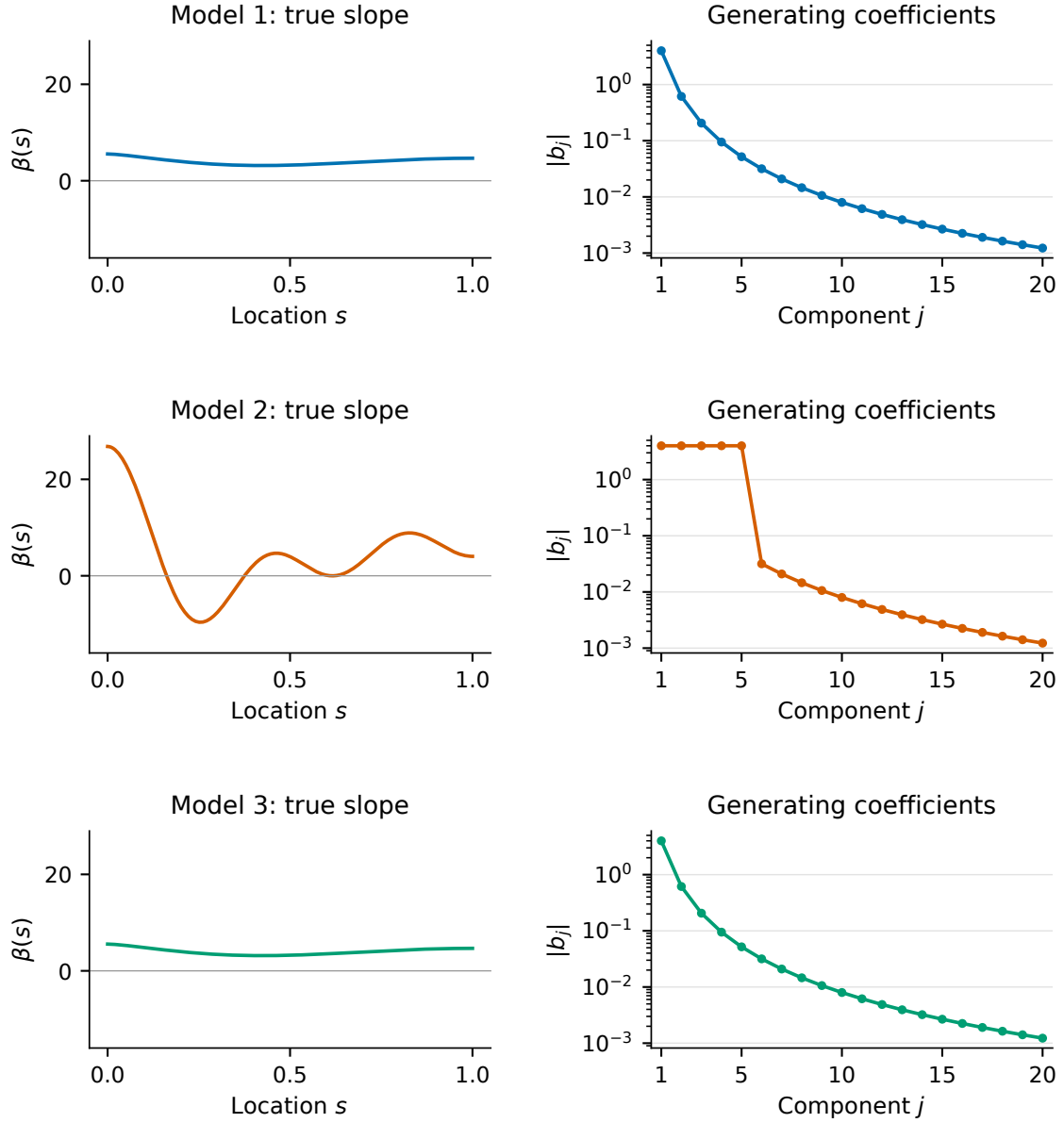


Figure 4.1: True slope functions and coefficient magnitudes in the three Monte Carlo models. The curves use $J = 100$ functions and are evaluated on the $T = 200$ grid. Only the first twenty coefficients are shown in the right-hand panels; all one hundred enter the data-generating process. Rows correspond to Models 1–3, with the slope functions on the left.

Table 4.1: Definition and controlled role of the three simulation models.

Model	Slope coefficients	Score variances	Controlled role
Model 1	$b^{(0)}$	$\lambda^{(0)}$	Baseline: rapidly decaying slope coefficients and a power-law score-variance sequence.
Model 2	$b^{(1)}$	$\lambda^{(0)}$	Slope-only change: the first five coefficients are set to 4, so coefficients 2–5 increase while the first remains unchanged; the predictor distribution is unchanged.
Model 3	$b^{(0)}$	$\lambda^{(1)}$	Predictor-only change: the first five score variances are set to 2, so variances 2–5 increase while the first remains unchanged; the true slope is unchanged.

Figure 4.2 isolates the predictor-side change. Models 1 and 2 share the power-law score-variance sequence $2j^{-1.1}$. Model 3 has a flat leading block of five score variances followed by the same tail. The second panel reports cumulative predictor variation, not predictive signal: these two quantities need not rank directions in the same way.

4.3 Signal strength and effective directionality

The scale of a slope alone does not determine prediction difficulty. Let

$$c_j = \langle \beta, \phi_j \rangle_T, \quad q_j = \lambda_j c_j^2. \quad (4.11)$$

Because the scores are independent and standard normal, q_j is the exact discrete contribution of component j to the variance of the noiseless response. In particular,

$$\mathbb{E}\|X\|_T^2 = \sum_{j=1}^J \lambda_j \|\phi_j\|_T^2, \quad (4.12)$$

$$\text{Var}\{\langle X, \beta \rangle_T\} = \sum_{j=1}^J q_j, \quad (4.13)$$

$$\text{Var}(Y) = \sum_{j=1}^J q_j + \sigma^2. \quad (4.14)$$

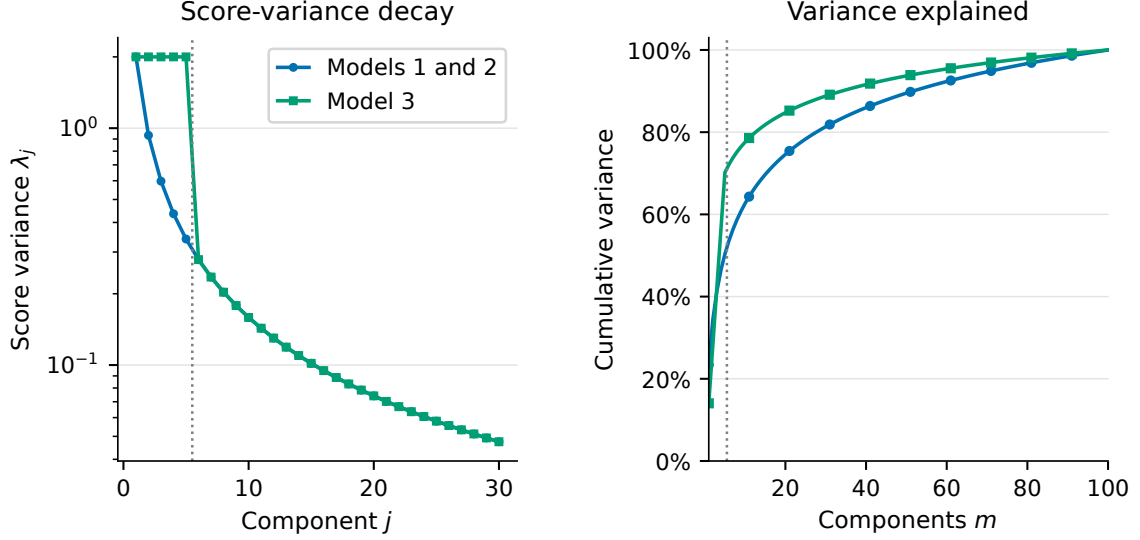


Figure 4.2: Designed covariance structure. The left panel displays the first thirty score variances—the covariance eigenvalues in the continuous design—on a logarithmic scale; the right panel shows their cumulative share over all $J = 100$ components. The dotted vertical line separates the leading block from the common tail.

Appendix Proposition A.28 proves these identities. The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{\sum_{j=1}^J q_j}{\sigma^2}. \quad (4.15)$$

Table 4.2 reports values computed from the finite $J = 100$, $T = 200$ design itself; they are not Monte Carlo estimates. Since $\sigma^2 = 1$, signal variance and SNR are numerically identical. The substantially larger signal variance in Model 2 also means that raw prediction errors should primarily be compared across methods within a model, not used as an unnormalised measure of which model is intrinsically easiest.

Table 4.2: Exact derived properties under the discrete data-generating process. The final column gives the first component counts reaching 90%, 95%, and 99% of predictive signal.

Model	$\ \beta\ _T^2$	$\mathbb{E}\ X\ _T^2$	$\text{Var}\{\langle X, \beta \rangle_T\}$	SNR	First-five share	$m_{90}/m_{95}/m_{99}$
Model 1	16.484	8.589	32.520	32.520	99.986%	1/1/2
Model 2	81.439	8.589	71.306	71.306	99.937%	4/5/5
Model 3	16.484	14.311	33.061	33.061	99.986%	1/1/2

The allocation is made explicit in Figure 4.3. Model 2 needs five components to reach 99% of the exact predictive signal, whereas Models 1 and 3 reach that level after two. This statement describes the population design; it does not predetermine how many empirical components a noisy, data-driven estimator will select.

Representative predictor curves are shown in Figure 4.4. Models 1 and 2 have the same predictor distribution and therefore share a panel. For this setup illustration only, the

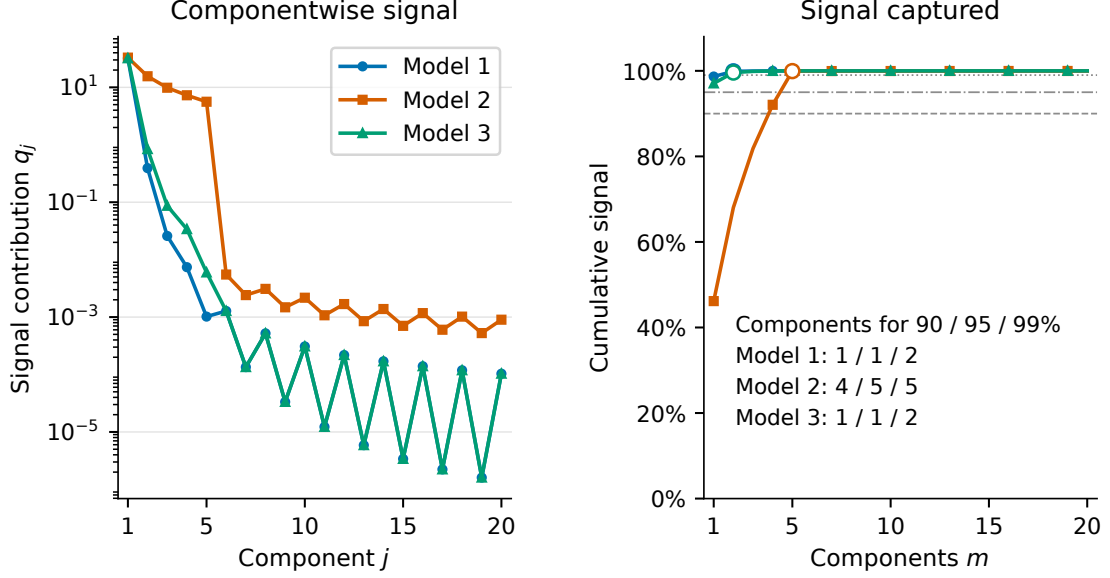


Figure 4.3: Componentwise and cumulative predictive-signal allocation, calculated as $q_j = \lambda_j \langle \beta, \phi_j \rangle_T^2$ under the exact discrete design. Open markers denote the first component count exceeding 99% of total signal.

same Gaussian score draws are used in both panels so that the visibly larger variation for Model 3 can be attributed to its modified spectrum. The shaded regions are exact pointwise 90% Gaussian bands under the corresponding finite data-generating process. They are not estimated confidence bands and are not used by any estimator.

These population contrasts define the signal each estimator must recover. The Monte Carlo protocol now determines how sampling variation and data-driven tuning act on those contrasts.

4.4 Monte Carlo protocol

The full experiment uses $R = 5,000$ independent replications for each of the three models. Within replication r of a given model, the following protocol is applied.

1. Generate a training sample of $n = 100$ curve–response pairs from (4.5)–(4.6).
2. Construct the empirical moments

$$\hat{K} = \frac{X^\top X}{nT}, \quad \hat{r} = \frac{X^\top y}{n}, \quad (4.16)$$

and fit all four methods to the same training sample.

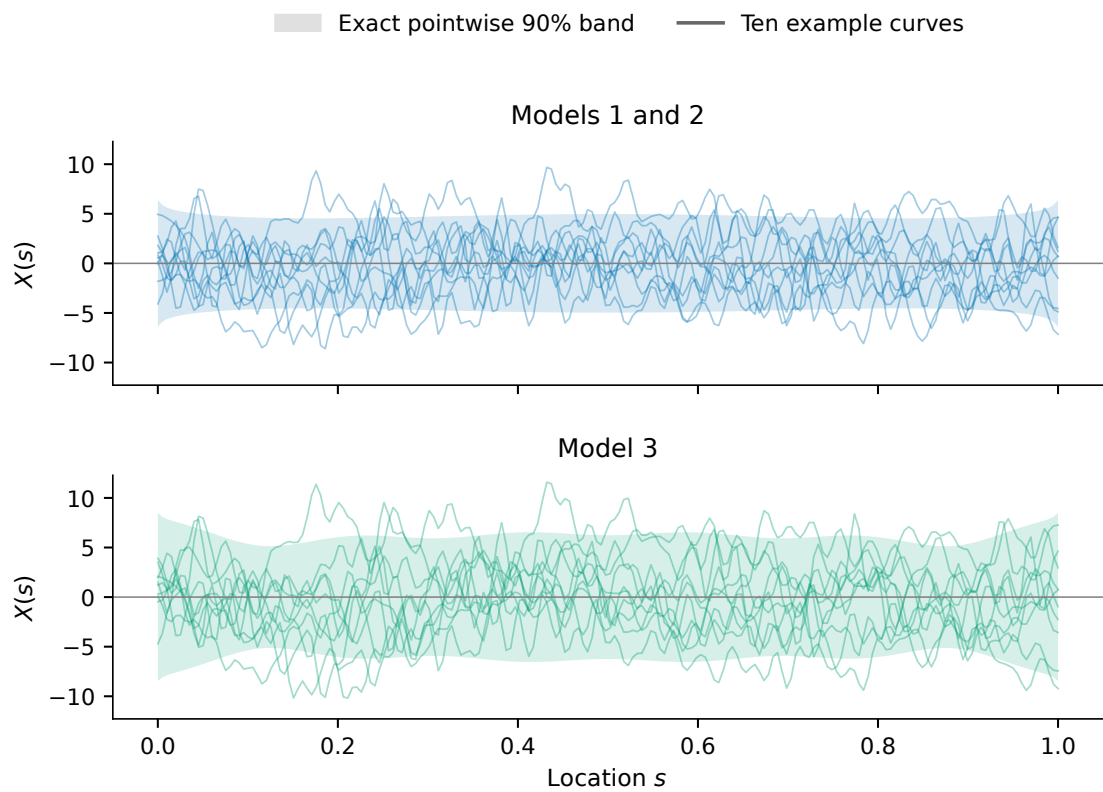


Figure 4.4: Representative functional predictors and their exact pointwise 90% bands: Models 1 and 2 above, Model 3 below. The shared score draws make the spectrum change visually comparable; the actual Monte Carlo models use independent random-number streams.

3. Generate an independent test sample of $n_{\text{test}} = 100$ from the same model. No test observation is used for tuning.
4. Store the selected dimension, slope and prediction errors, and method-specific numerical diagnostics.

The candidate set is $m \in \{1, \dots, 70\}$ for every method. Thus $m = 0$ is never reported as an estimator. For CG-FPLS, absence of a threshold crossing returns the $m = 70$ estimate. The common upper bound keeps the candidate complexities comparable while remaining below the empirical rank limit; the point selected along that path is nevertheless method-specific. Methods are paired within a replication, but the three model streams are independent.

4.5 Estimator-specific tuning

The method labels refer to complete procedures, including their own tuning rules. Table 4.3 provides an overview; the details below fix the criteria and constants.

Table 4.3: The four reported estimators and their tuning mechanisms.

Method	Restricted problem	Selection of m	Numerical construction
CG-FPLS	Moment-residual least squares	First discrepancy-threshold crossing; $\tau = 1.01$, $\delta = 0.1$	Released short recurrence; internal FPCR noise pilot
Raw FPLS	Response least squares on $\hat{\mathcal{K}}_m$	Five-fold prediction cross-validation	Unscaled covariance powers and normal equations
Arnoldi FPLS	Same response least-squares problem as raw FPLS	Five-fold prediction cross-validation	CGS2 Arnoldi basis and response-space QR; rank tolerance 10^{-10}
FPCR	Response least squares on leading empirical eigenvectors	Response-space GCV	Symmetric eigendecomposition and spectral cut-off

4.5.1 Five-fold tuning of raw and Arnoldi FPLS

One random partition $I_1, \dots, I_5$ is generated in each replication and is used by both FPLS implementations. For method $a \in \{\text{Raw}, \text{Arnoldi}\}$, let $\hat{\beta}_{m,a}^{(-k)}$ be the estimate

fitted without fold I_k . Its criterion is

$$\text{CV}_a(m) = \frac{1}{n} \sum_{k=1}^5 \sum_{i \in I_k} \left\{ Y_i - \langle X_i, \widehat{\beta}_{m,a}^{(-k)} \rangle_T \right\}^2. \quad (4.17)$$

Each method minimises its own curve. The common folds remove avoidable Monte Carlo variation, but separate minimisation allows finite-precision instability in the raw basis to affect the complete procedure honestly. A raw component count is eligible only when its fit is finite and valid in the full sample and in every training fold. Arnoldi uses two-pass classical Gram–Schmidt (CGS2) with rank tolerance 10^{-10} . If several eligible values attain the same minimum numerically, the smallest is retained.

4.5.2 Response-GCV tuning of FPCR

FPCR uses the response-space criterion already introduced in (3.8):

$$\text{GCV}_Y(m) = \frac{n^{-1} \|y - X \widehat{\beta}_m^{\text{FPCR}}/T\|_2^2}{(1 - m/n)^2}. \quad (4.18)$$

Only numerically positive empirical eigenvalues are eligible. Under the present $m_{\max} = 70 < n - 1$ design, the shared upper bound is the active path limit. The smallest minimiser of (4.18) is selected.

4.5.3 Discrepancy tuning of CG–FPLS

For CG–FPLS, define the discrete moment norm and predictor-energy estimate by

$$\|e_m\|_T = \left(\frac{1}{T} e_m^\top e_m \right)^{1/2}, \quad \widehat{E}_X = \frac{1}{n} \sum_{i=1}^n \frac{1}{T} X_i^\top X_i, \quad (4.19)$$

where $e_m = \widehat{r} - \widehat{K} \widehat{\beta}_m^{\text{CG}}$. The noise pilot is constructed from the same FPCR path but is selected internally by moment-space GCV,

$$\text{GCV}_M(q) = \frac{T^{-1} \|\widehat{r} - \widehat{K} \widehat{\beta}_q^{\text{FPCR}}\|_2^2}{(1 - q/T)^2}. \quad (4.20)$$

If $\widehat{q}$ minimises this criterion, then

$$\widehat{\sigma}^2 = \frac{1}{n} \|y - X \widehat{\beta}_{\widehat{q}}^{\text{FPCR}}/T\|_2^2. \quad (4.21)$$

This moment-GCV calculation is used only to estimate the threshold scale; it is neither the reported FPCR fit nor a fifth comparator.

For interpretation, an ideal version of the discrepancy scale would replace the unknown quantities below by the population error variance and predictor energy. The implemented rule is feasible instead: it uses the internally estimated $\widehat{\sigma}^2$ in (4.21) and the sample energy $\widehat{E}_X$. Thus $\widehat{\nu}_n$ is a plug-in threshold, not an oracle threshold supplied with the true σ^2 or the population covariance. The selected component count is the first $m \geq 1$ satisfying

$$\|e_m\|_T \leq \widehat{\nu}_n = 1.01 \left(\frac{2\widehat{\sigma}^2 \widehat{E}_X}{0.1n} \right)^{1/2}. \quad (4.22)$$

If no iterate crosses the threshold, $m = 70$ is used and the absence of a crossing is recorded. Together with the cross-validation and GCV rules above, this defines the four complete procedures whose consequences are evaluated through slope and prediction loss. The rule in (4.22) is the single CG selection procedure in the final workflow; no oracle or supplementary CG variant is included.

4.6 Performance measures and summaries

4.6.1 Slope estimation and independent prediction

For method a , the integrated squared error in one replication is the discrete loss

$$\text{ISE}_a = \|\widehat{\beta}_a - \beta\|_T^2 = \frac{1}{T} \sum_{t=1}^T \{\widehat{\beta}_a(s_t) - \beta(s_t)\}^2. \quad (4.23)$$

The independent test mean squared prediction error is

$$\text{MSPE}_a = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \{Y_i^{\text{test}} - \langle X_i^{\text{test}}, \widehat{\beta}_a \rangle_T\}^2. \quad (4.24)$$

MSPE includes irreducible response noise. Conditional on a training-sample estimate and averaging over a new test observation,

$$\mathbb{E}\{\text{MSPE}_a \mid \widehat{\beta}_a\} = \sigma^2 + \sum_{j=1}^J \lambda_j \langle \phi_j, \widehat{\beta}_a - \beta \rangle_T^2. \quad (4.25)$$

Thus the oracle noise floor is $\sigma^2 = 1$, while the second term is excess prediction risk. Equation (4.25) is the discrete counterpart of Appendix Proposition A.27.

For every model–method pair, the archive reports the mean and its Monte Carlo standard error, the median, the 90th and 99th percentiles, and the maximum of both ISE and MSPE. If $L_1, \dots, L_{R_f}$ are the finite values of a loss, the reported standard error is

$$\text{MCSE}(\bar{L}) = \frac{s_L}{\sqrt{R_f}}, \quad (4.26)$$

where s_L is their sample standard deviation. Medians and upper quantiles are reported alongside means because rare numerical failures can make the error distributions extremely right-skewed.

4.6.2 Bias–variance decomposition

The deterministic replay stage reconstructs each selected slope whenever the fit is numerically reproducible. Unstable raw fits are omitted only from replay-based function-valued diagnostics—the bias–variance decomposition, basis-coefficient distributions, pointwise distributions, and normal quantile–quantile plots. They remain in every primary ISE and MSPE summary. With $\bar{\beta}_a = R^{-1} \sum_{r=1}^R \hat{\beta}_{a,r}$, the empirical slope decomposition is

$$\frac{1}{R} \sum_{r=1}^R \|\hat{\beta}_{a,r} - \beta\|_T^2 = \|\bar{\beta}_a - \beta\|_T^2 + \frac{1}{R} \sum_{r=1}^R \|\hat{\beta}_{a,r} - \bar{\beta}_a\|_T^2. \quad (4.27)$$

The two right-hand terms are labelled squared bias and variance. Prediction error is decomposed analogously after weighting the coordinatewise bias and variance by λ_j , with σ^2 added as irreducible noise. Appendix Proposition A.29 proves the finite- R identity. The factor R^{-1} , rather than $(R-1)^{-1}$, is used so the two terms add exactly to the Monte Carlo mean squared error.

The primary risk summaries retain every finite replication. A secondary paired raw-stable subset is used only for a like-for-like bias–variance display when the selected raw power-basis fit can be reconstructed reliably. Whenever that subset is used, the same replications are retained for all four methods and the restriction is labelled explicitly. It does not replace the full-sample ISE or MSPE comparison.

4.6.3 Complexity, instability, and tail events

Selected component count is recorded for every fit and summarised by its mean and median. Complexity is not itself an accuracy measure, but it shows how aggressively each tuning rule regularises the inverse problem.

The simulation also distinguishes statistical error from numerical failure. For raw FPLS, let G_m be the raw Krylov power matrix and $Z_m = XG_m/T$ its response-space design. Let $\tilde{G}_m$ and $\tilde{Z}_m$ denote the same matrices after normalising every column to unit Euclidean length. The diagnostic is

$$\Delta_m^{\text{Raw}} = \max\{\kappa_2(\tilde{G}_m), \kappa_2(\tilde{Z}_m)\}. \quad (4.28)$$

The column normalisation removes trivial scale growth and focuses the measure on near-collinearity. A selected raw fit is flagged as numerically unstable when its loss is non-finite or

$$\Delta_m^{\text{Raw}} \geq \varepsilon_{\text{mach}}^{-1/2} \approx 6.71 \times 10^7. \quad (4.29)$$

For Arnoldi FPLS the corresponding diagnostic is the spectral-norm orthogonality defect

$$\Delta_m^{\text{Arnoldi}} = \|Q_m^\top Q_m - I_m\|_2. \quad (4.30)$$

A fit is flagged when its loss is non-finite or the defect is at least $\varepsilon_{\text{mach}}^{1/2} \approx 1.49 \times 10^{-8}$. CG-FPLS and FPCR are flagged only if ISE or MSPE is non-finite because their relevant path diagnostics are already governed by the recurrence and positive-eigenvalue checks.

Finally, a finite ISE or MSPE value is marked as an extreme tail event if it exceeds 100 times the positive median for its own model–method pair. This threshold is descriptive rather than a deletion rule: extreme observations remain in means, maxima, archives, and full-distribution figures. Reporting failure rates, instability rates, and tail counts alongside robust summaries prevents a numerically fragile procedure from appearing reliable merely because its typical replication is accurate. Because these classifications must themselves be reproducible, the random streams and replay rules are fixed as carefully as the losses and thresholds.

4.7 Randomness and reproducibility

The top-level seed is 2026. A NumPy `SeedSequence` deterministically spawns one independent child stream for each model. Within a model, the training sample is generated first, followed by the random five-fold partition and then the independent test sample. The order matters for exact replay. The four methods share the training and test data, and raw and Arnoldi FPLS additionally share the folds. Models do not share their realised samples.

The final plotting stage deterministically regenerates selected estimates from the seed,

model definition, and archived component counts before checking their ISE and MSPE against the frozen archive. This replay validation is an audit of internal consistency, not a second simulation or an exclusion rule. The exact command, file manifest, validation tolerances, and environment fields to record are kept in the accompanying reproducibility documentation rather than in the statistical argument.

Exact reproduction of the stabilised methods is expected on a common software stack. Small platform differences are more plausible for raw FPLS because ill-conditioned linear algebra is deliberately part of the method being diagnosed; the archived configuration and validation report make those differences visible.

Because the models isolate slope and covariance changes while every method receives the same samples, the results can separate statistical regularisation from numerical implementation. The fixed tuning and replay rules also ensure that adverse tail behaviour remains part of the comparison. Chapter 5 examines how those mechanisms appear in slope recovery and prediction.

Chapter 5

Estimation and Prediction Results

The fixed experiment from Chapter 4 is analysed without post hoc changes to its tuning rules or diagnostics. The results first separate slope recovery from independent prediction and then use selected dimensions, tail behaviour, and paired raw–Arnoldi fits to distinguish statistical regularisation from numerical stabilisation. The central issue is not a single method ranking, but whether the target geometry and numerical representation favour the same amount and form of regularisation.

5.1 Archive verification and reporting conventions

All 60,000 fits enter the primary ISE, MSPE, component-count, and tail summaries. Function-valued replay excludes only the 445 raw fits flagged by the fixed numerical-stability rule, never on the basis of their observed loss; the adverse values therefore remain part of the reported performance of the complete raw procedure.

Means, medians, and upper quantiles answer different questions in this experiment. The mean estimates the risk of using the complete procedure, including rare adverse outcomes. The median describes a typical replication, while the 90th and 99th percentiles describe progressively less common losses without allowing a single catastrophic value to dominate. For the numerically stable methods, the Monte Carlo standard errors of mean ISE range from 0.0018 to 0.0475, and those of mean MSPE range from 0.0024 to 0.0088. The exceptionally large standard errors attached to some raw-FPLS means are themselves evidence of a heavy numerical tail.

5.2 Overall estimation and prediction accuracy

Table 5.1 gives the primary full-archive results. All 5,000 replications enter each row, including every numerically flagged raw fit. The boxplots in Figure 5.1 display the same observations on logarithmic axes, making both central performance and the complete upper tails visible.

Table 5.1: Full-archive slope-estimation and prediction results. Means, medians, and 90th percentiles are calculated over all 5,000 replications for each model–method pair.

Model	Method	Slope ISE			Test MSPE		
		Mean	Median	p_{90}	Mean	Median	p_{90}
Model 1	CG–FPLS	0.633	0.597	0.910	1.270	1.230	1.617
	Raw FPLS	334.298	0.539	0.990	21.224	1.114	1.357
	Arnoldi FPLS	0.760	0.539	0.987	1.130	1.113	1.356
	FPCR	0.670	0.532	0.952	1.126	1.112	1.357
Model 2	CG–FPLS	5.506	5.272	7.354	1.999	1.861	2.753
	Raw FPLS	7.594	5.384	9.949	1.518	1.428	1.793
	Arnoldi FPLS	6.065	5.345	8.736	1.442	1.419	1.763
	FPCR	5.662	5.365	7.392	1.495	1.472	1.836
Model 3	CG–FPLS	0.447	0.426	0.608	1.221	1.176	1.534
	Raw FPLS	1.219	0.446	0.861	1.154	1.114	1.365
	Arnoldi FPLS	0.670	0.446	0.856	1.131	1.114	1.363
	FPCR	0.615	0.463	0.930	1.127	1.113	1.349

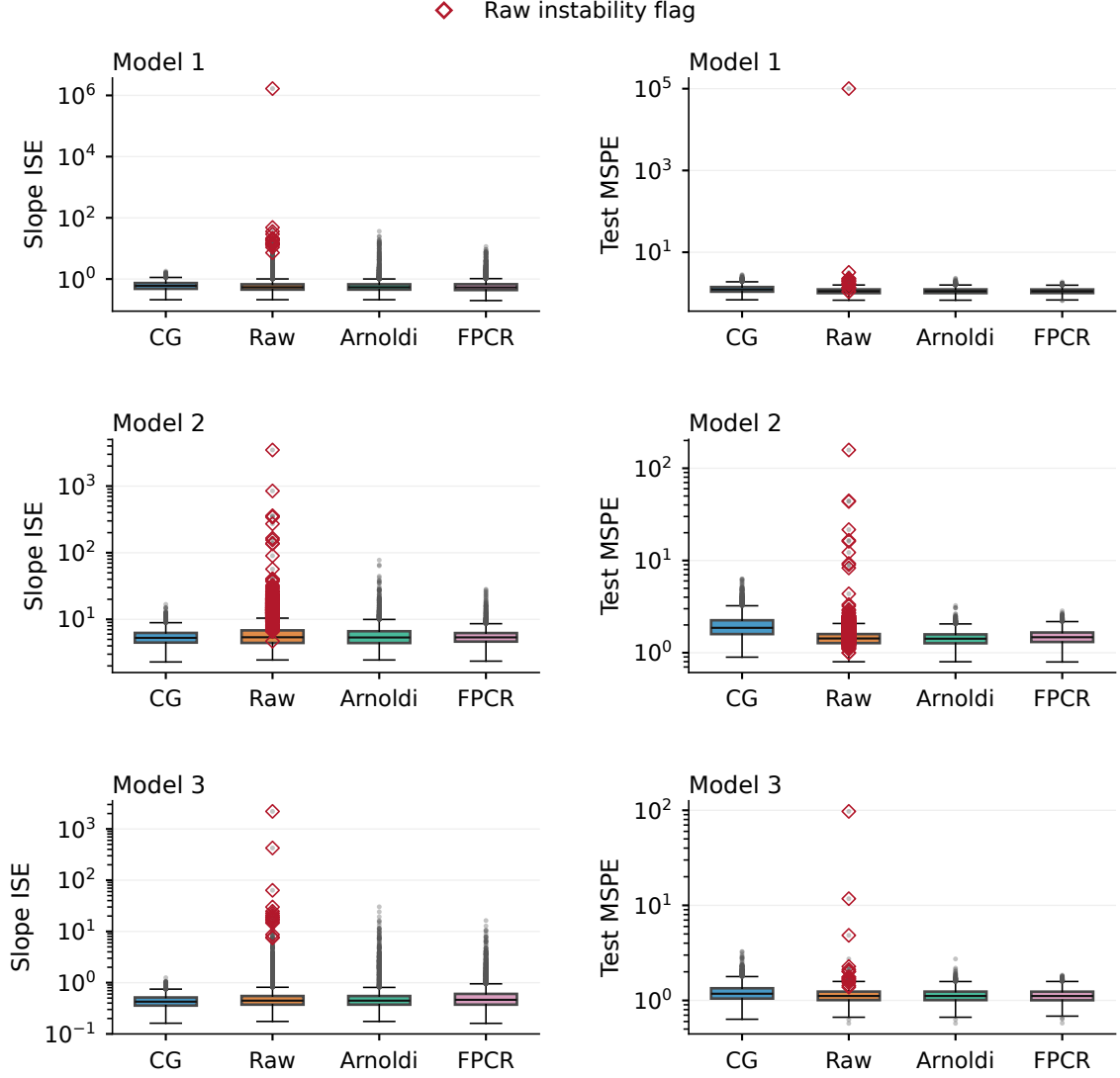


Figure 5.1: Complete empirical distributions of slope ISE (left column) and independent test MSPE (right column), with one row per model. Boxes use the conventional 1.5 interquartile-range whiskers, all displayed axes are logarithmic, and red open diamonds identify raw-FPLS fits satisfying the numerical-instability rule. The labels “CG”, “Raw”, and “Arnoldi” denote the released-code CG-FPLS, raw FPLS, and Arnoldi FPLS procedures. No tail observation is removed.

5.2.1 Model 1: rapidly concentrated predictive signal

In Model 1, the typical slope losses are close. FPCR has the smallest median ISE at 0.532, followed by raw and Arnoldi FPLS at approximately 0.539; the CG-FPLS median is 0.597. The ranking of the stable methods changes when the arithmetic mean is used. CG-FPLS has mean ISE 0.633, compared with 0.670 for FPCR and 0.760 for Arnoldi FPLS, because its aggressive stopping rule produces a comparatively concentrated slope-error distribution. The raw-FPLS mean of 334.298 does not describe its typical

fit: it is driven by a very small number of catastrophic power-basis solves.

Prediction favours FPCR and Arnoldi FPLS. Their median MSPEs are 1.112 and 1.113, respectively, only slightly above the irreducible noise floor of 1. Raw FPLS has the same central behaviour, whereas CG-FPLS has median MSPE 1.230. Thus, CG's lower unweighted mean slope error does not imply a lower covariance-weighted prediction risk: errors in the dominant directions receive greater weight in MSPE. This divergence supplies the concentrated- signal baseline for Model 2, where substantial slope signal extends across five leading directions.

5.2.2 Model 2: a less compressible slope

Model 2 sets the first five slope coefficients to 4 without changing the predictor score variances. Because the first coefficient already equals 4 in Model 1, the actual change is an increase in coefficients 2–5. Every procedure consequently tends to select a richer model and incurs substantially larger slope loss. CG-FPLS has the smallest median and mean ISE, 5.272 and 5.506. FPCR follows with mean ISE 5.662, while Arnoldi FPLS has mean ISE 6.065. The differences in their medians are much smaller: 5.272, 5.365, and 5.345, respectively. The bias–variance decomposition below shows that this is the clearest case in which the early CG rule trades additional bias for reduced slope variance in the ordinary L^2 norm.

The prediction ordering is different. Arnoldi FPLS has the smallest mean and median MSPE, 1.442 and 1.419, followed closely by the typical raw-FPLS fit. FPCR has mean MSPE 1.495, whereas CG-FPLS reaches 1.999. The directions suppressed by the early CG rule therefore matter materially for the response even though retaining them increases unweighted slope variance. Model 2 therefore makes the separation between slope recovery and prediction especially visible. Model 3 supplies the complementary intervention: the slope returns to the baseline while the leading predictor variances change.

5.2.3 Model 3: stronger leading predictor directions

Model 3 keeps the slope of Model 1 and sets the first five score variances to 2. Since the first score variance is already 2 in Model 1, only score variances 2–5 are raised. The resulting stronger information in these directions is accompanied by a reduction in CG-FPLS mean ISE from 0.633 in Model 1 to 0.447, the best value in Model 3. FPCR and Arnoldi FPLS have mean ISEs of 0.615 and 0.670. The median differences are more modest: 0.426 for CG-FPLS, approximately 0.446 for both Krylov least-squares

implementations, and 0.463 for FPCR.

Prediction again favours the less aggressively stopped procedures. FPCR has mean MSPE 1.127, Arnoldi FPLS 1.131, and CG-FPLS 1.221; their medians are 1.113, 1.114, and 1.176. Raising the leading score variances improves identification in the ordinary slope norm, but it also increases the prediction penalty attached to estimation error in those directions. The comparison between Models 1 and 3 therefore shows that a change in the score-variance sequence can affect ISE and MSPE differently even when the true slope is identical. The selected component counts are consequently needed to show how each tuning rule converts these changes in signal and covariance into regularisation.

5.3 Selected regularisation levels

Table 5.2 shows how each complete procedure converts the data into a regularisation level. The response-informed Krylov methods generally use far fewer components than FPCR. This does not mean that one Krylov component equals one principal component; it reflects the fact that a Krylov direction combines covariance information with the response moment.

Table 5.2: Distribution of the selected component count over 5,000 replications. The maximum records rare selections and should be read together with the numerical diagnostics.

Model	Method	Mean	Median	p_{90}	Maximum
Model 1	CG-FPLS	1.098	1	1	2
	Raw FPLS	2.134	2	2	57
	Arnoldi FPLS	2.010	2	2	14
	FPCR	5.840	5	11	42
Model 2	CG-FPLS	2.886	3	3	3
	Raw FPLS	7.820	5	6	70
	Arnoldi FPLS	4.808	5	6	16
	FPCR	18.350	17	28	60
Model 3	CG-FPLS	1.833	2	2	3
	Raw FPLS	2.956	3	3	68
	Arnoldi FPLS	2.845	3	3	11
	FPCR	7.057	6	11	46

The component counts respond systematically to the controlled changes. Model 2 exhibits an upward shift consistent with its predictive signal being less compressible after coefficients 2–5 are increased. This is not a one-to-one correspondence between basis coefficients and Krylov components, because each Krylov direction combines several basis directions. CG-FPLS selects three components in 88.6% of its Model 2 replications, compared with one component in 90.2% of Model 1 replications and two

components in 83.2% of Model 3 replications. FPCR selects many more components—a median of 17 in Model 2—because its ordering is determined by predictor variance rather than by the cross-moment with the response.

All 15,000 CG-FPLS fits cross the plug-in discrepancy threshold before the upper limit, so none is assigned $m = 70$ by the fallback rule. The internal moment-GCV pilot used to scale that threshold selects a median of 70 components in all three models and produces a mean residual variance estimate of approximately 0.300, although the data-generating error variance is 1. Because the pilot usually fits 70 components and divides its residual sum of squares by n without a degrees-of-freedom correction, a value near $1 - 70/100 = 0.30$ is consistent with an overfitting or degrees-of-freedom effect. The pilot is therefore not an accurate stand-alone noise estimator in this finite design. Since the threshold is proportional to $\hat{\sigma}$, this downward bias makes crossing harder and can only delay stopping along the same residual path. The observed one-to-three-component indices therefore reflect rapid residual contraction, but the risk results validate only the implemented plug-in procedure, not the oracle discrepancy theorem. The pilot remains part of the released-code procedure and is not retuned after observing this discrepancy.

The raw and Arnoldi component distributions coincide in their central region but diverge sharply in their maxima. Raw FPLS occasionally selects dimensions as high as 57, 70, and 68, whereas the Arnoldi maxima are 14, 16, and 11. Component counts alone cannot determine whether these extremes reflect statistical complexity or numerical failure, so the next section links them to the fixed stability diagnostics.

5.4 Complete tails and numerical stability

The similarity of medians in Table 5.1 must not be confused with equivalent reliability. Table 5.3 isolates the raw-FPLS stability diagnostic. A flagged fit is not automatically non-finite: all reported losses are finite. Instead, the flag records that the condition number of the normalised raw Krylov basis or of its response design exceeds the fixed threshold. Because raw FPLS then solves normal equations in those coordinates, its reconstructed slope is not regarded as numerically reliable for replay-based function diagnostics.

Model 2 is most exposed, and this exposure coincides with typical FPLS paths of approximately five rather than two or three Krylov directions. Its raw instability rate is 8.18%, compared with 0.40% and 0.32% in Models 1 and 3. In Model 1, only one raw ISE exceeds one hundred times its model-method median, but that single value reaches approximately 1.67 million. This is why raw FPLS has a median ISE of 0.539 and a mean of 334.298 in the same model. Robust summaries alone would conceal a

Table 5.3: Numerical instability and complete upper tails of raw FPLS. Stable-subset size is the number of replications available for paired function-valued replay.

Model	Flagged fits	Rate	Maximum ISE	Maximum MSPE	Stable subset
Model 1	20	0.40%	1.668×10^6	1.005×10^5	4,980
Model 2	409	8.18%	3.482×10^3	1.578×10^2	4,591
Model 3	16	0.32%	2.203×10^3	9.737×10^1	4,984

practically important failure mode, while the mean alone would conceal the method’s ordinarily competitive behaviour.

The combined Arnoldi–QR implementation removes the catastrophic numerical tail in this experiment, not all statistical tail variation. Arnoldi FPLS has no numerical-instability flags, but its maximum ISEs are 36.35, 77.41, and 30.23 across the three models. These values are compatible with ordinary sample and tuning variability and do not coincide with a loss of Arnoldi-basis orthogonality. The corresponding FPCR maxima are 11.53, 28.23, and 16.30, while the CG–FPLS maxima are 1.82, 16.79, and 1.25. The especially compact CG tails are consistent with its strong early-stopping regularisation; the bias–variance decomposition next shows how much variance reduction this achieves and what bias it incurs.

5.5 Bias–variance mechanisms

Figure 5.2 decomposes the losses of the three fully replayable methods over all replications. For raw FPLS, only the frozen archive total is displayed in the full-sample panels: a decomposition into a mean function and dispersion would require reconstructing the 445 platform-sensitive slopes. The same restriction applies to the replay-based basis-coefficient distributions, pointwise distributions, and normal quantile–quantile plots in Appendix B. Table 5.4 gives the numerical components for CG–FPLS, Arnoldi FPLS, and FPCR. Prediction bias and variance are weighted by the independent-score variances λ_j . This is the exact excess-prediction-risk geometry of the discrete data-generating process; the irreducible variance $\sigma^2 = 1$ must be added to their sum.

The mechanism is consistent in all three models. CG–FPLS substantially reduces slope variance but accepts more bias. In Model 1, for example, its slope variance is 0.526, compared with 0.752 for Arnoldi FPLS and 0.660 for FPCR, while its squared bias is 0.107 rather than approximately 0.01. This exchange lowers CG–FPLS mean ISE relative to the other replayable methods. In prediction geometry, however, the CG squared bias and variance are both larger: 0.110 and 0.160, compared with 0.013 and 0.116 for Arnoldi FPLS. Adding the common noise variance gives expected MSPEs of

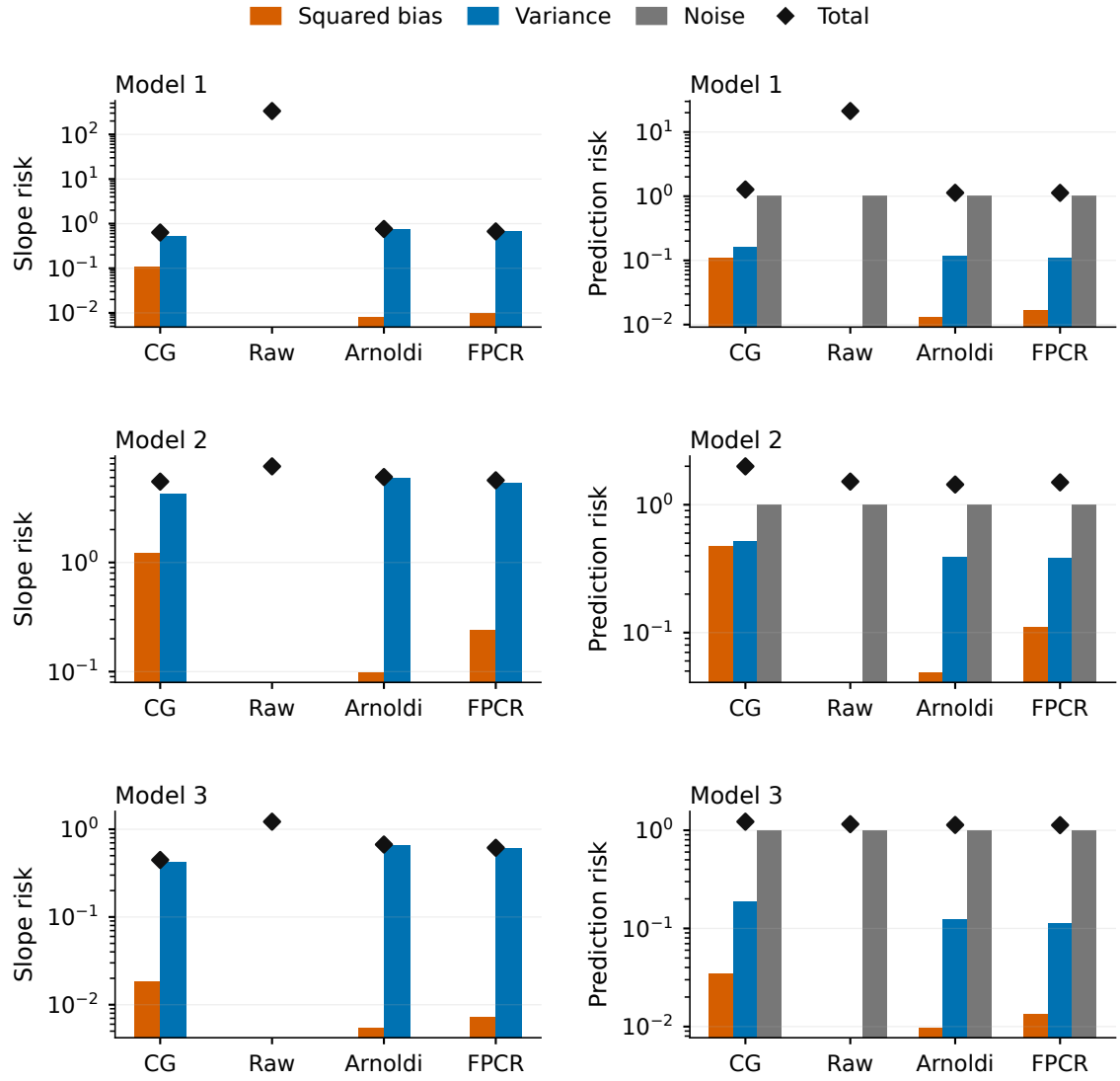


Figure 5.2: Monte Carlo bias–variance decomposition over all 5,000 replications. Each row is one model; slope risk is on the left and prediction risk on the right. Bars show squared bias, variance, and irreducible noise on logarithmic scales; black diamonds show total risk. Full raw-FPLS totals are retained from the archive, but their components are intentionally not reconstructed for numerically unstable fits.

Table 5.4: Bias–variance components for the fully replayable methods. The prediction columns exclude the irreducible noise variance of 1.

Model	Method	Slope risk		Excess prediction risk	
		Squared bias	Variance	Squared bias	Variance
Model 1	CG–FPLS	0.1069	0.5259	0.1102	0.1604
	Arnoldi FPLS	0.0082	0.7520	0.0132	0.1163
	FPCR	0.0098	0.6602	0.0164	0.1102
Model 2	CG–FPLS	1.2263	4.2796	0.4733	0.5224
	Arnoldi FPLS	0.0990	5.9656	0.0491	0.3903
	FPCR	0.2419	5.4197	0.1106	0.3857
Model 3	CG–FPLS	0.0183	0.4282	0.0346	0.1883
	Arnoldi FPLS	0.0055	0.6642	0.0098	0.1230
	FPCR	0.0072	0.6078	0.0135	0.1141

approximately 1.271 and 1.130.

Model 2 sharpens the same distinction. CG-FPLS reduces slope variance by about 1.69 relative to Arnoldi FPLS, more than compensating for its larger unweighted squared bias. After covariance weighting, CG has prediction squared bias 0.473 and variance 0.522, both above the Arnoldi values of 0.049 and 0.390. The procedure can therefore be preferable for the ordinary slope norm and inferior for response prediction without any contradiction. Model 3 repeats this pattern on a smaller scale. Early stopping is not uniformly “more biased but less variable” independently of the norm: the relevant bias and variance must be evaluated in the geometry of the target loss.

5.6 Raw and Arnoldi FPLS on a paired stable subset

The preceding decomposition combines differences in statistical regularisation with differences in numerical implementation. To isolate the latter, Figure 5.3 and Table 5.5 retain only replications in which the raw Krylov basis remains stable and apply that same subset to all four methods. This paired restriction is diagnostic only; it does not replace the primary full-archive comparison.

Table 5.5: Mean errors of raw and Arnoldi FPLS on paired numerically stable raw-FPLS replications.

Model	R_{stable}	Mean slope ISE		Mean test MSPE	
		Raw	Arnoldi	Raw	Arnoldi
Model 1	4,980	0.7085	0.7166	1.1279	1.1282
Model 2	4,591	5.5739	5.6121	1.4271	1.4282
Model 3	4,984	0.6384	0.6384	1.1302	1.1302

On the stable subsets, raw and Arnoldi FPLS are nearly indistinguishable. Their Model 3 mean ISEs differ by less than 3×10^{-7} and their mean MSPEs by less than 10^{-8} . Small differences in Models 1 and 2 arise because each implementation minimises its own finite-precision cross-validation curve, so a small number of selected dimensions differ even before the instability threshold is reached. The comparison supports the equivalence boundary established in Chapter 3: when the raw Krylov basis remains usable, it and the Arnoldi basis approximate the same response-least-squares solution; the practical advantage of Arnoldi is that this equivalence remains numerically accessible farther along the path when the orthogonal basis is paired with the response-space QR solve.

Stabilisation therefore extends how far the shared FPLS target remains numerically accessible; it does not create a different statistical target or guarantee a smaller loss in

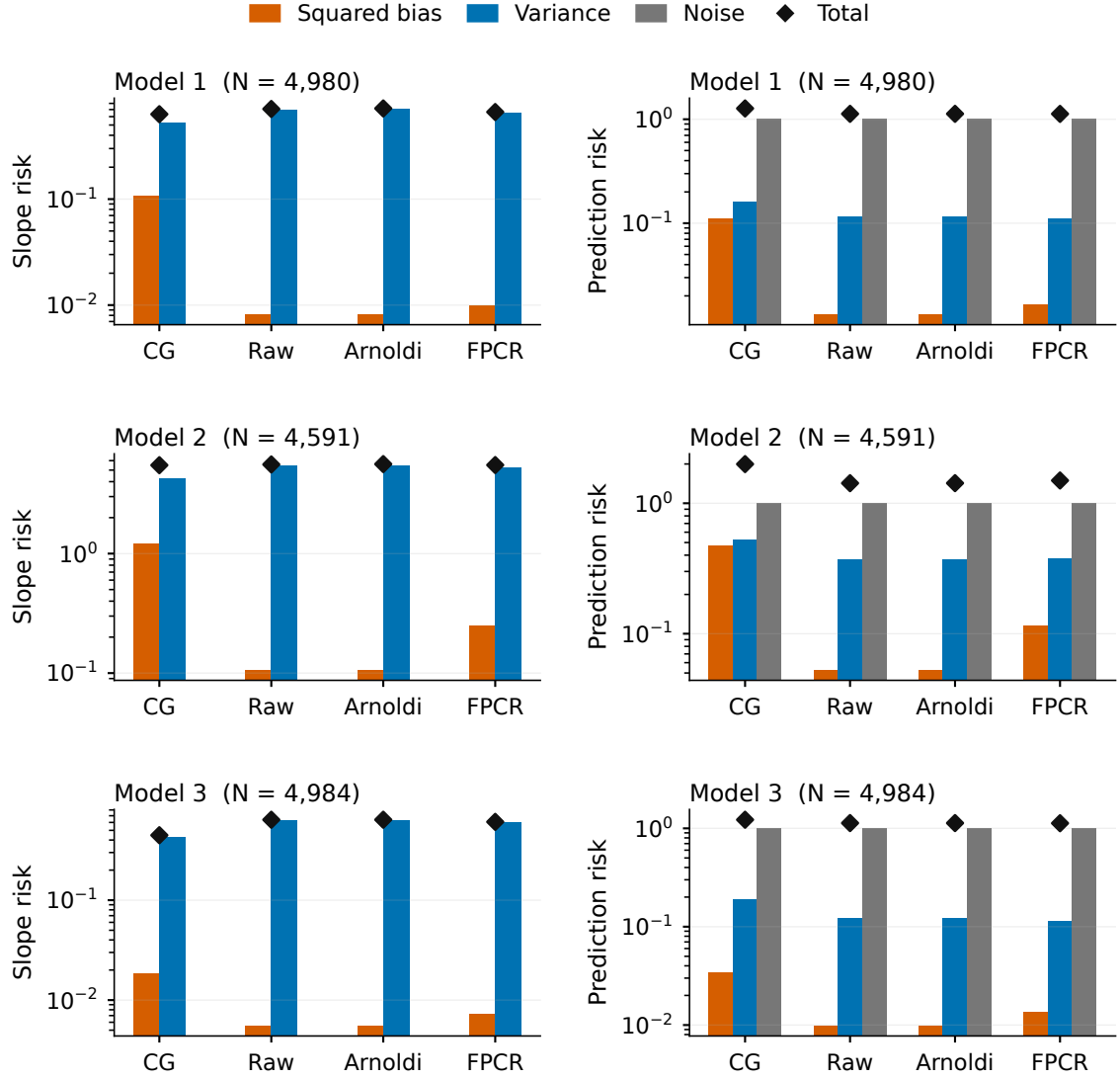


Figure 5.3: Bias-variance decomposition on the paired subset for which raw FPLS satisfies its numerical-stability rule. All four methods are evaluated on exactly the same retained replications within a model. Each row is one model, with its retained sample size shown; slope risk is on the left and prediction risk on the right. Bar colours and black total-risk diamonds have the same meaning as in Figure 5.2.

every sample. CGS2 supplies reliable coordinates for the same nested spaces, while QR avoids response-space normal equations. The coefficientwise and pointwise distributions considered next show what each tuning rule then retains or suppresses in the recovered slope.

5.7 Distribution of the recovered slope

The global error measures are complemented by coefficientwise and pointwise diagnostics in Appendix B. Figures B.1– B.3 show the first three basis coefficients, and Figures B.4– B.6 show the estimated slope at five locations. The raw-FPLS panels use only archived fits satisfying the raw stability rule; the other methods use all 5,000 replications.

These coefficients are recovered from each slope vector on the finite grid, rather than computed as continuum inner products. If $\Phi \in \mathbb{R}^{J \times T}$ has entries $\Phi_{jt} = \phi_j(s_t)$, the plotted coefficient vector is

$$\hat{\mathbf{b}} = (\Phi \Phi^\top)^{-1} \Phi \hat{\boldsymbol{\beta}}, \quad (5.1)$$

implemented by a Moore–Penrose pseudoinverse. On the endpoint-inclusive grid this least-squares recovery is not identical to simply taking discrete inner products with the nominal basis functions.

The coefficient distributions make the effect of early stopping concrete. In Model 1, CG–FPLS estimates the second coefficient with mean 0.366 when its true value is 0.616, and the third with mean 0.071 rather than 0.206. Its first coefficient is also shrunk, but much less severely: 3.867 rather than 4. Since 90.2% of CG fits stop at one component, the procedure captures the dominant supervised direction and deliberately leaves much of the weaker structure unresolved. Raw FPLS, Arnoldi FPLS, and FPCR have smaller bias in these later coefficients but wider functional variation.

The pattern changes with the design. In Model 2, where all first five slope coefficients equal 4, CG–FPLS means for the first three are 3.930, 3.848, and 3.887. In Model 3, the larger leading score variances are associated with CG–FPLS means of 0.599 and 0.198 for coefficients two and three, close to the true values 0.616 and 0.206. This is consistent with the marked improvement in CG slope ISE from Model 1 to Model 3.

The pointwise distributions tell the same story after the coefficients have been aggregated into functions. CG–FPLS is generally narrower but can be visibly shifted from the truth, particularly in Model 1. Model 2 has the widest distributions for every method, a pattern consistent with the less compressible, oscillatory slope and the use of several estimated directions. The normal quantile–quantile plots in Figures B.7– B.9 are approximately linear through their centres for most method–coefficient combinations.

Departures in the tails and bends for some CG coordinates reflect the mixture induced by a discrete, data-dependent stopping rule; they should not be interpreted as evidence that the complete functional estimator has a simple Gaussian distribution.

Across the loss, complexity, stability, and function-level evidence, two distinctions govern the comparison. Aggressive CG stopping stabilises ordinary slope estimation but can leave covariance-weighted signal unresolved; Arnoldi does not change the full-rank APLS target, but keeps that target numerically accessible when raw powers become nearly collinear. Any ranking therefore depends both on the statistical target and on whether the computation reliably realises the intended estimator.

These conclusions are conditional on dense error-free curves, Gaussian scores, $n = 100$, the three chosen designs, and the stated tuning rules. They identify mechanisms rather than a universal ranking. The compact spectroscopy application in Chapter 8 provides a first assessment beyond this controlled setting, while Appendix D retains the complete empirical diagnostics and sensitivity evidence.

That dependence changes again for inference. Estimation can benefit from the bias introduced by a small m , whereas Chapter 6 requires the CG residual to be negligible relative to $n^{-1/2}$ sampling uncertainty. The early stopping used here therefore cannot be transferred unchanged to hypothesis testing.

Chapter 6

Inference Methodology

Chapter 5 showed that the estimation performance of CG–FPLS can rest on aggressive stopping. For inference, the residual left by that stopping rule becomes a potential first-order distortion. The question is therefore not whether regularisation remains useful, but how far the Krylov path must be continued before its approximation error disappears relative to sampling uncertainty. Estimation balances approximation bias against sampling variance in $\hat{\beta}_m$. The inferential procedure of Babii et al. (2026) instead requires the fitted empirical moment $\hat{K}\hat{\beta}_m$ to reproduce $\hat{r}$ to an error smaller than the root- n sampling fluctuation. This chapter makes that distinction operational.

The analysis concerns the moment-residual CG–FPLS path defined in Section 3.7. It is not a new comparison of all four estimators from Chapter 5, and its conclusions do not automatically extend to raw FPLS, Arnoldi response-space FPLS, or FPCR. The analysis moves from the hypothesis and test statistic to the late-stopping requirement, null law, power behaviour, and inverted confidence set. Its implementation is then fixed before the numerical evidence is read in Chapter 7.

6.1 Inferential target and test statistic

Retain the centred scalar-on-function model

$$Y_i^c = \langle X_i^c, \beta \rangle + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad (6.1)$$

and the normal equation $K\beta = r$. The empirical moments $\hat{K}$ and $\hat{r}$ are those in (3.3); their grid implementation is given by (3.5). For a known candidate function b , the

nominal hypothesis is

$$H_0 : \beta = b \quad \text{against} \quad H_1 : \beta \neq b. \quad (6.2)$$

This is a global hypothesis about the slope function rather than a pointwise claim about $\beta(s)$ at one argument. Choosing $b = 0$ gives the important special case of no functional effect, but the procedure is not restricted to that null. In the Monte Carlo size experiment, b is the known true slope of the data-generating model.

The qualification from Section 2.5 remains essential. When K is not injective, the observations identify only $K\beta$ and cannot distinguish functions that differ by an element of $\text{Null}(K)$. The statistically meaningful unrestricted null is therefore

$$H_0 : K\beta = Kb. \quad (6.3)$$

If β and b are restricted to $\mathcal{H}_{\text{id}} = \text{Null}(K)^\perp$, this moment null is equivalent to (6.2). All references below to the slope null are understood on this identified space. In particular, no test based on the functional linear model can have power against a perturbation lying entirely in $\text{Null}(K)$.

Let $\hat{\beta}_m^{\text{CG}}$ minimise the empirical moment residual over the m th Krylov space as in (3.29). It is useful to make the iteration index explicit and write

$$\mathcal{T}_{n,m}(b) = n \left\| \hat{K}(\hat{\beta}_m^{\text{CG}} - b) \right\|^2. \quad (6.4)$$

For an inference-specific sequence $m = m_n$, the shorter notation $\mathcal{T}_n(b) = \mathcal{T}_{n,m_n}(b)$ recovers (1.5). The symbol T_n remains reserved for the sampling operator introduced in Chapter 3; the calligraphic symbol denotes the scalar test statistic.

Weighting the slope difference by $\hat{K}$ is decisive. Direct inference on $\|\hat{\beta}_m - b\|$ would inherit the unstable inversion of small covariance eigenvalues. By contrast, $\hat{K}(\hat{\beta}_m - b)$ lives in moment space, where the sampling fluctuation has a root- n limit under ordinary moment conditions. The procedure consequently tests the covariance-visible part of the slope without requiring an asymptotically regular estimator of the unweighted slope itself.

6.2 Late stopping and the direct-moment statistic

Define the null moment and the CG fitting residual by

$$u_n(b) = \hat{r} - \hat{K}b, \quad e_m = \hat{r} - \hat{K}\hat{\beta}_m^{\text{CG}}. \quad (6.5)$$

The exact decomposition previewed in (2.44) is

$$\widehat{K}(\widehat{\beta}_m^{\text{CG}} - b) = u_n(b) - e_m. \quad (6.6)$$

It separates statistical error in the empirical null moment from algorithmic error left by terminating the iterative solution. The corresponding direct-moment statistic is

$$\mathcal{S}_n(b) = n\|u_n(b)\|^2 = n\|\widehat{r} - \widehat{K}b\|^2. \quad (6.7)$$

Expanding the squared norm in (6.6) gives

$$\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b) = -2n\langle u_n(b), e_m \rangle + n\|e_m\|^2, \quad (6.8)$$

and hence

$$|\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b)| \leq 2\sqrt{n}\|u_n(b)\|\sqrt{n}\|e_m\| + n\|e_m\|^2. \quad (6.9)$$

Under the null, $\sqrt{n}\|u_n(b)\| = O_{\mathbb{P}}(1)$. It follows from (6.9) that the sufficient late-stopping condition is

$$\sqrt{n}\|\widehat{r} - \widehat{K}\widehat{\beta}_{m_n}^{\text{CG}}\| = o_{\mathbb{P}}(1). \quad (6.10)$$

When this condition holds, $\mathcal{T}_{n,m_n}(b) = \mathcal{S}_n(b) + o_{\mathbb{P}}(1)$. The detailed argument appears in Proposition A.40 and Theorem A.41.

Condition (6.10) is stricter than the estimation rule in (3.42). The latter stops when $\|e_m\|$ falls below a data-dependent constant multiple of $n^{-1/2}$; that is appropriate for regularisation, but does not by itself make $\sqrt{n}\|e_m\|$ vanish. Continuing the path can therefore worsen the unweighted slope estimate while improving the approximation relevant to the test. Inference asks for an accurate fitted moment, not an unregularised slope estimate.

The theory does not prescribe a universal component count. In finite samples the required index depends on the covariance spectrum, realised Krylov path, sample size, and numerical precision. The paper's null-distribution experiment and the present null-law and fixed-index stopping branches use $m_{\text{late}} = 70$ as a conservative diagnostic choice at $n = 100$. The power branch instead uses the direct-moment target, while the confidence illustration uses the separate choice $m_C = 10$. The mathematical requirement is (6.10), not the number 70. At the full empirical Krylov dimension, the residual is zero in exact arithmetic whenever $\widehat{r}$ lies in $\text{Range}(\widehat{K})$, but moving that far is neither necessary for first-order inference nor attractive for slope estimation. The stopping experiment therefore examines the entire path rather than treating a chosen index as self-validating.

6.3 Null distribution and critical-value calibration

Suppose the observations are independent and identically distributed, $\mathbb{E}\|X^c\|^4 < \infty$, and $\mathbb{E}(\varepsilon^2 | X) \leq \sigma_{\max}^2 < \infty$. Under $H_0 : K\beta = Kb$, and with the known zero means used in the simulations, the empirical null moment is exactly

$$u_n(b) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i X_i^c. \quad (6.11)$$

If population means are replaced by sample means in an empirical application, the corresponding centring term is asymptotically negligible under the stated moment conditions. Let the covariance operator of the random score εX^c be

$$V = \mathbb{E}\{\varepsilon^2(X^c \otimes X^c)\}, \quad Vh = \mathbb{E}[\varepsilon^2 \langle X^c, h \rangle X^c]. \quad (6.12)$$

The Hilbert-space central limit theorem yields

$$\sqrt{n} u_n(b) \rightsquigarrow G, \quad G \sim \mathcal{N}_{\mathcal{H}}(0, V). \quad (6.13)$$

If $\{\omega_j\}$ are the positive eigenvalues of V , the spectral representation of G and Parseval's identity imply

$$\mathcal{T}_n(b) \rightsquigarrow \|G\|^2 = \sum_{j \geq 1} \omega_j Z_j^2, \quad Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (6.14)$$

The series is well defined because V is trace class under the stated moment conditions. Unlike an ordinary chi-square statistic with known degrees of freedom, this distribution is not pivotal: its weights depend on the joint distribution of X and the regression error.

When the errors are conditionally homoskedastic, $\mathbb{E}(\varepsilon^2 | X) = \sigma^2$, then

$$V = \sigma^2 K, \quad \omega_j = \sigma^2 \lambda_j, \quad (6.15)$$

where $\{\lambda_j\}$ are the positive eigenvalues of K . Under heteroskedasticity, V and K need not share eigenfunctions. The homoskedastic simplification is proved in Proposition A.42.

Let $q_{1-\alpha}$ denote the $(1 - \alpha)$ quantile of the law in (6.14). The level- α rule is

$$\varphi_n(b) = \mathbf{1}\{\mathcal{T}_n(b) > q_{1-\alpha}\}. \quad (6.16)$$

Provided the limiting law is non-degenerate, it is continuous at its quantile, and the rejection probability converges to α under the null. The formal statement is Corollary A.43.

Three calibration concepts are relevant and play different roles.

Oracle weighted chi-square calibration. In this simulation the continuous-design score variances and error variance are known. The oracle benchmark draws independent Z_j and simulates

$$Q^{\text{oracle}} = \sum_{j=1}^J \sigma^2 \lambda_j Z_j^2, \quad (6.17)$$

then uses its empirical quantile. The reported experiment uses 50,000 such draws. The label *oracle* means that unknown population information is supplied by the data-generating design. It does not mean that the resulting test is exact at finite n . The implemented weights describe the continuous design; the endpoint-inclusive, equally weighted grid only approximates its orthonormality. To make the distinction precise, let $\Phi = (\phi_j(s_t))_{j,t} \in \mathbb{R}^{J \times T}$, $G_T = T^{-1} \Phi \Phi^\top$, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_J)$. Under the grid inner product, the non-zero covariance eigenvalues are those of $\Lambda^{1/2} G_T \Lambda^{1/2}$. Writing these eigenvalues as $\nu_{T,j}$, with zeros included if necessary, gives the fixed-grid limit

$$Q_T = \sum_{j=1}^J \sigma^2 \nu_{T,j} Z_j^2. \quad (6.18)$$

With $T = 200$ held fixed, increasing n does not remove the difference between $\nu_{T,j}$ and λ_j . For fixed J , grid refinement gives $G_T \rightarrow I_J$ and recovers the continuous-design reference. The stored experiment uses (6.17), so its calibration discrepancy combines finite-sample effects, grid approximation, and Monte Carlo error in the simulated reference distribution and quantile. It does not isolate finite-sample error relative to the exact fixed-grid limit.

Independent oracle draws are generated for each model. Consequently, two models with the same theoretical weight sequence can receive slightly different simulated quantiles; such a difference is calibration Monte Carlo error rather than a substantive change in the null law.

Empirical-null calibration. When repeated null samples are available in a Monte Carlo experiment, one can take the desired quantile of their realised null statistics. This avoids relying on the continuous-design weighted chi-square approximation, while retaining Monte Carlo uncertainty in the estimated critical value. It is used for the reported power curves, separately for each model and sample size. Such calibration is also simulation-specific: a real dataset does not provide thousands of independent samples generated under a known null.

Feasible calibration for data analysis. With observed data, the nuisance distribution must be estimated. For a specified null, define the observable null residual $\widehat{\varepsilon}_i(b) = Y_i^c - \langle X_i^c, b \rangle$. One route replaces V by the covariance estimator

$$\widehat{V} = \frac{1}{n} \sum_{i=1}^n \widehat{\varepsilon}_i(b)^2 (X_i^c \otimes X_i^c) \quad (6.19)$$

and simulates a weighted chi-square variable using its estimated eigenvalues. Alternatively, a non-parametric bootstrap can approximate the null law. These two feasible directions are discussed by Babii et al. (2026). The supplied Python program also contains an exploratory centred-multiplier option, but the stored archive uses only the oracle benchmark for null size and confidence-set calculations. Consequently, Chapter 7 makes no empirical claim about the unreported feasible calibrations.

6.4 Alternatives and power

Under a fixed alternative with $K(\beta - b) \neq 0$, consistency follows from

$$\frac{1}{n} \mathcal{T}_n(b) \xrightarrow{\mathbb{P}} \|K(\beta - b)\|^2 > 0. \quad (6.20)$$

Thus $\mathcal{T}_n(b)$ diverges and the probability of rejection approaches one. The qualification $K(\beta - b) \neq 0$ again excludes observationally invisible null-space directions.

Non-trivial asymptotic power is described by the triangular sequence

$$H_{1,n} : \quad \beta_n = b + n^{-1/2} \Delta, \quad \Delta \in \mathcal{H}. \quad (6.21)$$

Under the same late-stopping condition,

$$\mathcal{T}_n(b) \rightsquigarrow \|G + K\Delta\|^2. \quad (6.22)$$

Writing $V\phi_j = \omega_j\phi_j$ makes the shift explicit:

$$\|G + K\Delta\|^2 = \sum_{j \geq 1} \omega_j Z_j^2 + 2 \sum_{j \geq 1} \sqrt{\omega_j} Z_j \langle \phi_j, K\Delta \rangle + \|K\Delta\|^2. \quad (6.23)$$

Hence detectability is governed by the covariance-weighted displacement $K\Delta$, not by $\|\Delta\|$ alone. Under homoskedasticity, if $\Delta = \sum_j d_j v_j$ in the eigenbasis of K , then

$$\|K\Delta\|^2 = \sum_{j \geq 1} \lambda_j^2 d_j^2. \quad (6.24)$$

For alternatives of comparable coefficient norm, the deterministic shift is less attenuated in leading covariance directions and such departures are typically easier to detect than departures concentrated in weak directions. More generally, local power depends on the orientation of $K\Delta$ relative to the covariance operator V , not on $\|K\Delta\|$ alone. This is a pointwise local-power statement; it does not imply uniform power over all unit-norm directions in an ill-posed problem. A proof of the fixed and local claims is given in Proposition A.44.

The finite-sample power experiment uses the perturbation family

$$\beta_\delta(s) = \beta(s) + \delta\Delta(s), \quad \Delta(s) = s, \quad \delta \in \{-1, -0.95, \dots, 0.95, 1\}. \quad (6.25)$$

The tested candidate remains the baseline $b = \beta$. Thus $\delta = 0$ is the null and every fixed non-zero δ is a fixed alternative at a given sample size. It corresponds to the theoretical local sequence only if $\delta = c/\sqrt{n}$ for a fixed constant c . This distinction prevents the simulation grid from being mistaken for the literal triangular alternative in (6.21).

To make the large experiment computationally feasible, power is calculated with the direct statistic $\mathcal{S}_n(b)$ in (6.7), evaluated exactly in the generating score coordinates. It is not obtained by running 70 CG iterations for every alternative replication. The reported curves therefore describe the finite-sample power of the direct-moment target, which is the first-order limit of sufficiently late-stopped inference; they are not finite- $m = 70$ power curves. The stopping experiment separately assesses the approximation under the null designs. At the 5% level, empirical null quantiles are computed for each model and for $n \in \{100, 200\}$. Monte Carlo standard errors for the resulting rejection frequencies use the binomial formula in (4.26). These standard errors condition on the realised simulated critical values; they do not propagate the additional Monte Carlo uncertainty from estimating an oracle or empirical-null quantile.

6.5 Confidence sets by test inversion

Any level- α test can be inverted. On an identified parameter space $\Theta \subseteq \mathcal{H}_{\text{id}}$, define

$$\mathcal{C}_{1-\alpha} = \{b \in \Theta : \mathcal{T}_n(b) \leq q_{1-\alpha}\}. \quad (6.26)$$

If the test has asymptotic size α , then $\mathbb{P}_\beta\{\beta \in \mathcal{C}_{1-\alpha}\} \rightarrow 1 - \alpha$; see Corollary A.45. Without the identification restriction, the set is invariant along $\text{Null}(K)$ and describes an equivalence class rather than a unique slope. In a finite sample, $\hat{K}$ is itself finite rank, so unrestricted directions in $\text{Null}(\hat{K})$ also make the raw inverted set unbounded.

For visualisation, the computation restricts candidate functions to a p -dimensional space

$$\mathcal{H}_p = \text{span}\{h_1, \dots, h_p\}, \quad b_c(s) = \sum_{j=1}^p c_j h_j(s), \quad (6.27)$$

using the first $p = 5$ cosine functions from the simulation basis. Let H be the $T \times p$ matrix whose j th column contains h_j on the numerical grid, let

$$A = \widehat{K}H, \quad f = \widehat{K}\widehat{\beta}_{m_C}^{\text{CG}}, \quad (6.28)$$

and recall that the discrete squared L^2 norm is $T^{-1}\|\cdot\|_2^2$. The accepted coefficient set is exactly

$$C_{1-\alpha}^{(p)} = \left\{ c \in \mathbb{R}^p : \frac{n}{T} \|f - Ac\|_2^2 \leq q_{1-\alpha} \right\}. \quad (6.29)$$

When A has full column rank, let

$$\widehat{c} = (A^\top A)^{-1} A^\top f, \quad \rho^2 = \frac{Tq_{1-\alpha}}{n} - \|f - A\widehat{c}\|_2^2. \quad (6.30)$$

If $\rho^2 < 0$, the restricted set is empty. Otherwise it is the ellipsoid

$$(c - \widehat{c})^\top A^\top A (c - \widehat{c}) \leq \rho^2. \quad (6.31)$$

For the row vector $h(s)^\top = (h_1(s), \dots, h_p(s))$, its pointwise extrema are

$$h(s)^\top \widehat{c} \pm \left\{ \rho^2 h(s)^\top (A^\top A)^{-1} h(s) \right\}^{1/2}. \quad (6.32)$$

Proposition A.46 derives (6.31)–(6.32) and records the rank-deficient cases. This analytic calculation gives the exact envelope of the full ellipsoid in $\mathbb{R}^5$, without the notebook’s additional coefficient-box restriction. It replaces the reference notebook’s discrete search over 20^5 coefficient vectors in $[0, 4.5]^5$; the two algorithms therefore illustrate the same test-inversion idea but do not define identical finite grids.

The confidence experiment uses $m_C = 10$, retaining the reference notebook’s separate choice for this illustration, whereas the null-law experiment uses $m_{\text{late}} = 70$. Choosing ten components does not by itself prove (6.10); its adequacy must be judged from finite-sample coverage and residual diagnostics. Thus $m_C = 10$ reproduces a feature of the reference illustration; it is not presented as a generally valid inference-stopping recommendation. Its formal confidence interpretation therefore remains conditional on the moment residual being sufficiently small. Nor is the five-dimensional display the full functional confidence set: the data-generating slopes have non-zero coefficients beyond the fifth basis function. Three reported summaries are therefore related but different:

1. non-rejection when the test statistic is evaluated at the full true simulated slope;
2. acceptance of the true slope's five-term projection within $\mathcal{H}_5$; and
3. the lower and upper pointwise envelope of the accepted five-dimensional coefficient set.

The shaded region drawn in Chapter 7 is formed from the pointwise medians of 5,000 lower and upper envelopes. It is a Monte Carlo summary of many realised sets, not one observed-sample 95% confidence band. Visual containment of the true curve in that median region is consequently not a coverage statement. The full-slope acceptance indicator deliberately reuses the null-size data stream, but it is evaluated at $m_C = 10$ rather than $m_{\text{late}} = 70$. Its average is therefore closely comparable with, but not algebraically complementary to, the reported late-stopped size. Acceptance and rejection are exact complements only when the same stopping index and critical value are used; the reused samples do not provide an independent validation of the size result.

6.6 Implementation and diagnostic design

The inference study reuses the three data-generating models in Sections 4.1 and 4.2. It therefore inherits the same 100-term Gaussian score expansion, 200-point grid, Babii cosine convention, and homoskedastic error standard deviation one. Reusing the design connects the inferential evidence to the estimation mechanisms already isolated, while using a separate seed prevents accidental dependence between the two Monte Carlo archives.

Table 6.1 records the inference-specific choices. Each listed replication count is per model and, for power, per sample size and perturbation value. The size study includes 1%, 5%, and 10% nominal levels; the other branches use 5%.

For the stopping branch, the adaptive comparator uses the same estimation-oriented discrepancy formula and constants as Chapter 3, with $\tau = 1.01$ and $\delta_{\text{stop}} = 0.1$. Its internal FPCR moment-GCV variance pilot uses (4.20)–(4.21), but is capped at 20 components, compared with 70 in the estimation experiment. It also uses the stabilised moment-residual path described below in place of the estimation study's short recurrence. The stopping experiment therefore assesses an estimation-oriented variant; it does not reproduce the complete Chapter 5 procedure unchanged. The pilot cap can alter the variance estimate, discrepancy threshold, and selected index. Writing δ_{stop} distinguishes this tuning constant from the power perturbation δ in (6.25). The maximum-rank benchmark projects $\hat{r}$ onto the numerical range of $\hat{K}$. For every stopping rule, the

Table 6.1: Stored design of the CG–FPLS inference study.

Study branch	Statistic and stopping choice	Sample and replications	Calibration and purpose
Null law and size	$\mathcal{T}_{n,70}$	$n = 100$; $R = 5,000$	50,000-draw oracle weighted chi-square law at $\alpha \in \{0.01, 0.05, 0.10\}$; compare the finite-sample null law with the continuous-design reference.
Stopping diagnostics	$\mathcal{T}_{n,m}$ and $\mathcal{S}_n$	$n = 100$; $R = 5,000$	Oracle 5% critical value; compare $m \in \{1, 2, 3, 5, 10, 20, 40, 70\}$, the estimation-oriented adaptive variant, and the maximum-rank fit.
Power	Direct $\mathcal{S}_n$	$n \in \{100, 200\}$; $R = 5,000$ for every δ	Empirical 5% null quantile for each model and n ; evaluate (6.25).
Inverted-set illustration	$\mathcal{T}_{n,10}$; analytic $p = 5$ inversion	$n = 100$; $R = 5,000$	Oracle 5% critical value; record full-slope non-rejection, five-term projected acceptance, rank, emptiness, and envelope width.

diagnostics retain

$$\sqrt{n}\|e_m\|, \quad |\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b)|, \quad \mathbf{1}\{\mathcal{T}_{n,m}(b) > q_{0.95}\}, \quad (6.33)$$

as well as the selected or fixed index. Together these quantities show whether a path has approached the inferential target and what size distortion remains; merely reporting m would not.

The reported archive uses a numerically stabilised realisation of the moment-residual path. At each dimension, a two-pass reorthogonalisation forms an orthonormal basis Q_m for $\hat{\mathcal{K}}_m(\hat{K}, \hat{r})$, after which a QR solve computes

$$\hat{a}_m \in \arg \min_{a \in \mathbb{R}^m} \|\hat{r} - \hat{K}Q_m a\|_2, \quad \hat{\beta}_m^{\text{CG}} = Q_m \hat{a}_m. \quad (6.34)$$

In exact arithmetic this solves the same nested minimum-moment-residual problem as the released recurrence; the coordinate system and finite-precision path are stabilised because inference deliberately proceeds much farther than the estimation rule. This solver is distinct from the Arnoldi FPLS estimator in Chapter 3: both use an orthogonal Krylov basis, but Arnoldi FPLS minimises response error, whereas (6.34) retains the CG–FPLS moment-residual objective. Propositions A.35 and A.38 establish the relevant span and objective distinctions.

All random quantities derive reproducibly from the master seed 2025. Dedicated streams separate the stopping experiment and the alternative power samples, while several reuses are intentional: the confidence experiment regenerates the null-size samples, the $n = 100$ empirical power calibration reuses the direct null statistics, and the oracle draws serve all relevant branches. The archive retains the configuration, replication-level arrays, summary tables, and vector figures. The executable command and output manifest are recorded in the reproducibility notes rather than embedded in the methodological argument. This separation is deliberate: the chapter fixes the estimand, statistic, approximation, and calibration, while the archive makes their numerical implementation auditable.

This ordering is essential for interpreting the evidence that follows. Before rejection frequencies can be attributed to calibration or power, the iterative approximation must be shown to be negligible. Chapter 7 therefore begins with the stopping diagnostics and then turns to null size, power, and confidence-set behaviour.

Chapter 7

Inference Results

Chapter 6 separated two potential sources of discrepancy: the residual left by finite CG stopping and the approximation used to calibrate the resulting moment statistic. The results therefore begin with the Krylov path itself. Until $\mathcal{T}_{n,m}$ has approached its direct-moment target $\mathcal{S}_n$, departures from the weighted chi-square reference cannot be attributed primarily to calibration. Once this approximation has been assessed, power and confidence-set inversion show what the test detects and which candidate slopes it retains.

The results concern the inference-specific CG–FPLS path rather than a new comparison of all four estimators studied in Chapter 5. This narrower focus is important: raw FPLS, Arnoldi FPLS, and FPCR enter the thesis as estimation and prediction procedures, whereas the inferential theory examined here is built around the late-stopped CG moment residual.

7.1 Archive verification and reporting conventions

The frozen archive retains the configuration, replication-level statistics and diagnostics, five numerical summary tables, and five vector figures. Recomputing the summaries supported by the stored arrays recovers the corresponding reported quantities to floating-point precision. No stored numerical array contains a missing or infinite value, and no replication is removed from the analysis.

The four experimental branches retain the distinctions fixed in Table 6.1. Null calibration evaluates $\mathcal{T}_{n,70}$ at $n = 100$ against a 50,000-draw continuous-design oracle weighted chi-square reference. The stopping experiment compares fixed indices, the estimation-oriented adaptive variant, and the maximum-rank fit on a separate Monte Carlo stream. Power uses the direct statistic $\mathcal{S}_n$ and an empirical null critical value for each model and

sample size. The inverted-set illustration instead uses $\mathcal{T}_{n,10}$, oracle calibration, and an analytic five-dimensional coefficient ellipsoid. Each branch answers a distinct inferential question and is therefore interpreted using its own statistic and calibration.

Unless stated otherwise, probabilities are based on 5,000 replications per model and design point. Their reported Monte Carlo standard errors use (4.26). They condition on the realised simulated critical value and therefore do not include uncertainty from estimating a 50,000-draw oracle quantile or a 5,000-draw empirical null quantile. This additional uncertainty is not quantified in the archive. The resulting error bars are therefore conditional Monte Carlo summaries, not complete uncertainty intervals for the calibration procedure.

7.2 From estimation stopping to inferential stopping

The first question is whether an estimation-oriented discrepancy threshold also makes the CG fitting error negligible at the root- n scale. As specified in Section 6.6, the adaptive comparator retains the earlier threshold formula and constants but caps its variance pilot at 20 components rather than 70 and uses a stabilised moment-residual path rather than the short recurrence. The results below concern this variant, rather than an unchanged rerun of the Chapter 5 procedure. Figure 7.1 follows the path through three linked quantities: the scaled residual $\sqrt{n}\|e_m\|$, the difference $|\mathcal{T}_{n,m} - \mathcal{S}_n|$, and the resulting null rejection frequency.

Table 7.1 extracts the adaptive rule and two fixed indices from the complete paths. The adaptive rule selects a median of one, two, and one components in Models 1–3. More specifically, it selects $m = 1$ in 99.96% of Model 1 samples, $m = 2$ in 92.94% of Model 2 samples, and $m = 1$ in 63.88% of Model 3 samples. These choices are consistent with the rule’s estimation objective, but the resulting null rejection frequencies are 52.70%, 96.02%, and 71.94%. These large discrepancies show that the residual is not negligible in the first-order approximation and causes severe finite-sample distortion in these designs; theoretically, it can enter the statistic at first order rather than merely inducing a small calibration error.

The failure is not caused by the adaptive rule missing its own threshold. All 15,000 adaptive fits satisfy their realised estimation-oriented stopping threshold. On the $\sqrt{n}$ scale, the median thresholds are 11.82, 12.57, and 15.25, all above the corresponding adaptive median residuals in Table 7.1. The rule therefore stops as designed. The problem is that an order-one threshold is compatible with regularised estimation but not with the vanishing requirement in (6.10).

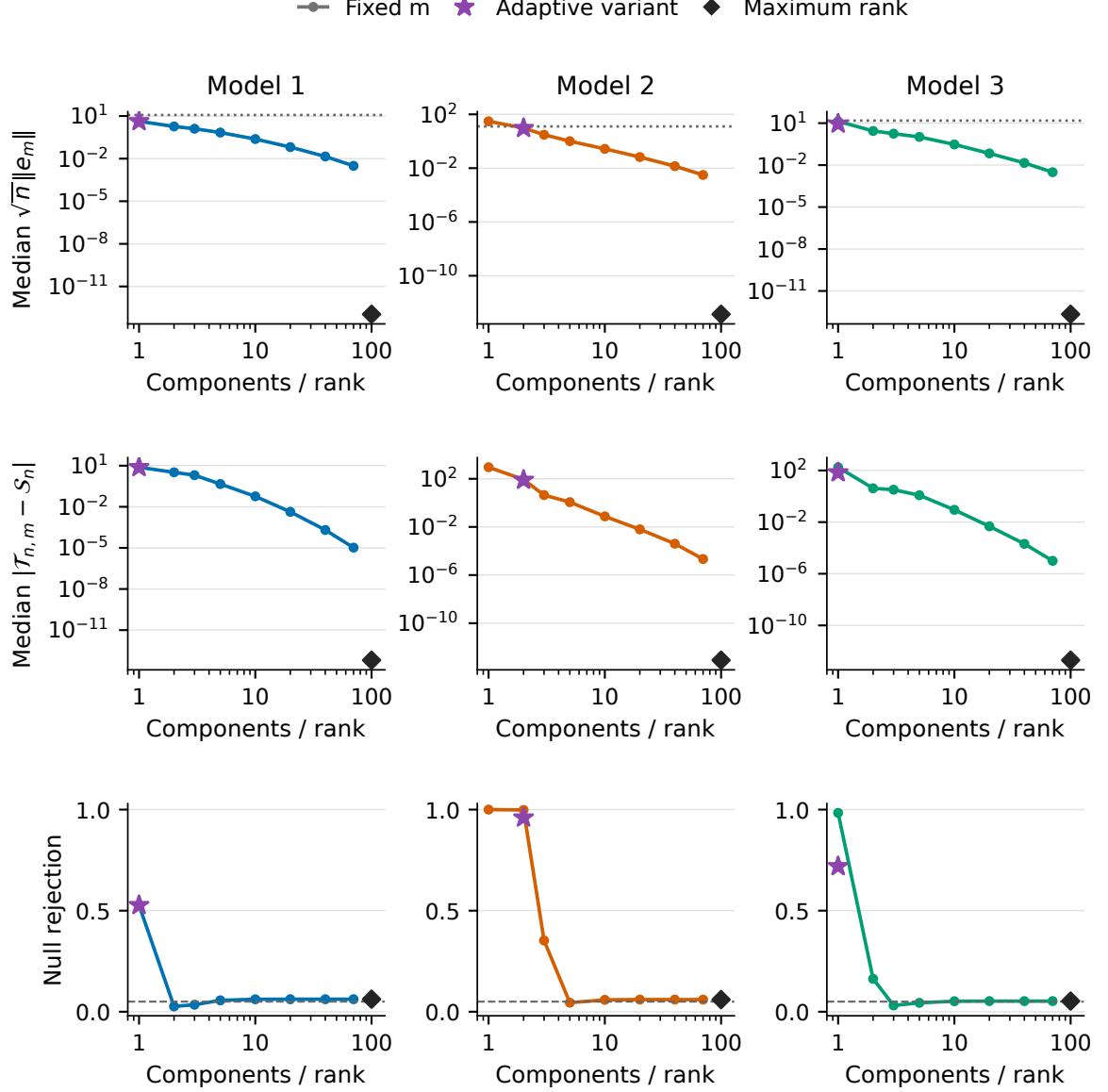


Figure 7.1: Effect of the stopping index on CG-FPLS inference at $n = 100$. The top row shows the median scaled moment residual $\sqrt{n}\|e_m\|$ from (6.5). The dotted line is the median estimation-oriented stopping threshold for $\sqrt{n}\|e_m\|$, not the late-stopping target. The middle row shows the median absolute difference between the finite- m and direct statistics, and the bottom row shows rejection under the null using the oracle 5% critical value. Circles denote fixed m , stars the adaptive variant with a 20-component pilot cap, and diamonds the maximum-rank fit. Each summary uses 5,000 replications from the stopping-diagnostic stream.

Table 7.1: Selected stopping diagnostics from the separate 5,000-replication experiment. The residual and statistic-difference columns report medians; the final column reports the empirical rejection percentage.

Model	Rule	Median m	Median $\sqrt{n}\ e_m\ $	Median $ \mathcal{T}_{n,m} - \mathcal{S}_n $	Rejection (%)
Model 1	Adaptive	1	4.2452	7.6719	52.70
	$m = 10$	10	0.2392	0.05770	6.16
	$m = 70$	70	0.003113	1.05×10^{-5}	6.20
Model 2	Adaptive	2	9.4103	79.9799	96.02
	$m = 10$	10	0.2711	0.07506	5.88
	$m = 70$	70	0.003109	2.16×10^{-5}	6.00
Model 3	Adaptive	1	9.0848	69.7139	71.94
	$m = 10$	10	0.3005	0.08947	5.20
	$m = 70$	70	0.003113	1.01×10^{-5}	5.26

Rejection frequency is not monotone in m . For example, fixed $m = 2$ yields 2.64% rejection in Model 1 but 99.82% in Model 2, while $m = 3$ yields 3.12% in Model 3. At $m = 5$, the three rejection frequencies happen to lie between 4.40% and 5.64%, although the median statistic discrepancy still ranges from 0.458 to 1.203. An apparently satisfactory size at one intermediate index can therefore be accidental; the diagnostic evidence for late stopping is the systematic decay of the residual and statistic difference, not monotonic movement of the rejection rate toward 5%.

By $m = 10$, the median statistic discrepancies have fallen below 0.09 and the three rejection frequencies are already close to their late-path values. This numerical proximity motivates the separate $m_C = 10$ confidence-set illustration in these designs, but it does not validate ten components as a general inference rule. At $m = 70$, all three median scaled residuals are approximately 0.0031 and the median discrepancies are of order 10^{-5} . Continuing to maximum rank reduces both quantities to numerical roundoff without changing any rejection decision relative to $m = 70$ in this stopping experiment.

The separately generated principal size stream provides a decision-level check. Across its 5,000 samples per model, the maximum absolute differences between $\mathcal{T}_{n,70}$ and $\mathcal{S}_n$ are 9.37×10^{-5} , 4.02×10^{-4} , and 1.18×10^{-4} , and the two statistics produce identical decisions at every stored 1%, 5%, and 10% critical value. Iteration error is therefore negligible for the recorded decisions. The remaining departure from nominal size combines finite-sample calibration effects, the fixed-grid approximation described in (6.18), and Monte Carlo error in the reference distribution and critical value. Increasing the CG index does not address those sources of discrepancy.

7.3 Finite-sample null law and empirical size

Figure 7.2 compares the finite-sample law of $\mathcal{T}_{n,70}$ with the continuous-design oracle weighted chi-square reference from (6.17). The distributional centres are close in all three models, while the upper quantiles reveal the calibration discrepancy most relevant to testing.

The empirical and reference means in Table 7.2 differ by only 0.05, 0.12, and 0.02 in Models 1–3. Dispersion is less accurately reproduced. The finite-sample variances exceed their simulated reference values by 11.6%, 20.9%, and 15.6%, respectively. Correspondingly, the empirical 95th percentiles are 3.47%, 3.87%, and 3.71% above the oracle quantiles. The Q–Q panels express the same pattern: central points follow the diagonal closely, but the finite-sample right tail is heavier than the continuous-design reference predicts. Because the sampled basis is not exactly orthonormal, this comparison also includes a grid discrepancy that persists when n increases at fixed T .

Table 7.2: Null-distribution and empirical-size summary at $n = 100$. Quantities labelled “reference” use the 50,000-draw continuous-design oracle weighted chi-square sample. Rejection entries are percentages based on 5,000 finite-sample replications.

Model	Mean		$q_{0.95}$		Rejection probability		
	Empirical	Reference	Empirical	Reference	1% level	5% level	10% level
Model 1	8.590	8.543	15.582	15.059	1.20	5.90	11.18
Model 2	8.666	8.542	15.762	15.176	1.48	5.84	11.72
Model 3	14.273	14.253	27.333	26.355	1.46	6.10	10.72

Models 1 and 2 have the same population score variances and error variance, so their direct-moment null laws and weighted chi-square limits are identical even though their slopes differ. Under the null, the slope cancels exactly from the direct score average in (6.11). Their small differences in the table come from independent finite-sample and oracle simulation draws; the negligible $m = 70$ residual can add only a numerical difference to this comparison. Model 3 has a different covariance design and consequently a wider reference law, with a 95th percentile near 26.36 rather than 15.1.

At the principal 5% level, the oracle-calibrated rejection frequencies are 5.90%, 5.84%, and 6.10%, with conditional Monte Carlo standard errors of approximately 0.33–0.34 percentage points. The common direction of the differences indicates mild liberal behaviour at $n = 100$, although inference about exact calibration would also need to account for uncertainty in the simulated critical values. The 1% and 10% results show the same broad tendency, with magnitudes that vary across levels and models. Figure C.1 in Appendix C displays the 5% result with conditional Monte Carlo error bars.

The direction of the distortion agrees with the quantile comparison: the oracle reference

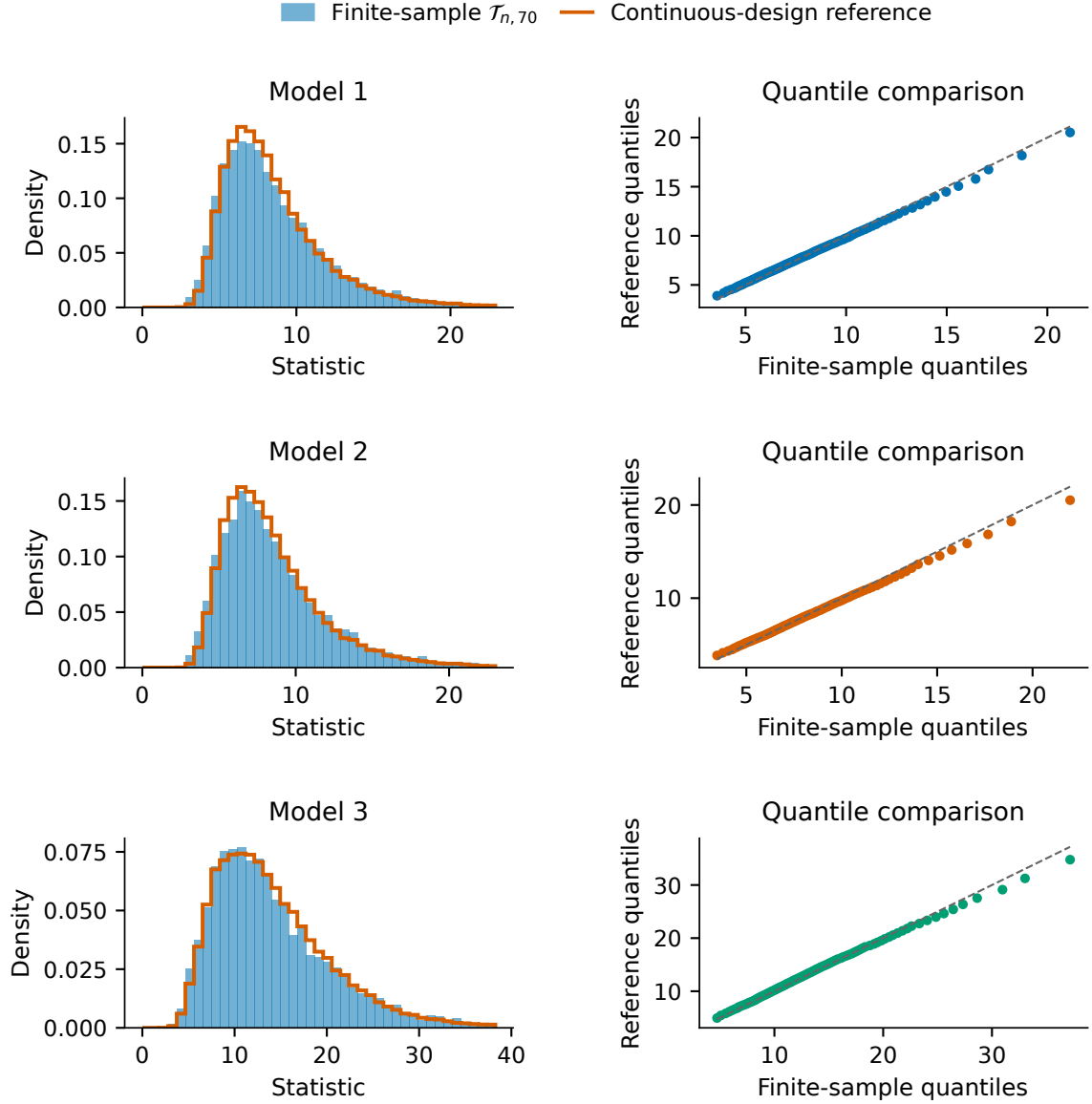


Figure 7.2: Finite-sample null calibration of the late-stopped CG-FPLS test. Rows correspond to Models 1–3. The left column overlays the empirical density of $\mathcal{T}_{n,70}$ from 5,000 samples at $n = 100$ with 50,000 draws from the continuous-design oracle weighted chi-square reference. The right column plots finite-sample quantiles horizontally and reference quantiles vertically against the identity line. Different density scales across models reflect the different null dispersions. Agreement is strongest in the centre; points below the line in the upper tail indicate larger finite-sample quantiles.

understates the finite-sample right tail and therefore rejects slightly too often. Because the $m = 70$ and direct statistics are already numerically indistinguishable, additional CG iterations would not correct this feature. For the power experiment, the analysis instead uses empirical null critical values so that detectability is not confounded with this modest oracle-calibration error.

7.4 Power against fixed perturbations

The power experiment tests the baseline candidate $b = \beta$ while generating responses from $\beta_\delta(s) = \beta(s) + \delta s$. Each non-zero value of δ is a fixed alternative at the stated sample size; the curves should not be read as a direct simulation of the root- n local sequence in (6.21). Moreover, the plotted statistic is $\mathcal{S}_n$, not a finite- $m = 70$ CG statistic. Its critical value is the empirical 95th percentile of a separate null distribution for each model and n , as specified in Section 6.4.

At $\delta = 0$, rejection remains close to 5% in all six model–sample-size combinations, as shown in Table 7.3. These values are not forced to equal 5%: the empirical critical value is obtained from one null sample, whereas the plotted $\delta = 0$ point uses another. The curves are approximately symmetric around zero, as expected for this two-sided quadratic statistic under the symmetric Gaussian design. Across the complete grid, the largest difference between rejection at δ and $-\delta$ is 1.38 percentage points. Because the two rejection rates come from separate alternative samples, the standard error of their difference can be about one percentage point; the observed asymmetry is modest relative to Monte Carlo variation.

Table 7.3: Selected empirical rejection probabilities from the power curves (percent). The displayed non-zero values use the positive branch of the approximately symmetric alternative grid.

Model	$n = 100$			$n = 200$		
	$\delta = 0$	$\delta = 0.25$	$\delta = 0.50$	$\delta = 0$	$\delta = 0.25$	$\delta = 0.50$
Model 1	5.08	37.88	89.84	5.08	64.82	99.74
Model 2	5.24	36.70	89.34	4.58	64.80	99.54
Model 3	4.78	23.60	75.58	4.90	44.56	97.26

Doubling the sample size materially steepens every curve. At $\delta = 0.25$, power rises from approximately 37% to 65% in Models 1 and 2 and from 23.6% to 44.6% in Model 3. At $\delta = 0.50$, Models 1 and 2 move from about 90% power to more than 99.5%, while Model 3 rises from 75.6% to 97.3%. These are fixed-alternative comparisons: increasing n separates a fixed covariance-weighted displacement more clearly from the null score variation.

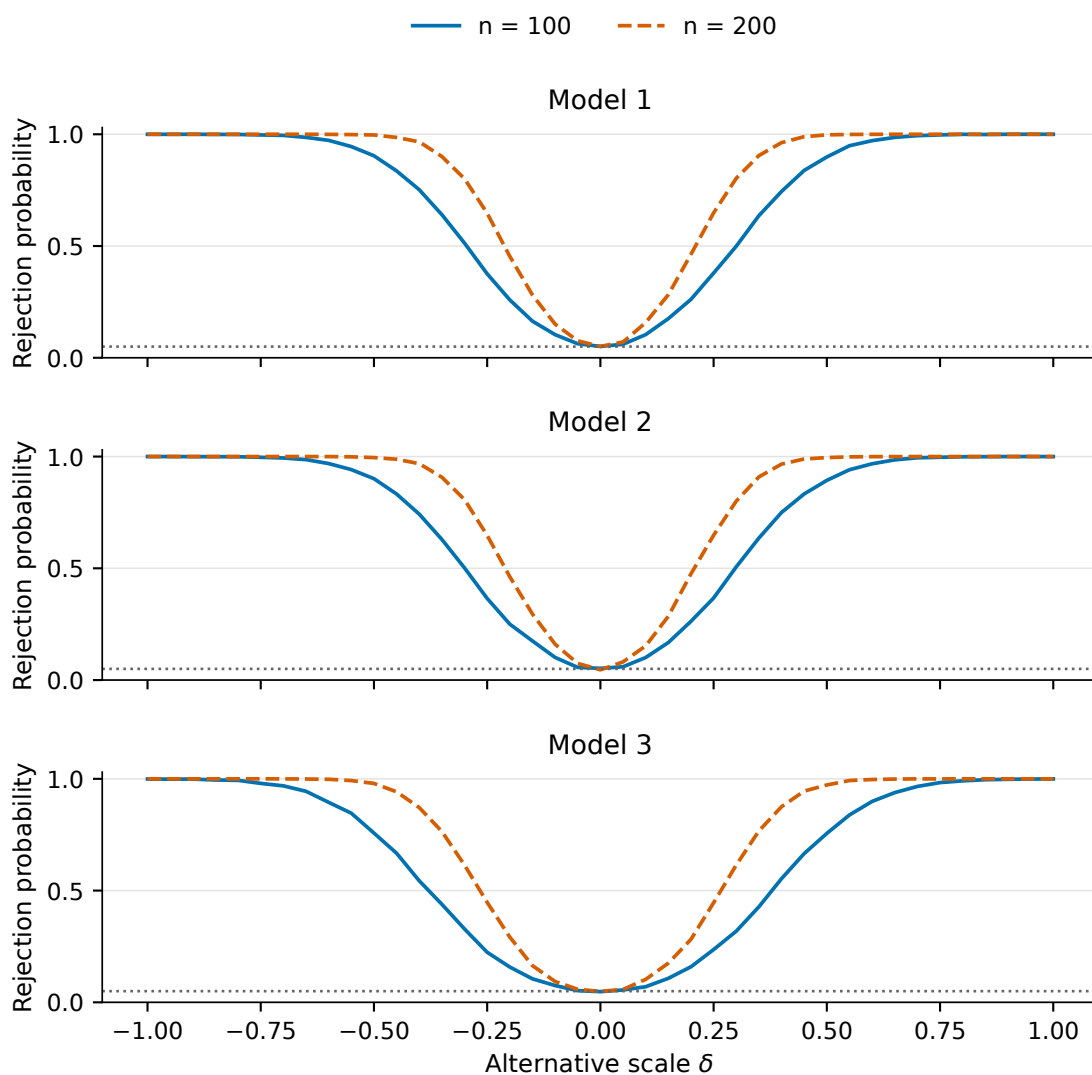


Figure 7.3: Empirical power against $\beta_\delta(s) = \beta(s) + \delta s$. Solid and dashed curves correspond to $n = 100$ and $n = 200$, respectively, and the dotted line marks 5%. Every point uses 5,000 alternative samples and the direct-moment statistic $\mathcal{S}_n$ with a model- and sample-size-specific empirical null critical value. The alternatives use fixed δ ; the figure is not a finite- $m = 70$ power calculation or a literal root- n local-power experiment.

Models 1 and 2 again have nearly coincident curves. They share the same covariance operator and perturbation direction; because the null candidate is the baseline $b = \beta$, subtracting $\langle X, b \rangle$ leaves $\delta \langle X, s \rangle + \varepsilon$ in both models. The small plotted differences are Monte Carlo variation. Model 3 is less powerful over moderate values of $|\delta|$ for this particular direction. That comparison reflects the joint effect of $K\Delta$ and the wider null law; it is not evidence that Model 3 is uniformly less informative against every functional alternative.

The power results are consistent with the theoretical consistency result and demonstrate a clear finite-sample gain for the direct-moment target along $\Delta(s) = s$. They do not separately verify the finite- m CG approximation under every alternative, and empirical-null calibration is not feasible from a single observed dataset. Both limitations matter when the simulation result is transferred to an application.

7.5 Finite-dimensional inverted-set illustration

The final experiment inverts $\mathcal{T}_{n,10}$ over the first five simulation basis functions. Figure 7.4 displays the resulting functional envelopes, while Table 7.4 separates three notions that a plot alone cannot distinguish: non-rejection at the complete simulated slope, acceptance of its five-term truncation, and the pointwise width of the five-dimensional inverted set.

Table 7.4: Summary of the analytic five-dimensional inverted sets. The first two entries are acceptance frequencies in percent. For each replication, width is averaged over the 200-point functional grid; the table then reports the mean and median of these replication-level widths. The final entry gives empty and unbounded frequencies in percent. Every quantity uses 5,000 replications.

Model	Full-slope non-rejection	Five-term acceptance	Mean width	Median width	Median rank	Empty/ unbounded
Model 1	94.20	94.32	2.810	2.798	5	0/0
Model 2	94.20	94.12	2.824	2.817	5	0/0
Model 3	94.02	93.94	1.118	1.114	5	0/0

All 15,000 replication-specific inverted sets are non-empty, bounded, and have rank five. The full-slope non-rejection frequencies are 94.20%, 94.20%, and 94.02%, while the five-term candidate is accepted in 94.32%, 94.12%, and 93.94% of samples. The latter is the acceptance frequency of the five-term truncated candidate in the candidate space underlying the display. It should nevertheless be interpreted with care: the data are generated from the complete slope, whose non-zero tail is not treated as a nuisance component in a separate projection model.

The full-slope values are paired with, but are not exactly one minus, the principal

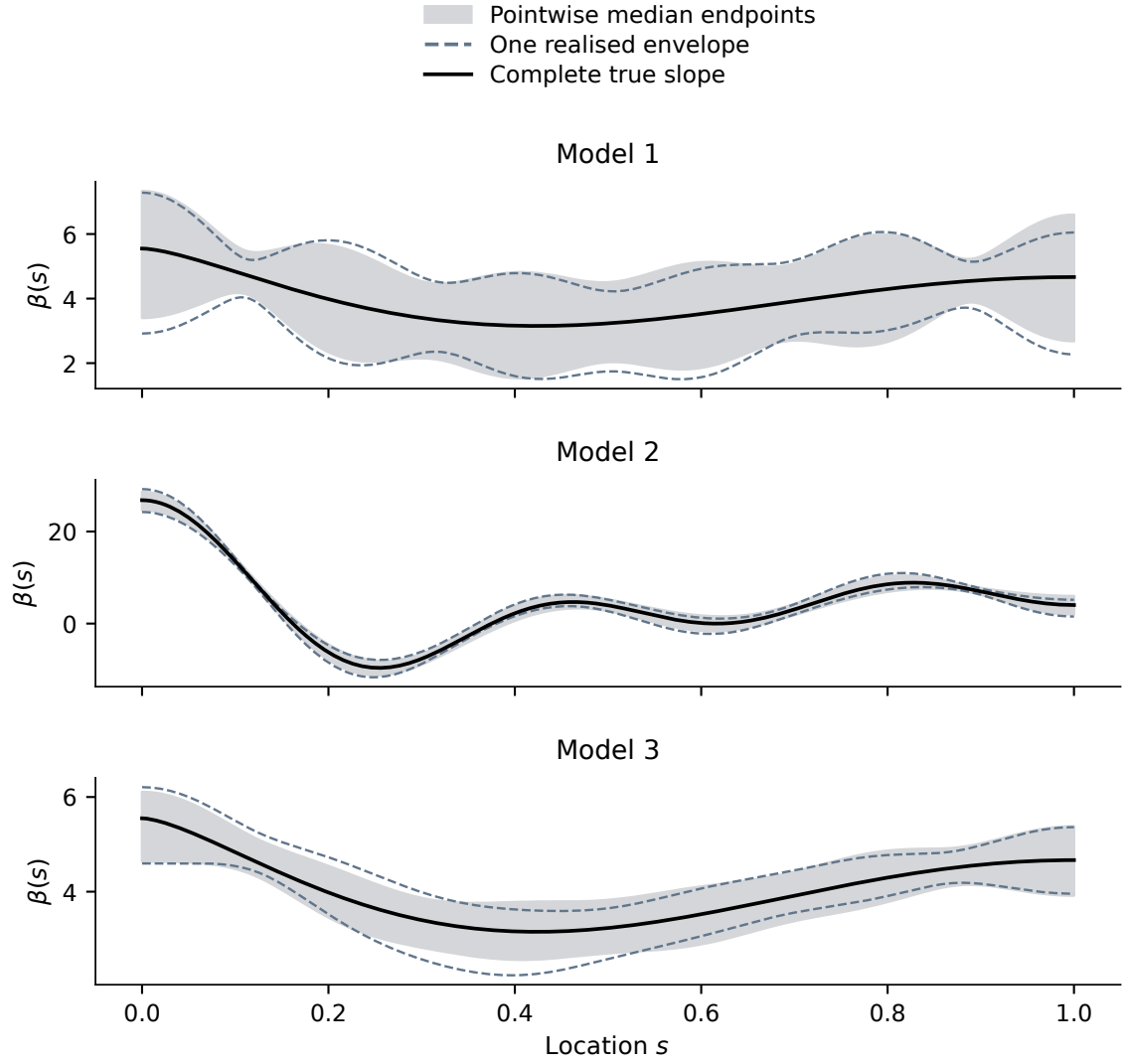


Figure 7.4: Finite-dimensional inverted-set illustration at $n = 100$ using $m_C = 10$, a five-dimensional analytic ellipsoid, and the oracle 5% critical value. The black curve is the complete true slope. The grey region joins the pointwise medians of the lower and upper endpoints over 5,000 separately realised envelopes; it is not one realised 95% confidence band. Dashed curves show the replication whose mean envelope width is closest to the median width. All grey and dashed envelopes range only over the five-dimensional basis span. Pointwise visual containment does not imply simultaneous membership in the coefficient ellipsoid.

size frequencies. The confidence branch regenerates the same samples and uses the same oracle critical values, but evaluates $\mathcal{T}_{n,10}$ rather than $\mathcal{T}_{n,70}$. The two stopping indices change only 5, 6, and 6 of the 5,000 classifications in Models 1–3, producing 94.20%, 94.20%, and 94.02% non-rejection at $m_C = 10$, compared with 94.10%, 94.16%, and 93.90% obtained by complementing the reported $m = 70$ sizes. The closeness is consistent with the stopping diagnostics, but the shared samples make this a paired check rather than independent coverage evidence.

Models 1 and 2 have almost identical mean widths, about 2.81 and 2.82, consistent with their shared predictor covariance geometry. Model 3’s mean width is 1.12, approximately 60% smaller in this five-dimensional representation. The stronger leading score variances provide more information for the displayed coefficient directions, even after allowing for Model 3’s larger critical value. This is a statement about the chosen leading subspace and parameterisation, not uniformly narrower uncertainty for the entire infinite-dimensional slope.

The display is descriptive: neither the pointwise median envelope nor the representative boundaries constitute a realised simultaneous confidence band. Moreover, the interpretation of the $m_C = 10$ inversion remains specific to these designs and conditional on the finite-iteration residual being sufficiently small.

Taken together, the experiments identify a coherent sequence. The estimation-oriented adaptive variant stops with a moment residual that is large on the root- n scale and, in these designs, severely distorts the test. Continuing the stabilised Krylov path makes that algorithmic error negligible; at $m = 70$, the test calibrated against the continuous-design oracle reference is only mildly liberal at $n = 100$. The remaining discrepancy includes finite-sample, grid, and reference-simulation effects. Empirically calibrated direct-moment power then increases sharply with sample size, and inversion yields non-empty, bounded, full-rank finite-dimensional sets in all three controlled designs.

The scope is correspondingly limited. The evidence uses dense Gaussian curves, homoskedastic errors, a single perturbation direction, and either oracle or simulation-specific empirical calibration. No stored result establishes the performance of a feasible plug-in or bootstrap calibration on observed data. Chapter 8 therefore shifts from oracle-assisted inferential benchmarking to prediction with the four regularisation procedures on spectroscopy curves. A formal test in that application would require a feasible calibration whose finite-sample behaviour has not been evaluated here. The principal empirical comparison is kept concise in Chapter 8; the complete preprocessing, tuning, stability, and sensitivity record is reserved for Appendix D.

Chapter 8

Empirical Application: Pharmaceutical-Tablet Spectroscopy

The controlled experiments in Chapters 5 and 7 separate statistical regularisation from numerical approximation because the data-generating slope and covariance operator are known. This chapter asks whether the resulting mechanisms remain visible in an observed functional regression problem. Near-infrared (NIR) spectra are used to predict the active-ingredient assay of pharmaceutical tablets with the four complete procedures studied in the thesis: CG-FPLS, raw FPLS, Arnoldi FPLS, and FPCR. The application concerns prediction and numerical stability. The physical slope is unknown, so it cannot assess slope recovery or support the oracle-assisted inference studied in Chapters 6–7.

The empirical protocol was frozen before any of the four procedures was fitted to the supplied test observations. That test set is nevertheless described as a *designated benchmark*, not a prospectively blind holdout: its responses and preliminary ordinary PCR/PLS performance had already been inspected while selecting a dataset. This qualification is material to the scope of the evidence. The chapter therefore reports a conditional method comparison on the supplied tablets rather than an estimate of future deployment performance.

8.1 Data and prediction problem

The data come from the 2002 pharmaceutical-tablet NIR “Shoot-out” described by Hopkins (2003) and distributed in MATLAB format by Eigenvector Research (2026). The archive contains spectra from two spectrometers and three supplied splits: 155 calibration tablets, 40 validation tablets, and 460 test tablets. These add to 655 tablets, although the online catalogue reports 654. All 655 complete archive rows are retained,

and the discrepancy is recorded rather than resolved through an undocumented deletion. The two instruments measure the same tablets; they are paired measurements and are never stacked as independent observations.

The primary predictor is the instrument-1 spectrum at all $T = 650$ wavelengths from 600 to 1898 nm in 2-nm increments. The sole response is assay, the archive’s third response column. Its physical unit is not stated in the distributed object, so prediction errors are reported in the supplied assay units. The calibration and validation rows form a 195-tablet development sample, while the 460 test rows form the designated benchmark. Figure 8.1 shows substantial common spectral structure, a visibly less regular high-wavelength tail, and the unusually narrow assay range of the supplied validation split.

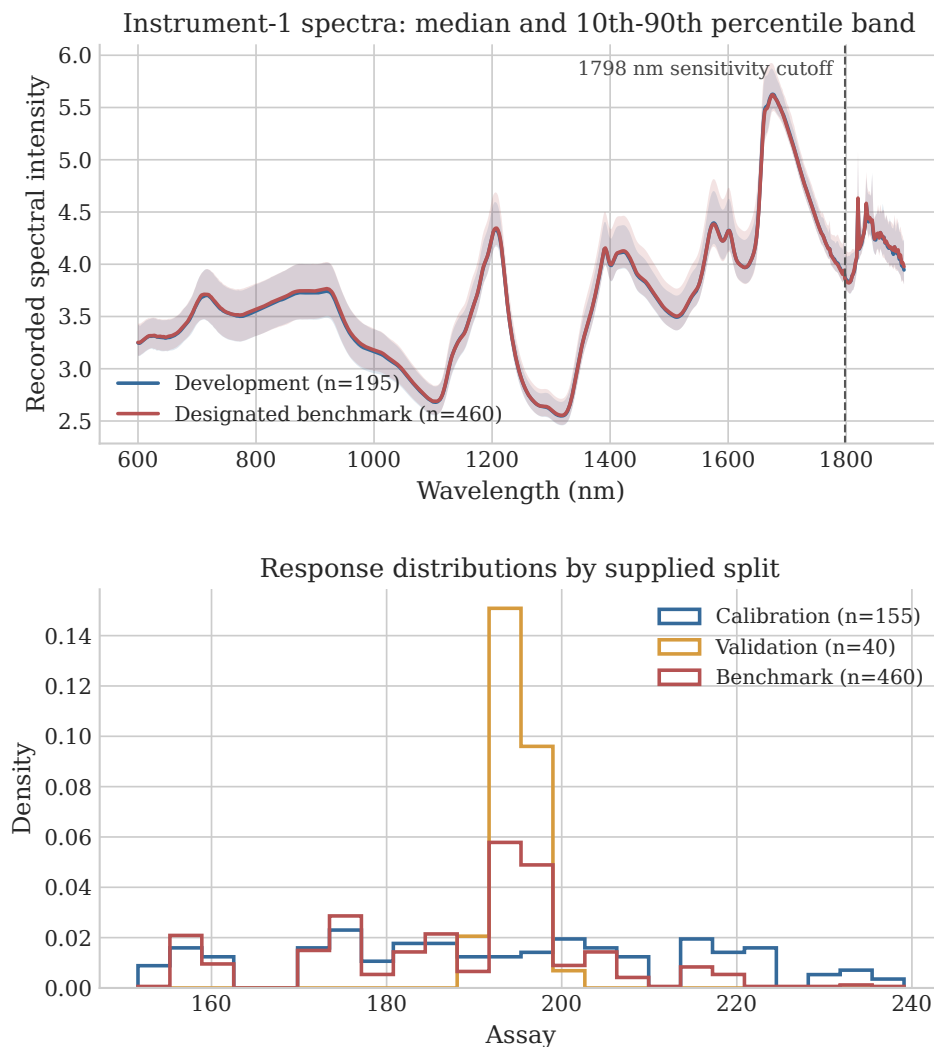


Figure 8.1: Instrument-1 spectra and supplied assay distributions. The spectral panel gives the median and 10th–90th percentile band for the development and designated-benchmark samples; the dashed line marks the prespecified 1798 nm boundary used only in a sensitivity analysis. The response panel separates the calibration, validation, and benchmark splits.

This is an empirically ill-posed functional regression problem. There are 650 grid coordinates but only 195 development observations, and sample centring limits the fitted covariance rank to at most 194. At least 456 grid directions are consequently unidentified from the development sample. The equal-grid convention from Section 3.1 is retained:

$$\langle f, g \rangle_T = T^{-1} f^\top g, \quad \hat{K} = \frac{X^\top X}{nT}, \quad \hat{r} = \frac{X^\top y}{n}. \quad (8.1)$$

The resulting prediction for a new spectrum x is

$$\hat{Y}(x) = \bar{Y}_{\text{train}} + \frac{1}{T}(x - \bar{X}_{\text{train}})^\top \hat{\beta}. \quad (8.2)$$

This preserves the scale of the existing implementation and makes the training-sample intercept explicit.

8.2 Frozen evaluation protocol

The primary spectra are otherwise unprocessed. Every wavelength and the response are centred within each training fold, and those training means are then applied unchanged to the corresponding validation observations. There is no primary variance scaling, smoothing, differentiation, standard-normal-variate transformation, multiplicative-scatter correction, baseline correction, or wavelength selection. This restrained primary analysis keeps preprocessing from becoming an unreported regulariser. Prespecified trimmed-spectrum, smoothed-spectrum, and instrument-2 analyses are reported as sensitivities in Appendix D.

The complete procedures retain their native regularisation rules. Raw and Arnoldi FPLS independently minimise prediction mean squared error over at most 70 requested Krylov dimensions, using the same five inner folds. Raw FPLS uses unscaled covariance powers and response-space normal equations; Arnoldi uses twice reorthogonalised Gram-Schmidt and response-space QR. FPCR uses response-space GCV over its spectral cut-offs. CG-FPLS uses the released-code conjugate-residual recurrence and discrepancy rule with $\tau = 1.01$ and $\delta = 0.10$, including its internal moment-GCV variance pilot. Component counts therefore do not represent common effective degrees of freedom. The comparison concerns four complete regularisation procedures, consistent with the three-axis framework in Section 3.10.

Development performance is assessed by five repeats of five-fold outer cross-validation. Folds are constructed separately within the supplied calibration and validation rows and balanced over assay, so every outer fold contains 31 calibration and 8 validation tablets. Raw and Arnoldi receive a new common five-fold inner partition inside each

outer training sample. The five repeat-level out-of-fold summaries describe development behaviour; the 25 overlapping fits are not treated as independent replications.

After development checks and code review, each method is fitted once to all 195 development tablets. A single frozen, common inner partition tunes Raw and Arnoldi, and the resulting models predict the 460 benchmark tablets once. The primary endpoint is RMSE. MAE, signed bias (prediction minus observation), and $R^2 = 1 - \text{SSE}/\text{SST}$ are secondary. Conditional uncertainty is summarised by 10,000 paired iid bootstrap resamples of the benchmark tablets. The same resampled tablet indices are used for every method, but the models are not refitted. Circular moving-block bootstraps of lengths 5, 10, and 20 assess possible row-order dependence. Exact folds and bootstrap indices are retained in the frozen archive.

8.3 Benchmark prediction

All four final fits and all benchmark predictions are finite. Table 8.1 reports the prespecified metrics. Every method explains substantial variation in the supplied benchmark assay, with R^2 between 0.887 and 0.914. Raw FPLS has the smallest point RMSE, 4.607, followed by Arnoldi FPLS at 4.643. FPCR and CG-FPLS have RMSEs of 5.185 and 5.293, respectively.

Table 8.1: Primary designated-benchmark results for full-spectrum instrument 1 ($n = 460$). Brackets give 95% percentile intervals from 10,000 paired iid bootstrap resamples of the benchmark tablets. The intervals shown beside RMSE are marginal method intervals, whereas method comparisons use paired contrasts in Table 8.2.

Method	m	RMSE [95% interval]	MAE	Bias	R^2
CG-FPLS	3	5.293 [4.367, 6.477]	3.673	0.177	0.887
Raw FPLS	55	4.607 [3.645, 5.785]	3.036	-0.303	0.914
Arnoldi FPLS	5	4.643 [3.693, 5.805]	3.086	-0.301	0.913
FPCR	53	5.185 [4.299, 6.255]	3.547	-0.628	0.892

Figure 8.2 shows that the four methods largely make the same errors. The raw and Arnoldi predictions are especially close: their correlation is 0.999934, the root mean squared difference between paired predictions is 0.174 assay units, and the maximum difference is 0.445. This pairing explains why the two wide marginal RMSE intervals in Figure 8.3 can overlap almost completely while the paired contrast is estimated much more precisely.

The focal paired difference in Table 8.2 is $\text{RMSE}_{\text{raw}} - \text{RMSE}_{\text{Arnoldi}} = -0.036$ assay units. Its iid percentile interval excludes zero, as do all three moving-block sensitivity intervals. The numerical difference is nevertheless only 0.78% of the Arnoldi RMSE. It

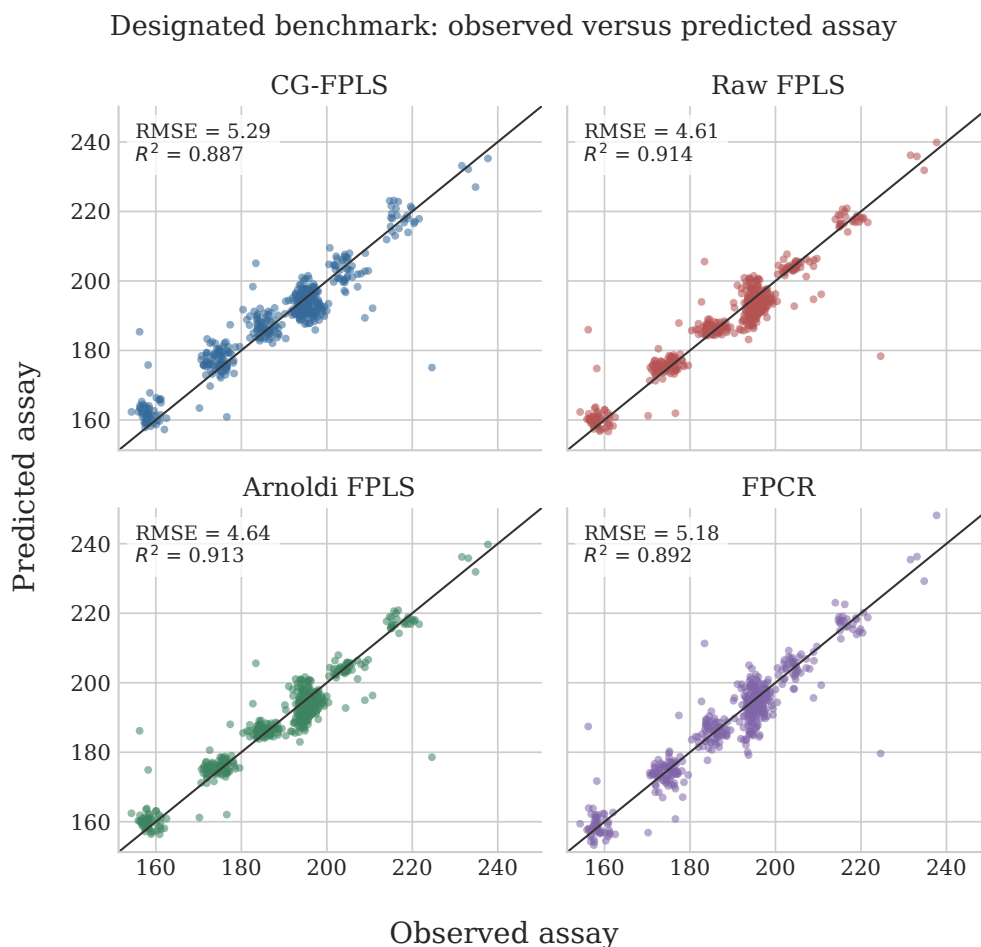


Figure 8.2: Observed and predicted assay on the designated benchmark. The diagonal is the one-to-one line. The four methods capture the main linear signal and share several difficult tablets near the extremes of the assay range.

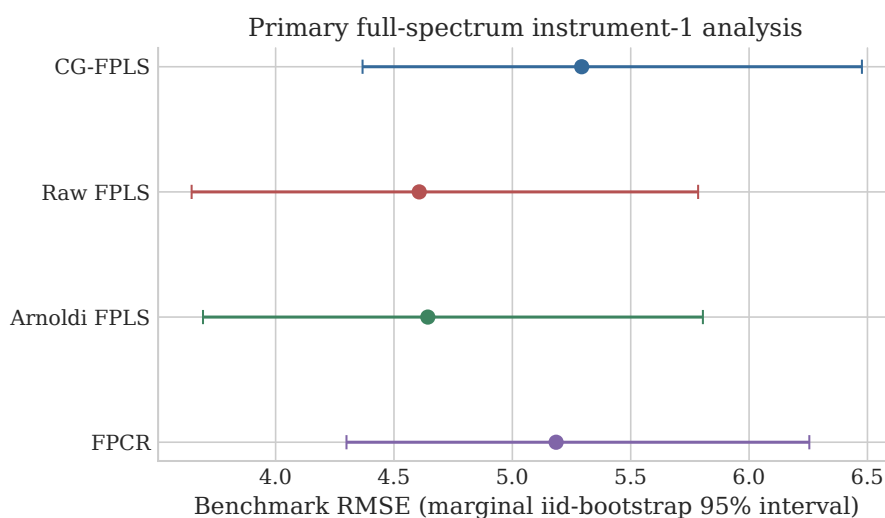


Figure 8.3: Benchmark RMSE point estimates and marginal 95% percentile intervals from the paired iid bootstrap. Overlap of these marginal intervals is not a test of a paired method contrast; the declared paired differences are reported in Table 8.2.

is small relative to the benchmark prediction errors; its practical importance cannot be determined without an assay scale and an application-specific tolerance. The interval describes a conditional difference between two already fitted models on these tablets and does not establish general superiority of the raw representation. Arnoldi’s advantages over CG–FPLS and FPCR are larger, approximately 0.65 and 0.54 RMSE units, and also persist under the principal block-length-10 bootstrap.

Table 8.2: Declared benchmark RMSE contrasts. Positive values favour Arnoldi FPLS. The iid and circular block-length-10 columns give 95% percentile intervals from 10,000 paired resamples. Models are held fixed throughout the bootstrap.

Contrast	Difference	Iid interval	Block-10 interval
Raw – Arnoldi	-0.036	[-0.057, -0.017]	[-0.062, -0.014]
CG – Arnoldi	0.650	[0.397, 0.921]	[0.257, 1.017]
FPCR – Arnoldi	0.542	[0.286, 0.844]	[0.201, 0.922]

8.4 Selection and numerical stability

Prediction alone obscures a much sharper difference between the two leading procedures. Across the 25 development outer fits, Arnoldi selects between four and six components, with median five and no numerical flag. Raw FPLS selects between four and 70 requested components, with median 44, reaches the cap twice, and is flagged in 24 of 25 fits. Its mean repeated-CV RMSE is also more variable across the five repeats: 5.082 ± 0.517 , compared with 4.917 ± 0.131 for Arnoldi. One adverse raw repeat accounts for much of that difference, and a contiguous-fold sensitivity reverses the leading pair. These dependent CV summaries therefore diagnose procedure stability rather than establish another inferential ranking.

The final raw fit selects nominal dimension 55. At that point both its normalised power-basis condition number and its response-design condition number are recorded at the reporting ceiling, 4.50×10^{15} , and the fit is flagged. The fixed $1/\sqrt{\epsilon}$ threshold is already crossed by $m = 6$. Finite predictions do not make those raw coordinates numerically trustworthy. Arnoldi selects five components, has basis condition number one, response-design condition number 11.95, and orthogonality defect 7.4×10^{-16} . The central empirical finding is consequently not that one representation wins on predictive loss: Arnoldi realises essentially the same predictive relation as Raw through stable five-dimensional coordinates. The complete paths are shown in Appendix D.

CG–FPLS stops at three components in every development fit and in the final fit. This stable but strongly regularising behaviour accompanies its larger prediction error. The finding concerns the implemented discrepancy rule, including its variance pilot, rather

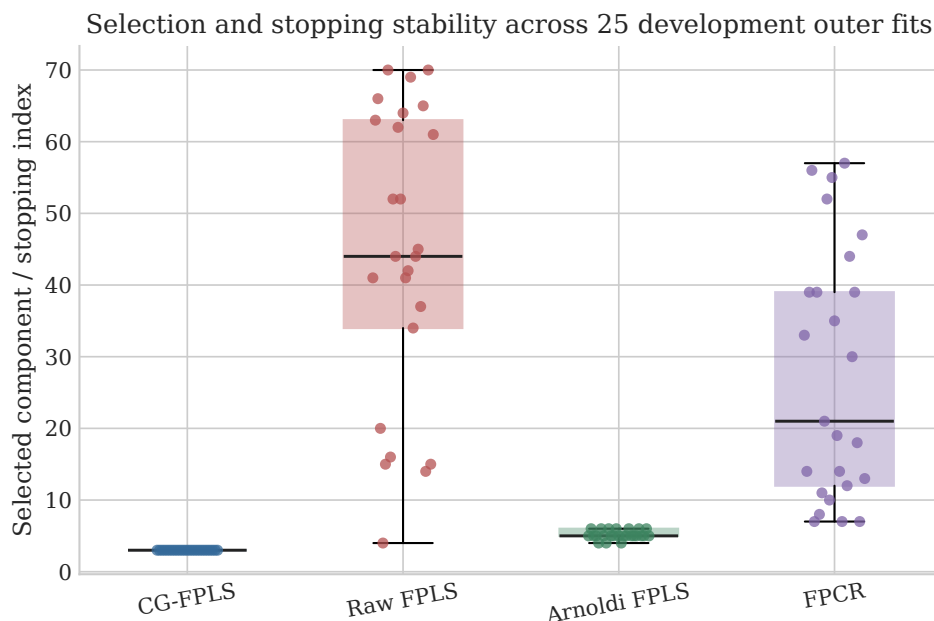


Figure 8.4: Selection and stopping indices across the 25 development outer fits. Points show individual fits and boxes summarise their distributions. Raw and Arnoldi use inner cross-validation, FPCR uses GCV, and CG-FPLS uses discrepancy stopping; counts are therefore not common effective degrees of freedom.

than a general inferiority of conjugate- gradient regularisation. FPCR selects between seven and 57 components in development and 53 in the final fit, illustrating again that nominal component counts cannot be compared mechanically across different search spaces and tuning criteria.

8.5 Robustness, interpretation, and scope

The prespecified sensitivities confirm that the detailed ranking is conditional on the measurement and preprocessing choices. Truncating instrument 1 at 1798 nm reduces the RMSEs of CG-FPLS, Arnoldi, and FPCR and compresses all four results to 4.63–4.77; Arnoldi then has the smallest point RMSE. Fixed Savitzky–Golay smoothing favours Raw at 4.66 but does not improve every procedure. Full-spectrum instrument 2 produces larger RMSEs for all four methods, with Raw and Arnoldi again the leading pair. Instrument 2 is a paired robustness analysis of the same tablets, not an external replication or a calibration-transfer experiment. These patterns show preprocessing and instrument dependence, but do not establish that the visible high-wavelength spikes caused particular prediction errors.

The residual displays in Appendix D contain a few large, common errors and regression toward the centre of the assay range. All such tablets remain in the analysis. Without

batch, acquisition-order, or laboratory metadata, row-order patterns cannot be assigned a physical cause. Likewise, the fitted coefficient vectors identify neither a unique functional slope nor chemically causal wavelengths in this 650-coordinate, rank-194 problem.

Three limitations govern the result. First, the designated benchmark had been inspected during dataset selection, so the reported errors are not a prospectively blind validation. Second, paired-bootstrap intervals condition on the fitted models and the supplied benchmark tablets; they omit development- sample, preprocessing, and tuning uncertainty. Third, the study contains one response from one tablet archive and no independent batch, laboratory, or acquisition-period validation. Subject to those limits, it transfers two mechanisms from the controlled experiments to an observed problem: aggressive early stopping can sacrifice prediction, and algebraically related Krylov fits can have almost identical predictions while their finite-precision representations have radically different reliability.

Chapter 9

Conclusion and Outlook

Functional linear regression turns a familiar linear prediction problem into an inverse problem. The covariance operator suppresses weak directions, its empirical version is finite rank, and any attempt to invert it must decide both which information to retain and how to represent the resulting computation. This thesis has examined those decisions through spectral truncation, raw and orthogonalised Krylov least squares, and direct conjugate-gradient iteration. Its main conclusion is that regularisation cannot be separated from purpose or from numerical realisation. A rule that is effective for slope estimation can be inadequate for prediction or inference, and two procedures that target the same exact-arithmetic subspace can have radically different reliability in floating-point coordinates.

The argument combines operator geometry, fully specified algorithms, controlled Monte Carlo experiments, a separate inference study, and one observed spectroscopic application. The evidence is deliberately conditional. It does not produce a universally best method; it identifies when target geometry, stopping, and numerical representation change the conclusion.

9.1 Answers to the research questions

9.1.1 Spectral and iterative regularisation

The first question asked how spectral and iterative regularisation compare as the covariance structure and slope change. The simulations show that the answer depends on the loss used to define success. CG-FPLS stops very early under the implemented discrepancy rule. That aggressive regularisation often reduces variance in the ordinary L^2 slope norm, giving it the smallest mean ISE in Models 2 and 3 and a concentrated

slope-error distribution. The same stopping can leave response-relevant directions unresolved. Its mean prediction error exceeds that of Arnoldi FPLS in all three models, most clearly in Model 2, where their mean MSPEs are 1.999 and 1.442.

FPCR and the response-informed Krylov procedures order information differently. FPCR follows predictor variance, whereas FPLS builds spaces from both $\hat{K}$ and $\hat{r}$. The response-informed methods can therefore reach a useful predictive relation in relatively few directions, but no fixed component count has the same meaning across the two geometries. In Models 1 and 3, FPCR and Arnoldi have almost identical mean prediction errors; in Model 2, Arnoldi has the smallest stable-method mean and median MSPE. The simulation result is thus a target-dependent trade-off rather than a categorical advantage of spectral or iterative regularisation.

The tablet application repeats the prediction-side cost of the implemented CG rule, which stops at three steps: CG-FPLS has benchmark RMSE 5.293, compared with 4.607 for Raw and 4.643 for Arnoldi. Because the physical slope is unknown, the application cannot determine whether early stopping improves ordinary slope recovery. It confirms only that, for this assay prediction problem, the remaining response information matters enough to improve benchmark prediction.

9.1.2 Finite-precision Krylov representation

The second question concerned the reliability of theoretically equivalent FPLS formulations. At a common valid dimension, raw and Arnoldi FPLS span the same Krylov space and target the same restricted response least-squares fit. Their paired stable-subset simulation behaviour is correspondingly almost identical. The equivalence fails as an implementation guarantee, however, because the raw powers become nearly dependent. Raw FPLS is flagged in 0.40%, 8.18%, and 0.32% of the 5,000 simulations in Models 1–3. Those rare fits create catastrophic ISE and MSPE tails that are absent after orthogonalisation. Medians alone conceal this reliability difference.

The observed spectra make the distinction even sharper. Twenty-four of 25 raw development fits and the final raw fit are flagged. Its selected dimensions range from four to 70, and the final $m = 55$ basis and design condition numbers reach 4.50×10^{15} . Arnoldi selections stay between four and six; the final $m = 5$ basis condition is one and its orthogonality defect is 7.4×10^{-16} . Yet the two final prediction vectors correlate at 0.999934 and differ by only 0.174 assay units in RMS. Their benchmark RMSE difference is just 0.036.

This is the clearest answer to the second question. Orthogonalisation does not promise a lower loss in each realised dataset, nor does it define a different exact fixed- m target.

It keeps the intended Krylov fit accessible and its selection path interpretable when the raw power coordinates cease to be trustworthy. A finite prediction score cannot by itself certify numerical reliability.

9.1.3 Stopping for estimation and inference

The third question asked why an inference-compatible CG index can differ from one selected for estimation. The distinction follows from the quantity that regularisation is allowed to leave behind. Estimation uses early stopping to control amplification of sampling noise. The direct-moment test instead requires the residual $\hat{r} - \hat{K}\hat{\beta}_m$ to be negligible on the root- n sampling scale. Crossing a fixed estimation-oriented discrepancy threshold does not imply that stronger requirement.

The stopping experiment illustrates the consequence with an estimation-oriented variant. It uses the earlier discrepancy formula and constants, but caps the variance pilot at 20 components rather than 70 and uses the stabilised moment-residual path rather than the short recurrence. It therefore does not reproduce the complete estimation procedure unchanged. This comparator selects one or two components in most samples, satisfies its own realised threshold, yet produces 5%-level null rejection frequencies of 52.70%, 96.02%, and 71.94% in Models 1–3. Its finite-iteration residual remains large on the root- n scale. Continuing the path reduces both the scaled residual and the difference from the direct-moment statistic. At $m = 70$, the two versions make identical stored decisions, with rejection frequencies of 6.20%, 6.00%, and 5.26% in that stopping-diagnostic experiment. In the separate principal size stream, the direct-moment oracle-calibrated rejection frequencies are 5.90%, 5.84%, and 6.10%. Both sets show modest liberal behaviour at $n = 100$. Once the iterative residual is negligible, the discrepancy from the continuous-design oracle reference combines finite-sample calibration effects, the fixed-grid approximation, and simulation error in the reference distribution and critical value.

The inference study also establishes the narrower positive results that the design supports. Empirically calibrated direct-moment power rises sharply when n increases from 100 to 200; the finite-dimensional inversions are non-empty, bounded, and full rank in every replication. Neither result makes $m = 70$ a universal stopping rule. The general lesson is to diagnose the root- n moment residual and to validate a feasible calibration, rather than reuse an estimation index for formal inference.

9.1.4 Regularisation in observed spectroscopy

The fourth question asked whether the same trade-offs appear with observed functional predictors. All four procedures recover substantial assay signal on the supplied benchmark, with R^2 from 0.887 to 0.914. Raw and Arnoldi form the leading predictive pair. The paired iid interval for Raw-minus- Arnoldi RMSE is $[-0.057, -0.017]$, but the point difference is less than 1% of Arnoldi’s RMSE and their tablet-level errors move almost together. It is a conditional difference that is small relative to the benchmark prediction errors. Its practical importance cannot be determined without an assay scale and an application-specific tolerance.

The stable empirical conclusion is therefore not that Raw wins. Arnoldi realises essentially the same predictive relation with a five-component, orthonormal representation, while the nominal 55-component raw fit is severely ill-conditioned. The primary disadvantage of CG is consistent with the cost of aggressive stopping seen in simulation, and FPCR’s larger selected index illustrates the different ordering imposed by spectral truncation. Trimming, smoothing, and changing instrument alter both the errors and selected dimensions; the Raw–Arnoldi point ordering itself changes across variants. The application transfers the mechanisms developed in the controlled study without converting them into a universal ranking.

9.2 Integrated implications for practice

The first practical implication is to specify the target before choosing a regularisation rule. Ordinary slope error, covariance-weighted prediction, and a moment-based inferential approximation reward different compromises. An early index that suppresses slope variance can still omit predictive signal, while a useful prediction index can leave an inferential residual that is much too large. Reporting a single generic measure of fit does not resolve this distinction.

Second, the basis is part of the procedure in finite precision. Krylov-space notation identifies a mathematical subspace, but software must still produce coordinates in that space and solve the induced least-squares problem. Raw powers can be adequate at low dimension and fail abruptly later. Arnoldi with reorthogonalisation and QR supplies a defensible default when the aim is to follow the APLS path, especially when its prediction is effectively tied with the raw fit. Condition numbers, orthogonality defects, effective dimensions, and invalid candidates should accompany loss summaries.

Third, component counts must be interpreted within their algorithms. A principal component, a raw or Arnoldi Krylov direction, and a CG stopping index do not constitute

equal units of effective complexity. The comparison in this thesis intentionally preserves each procedure’s native objective and tuning rule. It therefore evaluates complete procedures, not isolated regularisers at an artificial common index.

Finally, reproducibility must extend below an aggregated results table. The estimation archive retains replication-level losses, selected dimensions, instability diagnostics, and cached quantities supporting the distributional figures. The inference archive retains statistics, confidence-set summaries, diagnostics, and configuration information. The spectroscopy archive additionally preserves observation-level predictions, fitted coefficients, tuning paths, exact resampling indices, source snapshots, environment records, and checksums. These records support checks of the reported statistical and numerical behaviour, with the scope of exact replay determined by what each archive retains.

9.3 Limitations

The estimation Monte Carlo evidence covers three dense, error-free functional designs with Gaussian scores, homoskedastic errors, $n = 100$, and fixed implementation rules. These choices isolate mechanisms but do not span irregular sampling, measurement error, heavy-tailed or dependent errors, sparse curves, or a broad range of eigenvalue and slope classes. Monte Carlo rankings are conditional on the stated tuning procedures and should not be interpreted as minimax or uniform comparisons.

The inference analysis is still further from a ready observational workflow. Its principal calibration uses continuous-design population quantities known from the simulation; their covariance weights differ slightly from those of the fixed grid. Its power calculations use simulation-specific empirical null critical values. Only one perturbation direction is examined, and the confidence-set display restricts candidates to five basis coefficients. No stored result validates a feasible plug-in, multiplier, or bootstrap calibration on observed data. This is why the tablet analysis makes no formal functional inference claim.

The empirical evidence comes from one response in one public archive. There is no independent tablet batch, laboratory, site, or acquisition-period test; instrument 2 repeats measurements on the same tablets. Benchmark responses had already influenced dataset selection, so the supplied test split is not a prospectively blind holdout. Its bootstrap intervals hold all fitted models fixed and omit variation from development sampling, preprocessing, and tuning. No batch identifiers explain possible archive-order dependence, assay units are not explicit in the MAT object, and the true physical slope is unavailable. The coefficients consequently support replay and numerical comparison,

not causal chemical interpretation. Finally, the publicly downloadable archive contains no explicit open-data licence, limiting redistribution of the raw file.

9.4 Directions for further work

The most immediate methodological task is feasible calibration of the late- stopped statistic. Plug-in, multiplier, or bootstrap approximations should be studied under the same finite-sample discipline used here, with critical-value uncertainty separated from residual iteration error. A related goal is a data-dependent inference stopping rule that directly controls the scaled moment residual, rather than selecting a prediction index and hoping it is late enough.

On the numerical side, stability-aware selection could combine predictive loss with an explicit admissibility rule for conditioning and rank. When several candidates have practically indistinguishable validation error, such a rule could prefer the simplest stable representation. Comparisons with ridge, spline, and other functional regularisers would show whether the target and conditioning lessons persist beyond truncation and Krylov methods.

For spectroscopy, the decisive next step is genuine external validation across tablet batches, instruments, laboratories, and acquisition periods. Known groups would permit batch-aware or order-aware resampling, while a bootstrap that refits preprocessing, tuning, and models would address a broader uncertainty target than the fixed-model intervals used here. Measurement- error models and irregular-grid methods would further connect the idealised functional model to practical acquisition. Such studies should freeze the primary response and preprocessing rules before opening the external outcomes.

9.5 Final perspective

Regularisation in functional linear regression is simultaneously statistical, computational, and purpose-dependent. Early stopping can improve estimation yet invalidate inference; theoretically equivalent Krylov spaces can behave very differently in finite-precision coordinates; and a strong realised prediction score need not certify a stable estimator. Reliable practice therefore requires matching the stopping rule to the target, using numerically defensible representations, and preserving the diagnostics needed to distinguish predictive success from computational fragility.

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Appendix A

Definitions and Supporting Proofs

This appendix collects the definitions and mathematical arguments used from Chapter 1 through the estimation and inference chapters. Its purpose is to make the main text self-contained without interrupting the exposition with long proofs. A definition fixes the meaning of a term and therefore is not a statement that can be proved. What can be proved are the properties and consequences that follow from those definitions. Accordingly, Section A.1 records the terminology, and the remaining sections prove the results invoked in the thesis. The presentation follows standard Hilbert-space and inverse-problem arguments; see Lin (2021), Hsing and Eubank (2015), and Engl et al. (1996) for broader treatments.

Throughout, $\mathcal{H}$ is a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. In the functional regression model, $\mathcal{H} = L^2(\mathcal{I})$ for a compact interval $\mathcal{I}$. All equalities between elements of $L^2(\mathcal{I})$ are understood up to sets of Lebesgue measure zero. When the covariance operator K is not injective, slope parameters are reported through their minimum-norm representatives in the identified space

$$\mathcal{H}_{\text{id}} = \text{Null}(K)^\perp = \overline{\text{Range}(K)}.$$

Statements about a globally identified slope implicitly restrict the parameter space to $\mathcal{H}_{\text{id}}$, unless injectivity of K is assumed explicitly.

A.1 Definitions and notation

Definition A.1 (Inner product, norm, and orthogonality). An *inner product* on a real vector space is a symmetric bilinear map $\langle \cdot, \cdot \rangle$ satisfying $\langle f, f \rangle > 0$ for every $f \neq 0$. Its induced norm is $\|f\| = \langle f, f \rangle^{1/2}$. Elements f and g are *orthogonal*, written $f \perp g$, when

$$\langle f, g \rangle = 0.$$

Definition A.2 (Hilbert space and orthonormal basis). A *Hilbert space* is an inner-product space that is complete in its induced norm. A sequence $\{e_j\}_{j \geq 1}$ is *orthonormal* if $\langle e_j, e_k \rangle$ equals one for $j = k$ and zero otherwise. It is an *orthonormal basis* if the closure of its linear span is the whole space.

Definition A.3 (ℓ^p spaces, conjugate exponents, and dual balls). For $1 \leq p < \infty$, the sequence space ℓ^p consists of all real sequences $x = (x_j)_{j \geq 1}$ for which

$$\|x\|_p = \left(\sum_{j \geq 1} |x_j|^p \right)^{1/p} < \infty.$$

The space ℓ^∞ consists of bounded sequences and has norm $\|x\|_\infty = \sup_j |x_j|$. Exponents $p, q \in [1, \infty]$ are *Hölder conjugates* when $p^{-1} + q^{-1} = 1$, using the convention $\infty^{-1} = 0$. The continuous dual E^* of a normed space E is the space of its continuous linear functionals, equipped with

$$\|\varphi\|_* = \sup_{\|x\|_E \leq 1} |\varphi(x)|.$$

In $\mathbb{R}^d$, write $B_p^d = \{x : \|x\|_p \leq 1\}$. Its polar under the standard pairing is

$$(B_p^d)^\circ = \left\{ y : \sup_{x \in B_p^d} |y^\top x| \leq 1 \right\}.$$

The norm-induced distance is $d_p(x, z) = \|x - z\|_p$; its radius- r ball centred at z is $z + rB_p^d$.

Definition A.4 (Linear operators). For a linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$, its null space and range are

$$\text{Null}(A) = \{f : Af = 0\}, \quad \text{Range}(A) = \{Af : f \in \mathcal{H}\}.$$

The operator is *injective* when $\text{Null}(A) = \{0\}$ and is *bounded* when its operator norm is finite, where

$$\|A\|_{\text{op}} = \sup_{\|f\| \leq 1} \|Af\|.$$

A bounded operator is *self-adjoint* if $\langle Af, g \rangle = \langle f, Ag \rangle$ for all f, g , and *non-negative* if $\langle Af, f \rangle \geq 0$ for every f . An *eigenpair* (λ, v) satisfies $Av = \lambda v$ with $v \neq 0$.

Definition A.5 (Compact and trace-class operators). A bounded operator A is *compact* if it maps every bounded sequence to a sequence whose image has a convergent subsequence. A non-negative self-adjoint operator is *trace class* if, for one and hence every orthonormal basis $\{e_j\}$, $\text{tr}(A) = \sum_j \langle Ae_j, e_j \rangle < \infty$. Every trace-class operator is compact.

Definition A.6 (Rank-one and covariance operators). For $u, v \in \mathcal{H}$, the rank-one operator $u \otimes v$ is defined by $(u \otimes v)f = \langle v, f \rangle u$. If X is a random element with mean μ_X and $\mathbb{E}\|X\|^2 < \infty$, its covariance operator is

$$K = \mathbb{E}\{(X - \mu_X) \otimes (X - \mu_X)\}.$$

Definition A.7 (Well-posedness, ill-posedness, and regularisation). An inverse problem is *well posed* in the sense relevant here if a solution exists, is unique on the chosen parameter space, and depends continuously on the observed moments. Failure of one of these conditions makes the problem *ill posed*. A *regularisation method* replaces an unstable inverse by a family of stable approximations indexed by a tuning parameter, truncation level, or stopping index.

Definition A.8 (Condition number and Krylov subspace). For a tall matrix $A \in \mathbb{R}^{p \times m}$ with $p \geq m$, its spectral condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

when A has full column rank, and $\kappa_2(A) = \infty$ otherwise. For a symmetric positive-definite square matrix this reduces to $\kappa_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$. For an operator A , a right-hand side r , and an integer $m \geq 1$, the order- m Krylov subspace is

$$\mathcal{K}_m(A, r) = \text{span}\{r, Ar, \dots, A^{m-1}r\}.$$

Definition A.9 (Sampling operator, adjoint, and moment residual). For centred functional observations $X_1^c, \dots, X_n^c$, the sampling operator is $(T_n b)_i = \langle X_i^c, b \rangle$. Relative to the sample inner product $\langle a, c \rangle_n = n^{-1} a^\top c$, its adjoint is $T_n^* a = n^{-1} \sum_i a_i X_i^c$. Hence $\hat{K} = T_n^* T_n$, $\hat{r} = T_n^* y$, and the empirical moment residual at b is $e(b) = \hat{r} - \hat{K}b = T_n^*(y - T_n b)$.

Definition A.10 (PLS1, NIPALS, and APLS). *PLS1* denotes partial least squares with one scalar response. *NIPALS* is a classical sequential PLS algorithm that constructs a response-covariant score, estimates predictor and response loadings, and deflates both residual objects after each component. In the present functional setting, *APLS* denotes the alternative representation that minimises response least squares over $\mathcal{K}_m(\hat{K}, \hat{r})$. Under the full-rank conditions stated below, these are two representations of the same fixed-component standard functional PLS fit.

Definition A.11 (Arnoldi basis and conjugacy). An Arnoldi basis of $\mathcal{K}_m(A, r)$ is an orthonormal sequence generated by repeatedly applying A to the latest basis vector and orthogonalising the result against the existing basis. Non-zero vectors $p_0, \dots, p_k$ are *A-conjugate* if $\langle A p_j, p_\ell \rangle = 0$ whenever $j \neq \ell$.

Definition A.12 (Stochastic order and weak convergence). For random quantities Z_n , the notation $Z_n = O_{\mathbb{P}}(1)$ means that $\{Z_n\}$ is bounded in probability, while $Z_n = o_{\mathbb{P}}(1)$ means $Z_n \rightarrow 0$ in probability. The notation $Z_n \rightsquigarrow Z$ denotes convergence in distribution. These definitions also apply to norms of random elements in $\mathcal{H}$.

A.2 Hilbert-space geometry

Theorem A.13 (Hölder duality for ℓ^p). *Let $1 \leq p < \infty$ and let q be its Hölder conjugate. If $x \in \ell^p$ and $y \in \ell^q$, then*

$$\left| \sum_{j \geq 1} x_j y_j \right| \leq \|x\|_p \|y\|_q. \quad (\text{A.1})$$

The map $y \mapsto \varphi_y$, where $\varphi_y(x) = \sum_{j \geq 1} x_j y_j$, is an isometric isomorphism from ℓ^q onto $(\ell^p)^$. In finite dimension this implies*

$$(B_p^d)^\circ = B_q^d, \quad 1 \leq p \leq \infty.$$

Consequently, ℓ^2 is self-dual and $(B_2^d)^\circ = B_2^d$.

Proof. First suppose $1 < p < \infty$. Young's inequality $ab \leq a^p/p + b^q/q$ for $a, b \geq 0$ gives, after normalising non-zero x and y by their respective norms,

$$\sum_{j \geq 1} \frac{|x_j|}{\|x\|_p} \frac{|y_j|}{\|y\|_q} \leq \frac{1}{p} \sum_{j \geq 1} \frac{|x_j|^p}{\|x\|_p^p} + \frac{1}{q} \sum_{j \geq 1} \frac{|y_j|^q}{\|y\|_q^q} = 1.$$

This proves (A.1); if either sequence is zero the claim is immediate. When $p = 1$ and $q = \infty$, the same conclusion follows directly from $\sum_j |x_j y_j| \leq (\sup_j |y_j|) \sum_j |x_j|$.

Hölder's inequality shows that every $y \in \ell^q$ defines a continuous functional with $\|\varphi_y\|_* \leq \|y\|_q$. For $1 < p < \infty$ and $y \neq 0$, take N such that $\sum_{k=1}^N |y_k|^q > 0$; this holds for all sufficiently large N . Equality is obtained as a limit of the finite-support vectors

$$x_j^{(N)} = \begin{cases} \frac{\text{sgn}(y_j) |y_j|^{q-1}}{\left(\sum_{k=1}^N |y_k|^q \right)^{(q-1)/q}}, & j \leq N, \\ 0, & j > N, \end{cases}$$

for which $\|x^{(N)}\|_p = 1$ and $\varphi_y(x^{(N)}) = (\sum_{j=1}^N |y_j|^q)^{1/q}$. For $p = 1$, choose signed coordinate vectors whose coordinates approach $\|y\|_\infty$. Hence $\|\varphi_y\|_* = \|y\|_q$ in both cases.

Conversely, let $\varphi \in (\ell^p)^*$ and put $y_j = \varphi(e_j)$ for the standard coordinate vectors. If

$p = 1$, then $|y_j| \leq \|\varphi\|_*$, so $y \in \ell^\infty$. If $1 < p < \infty$, apply φ to the preceding norm-one construction based on $(y_1, \dots, y_N)$ whenever its q th-power sum is positive. When that sum is zero, the following inequality is immediate. Thus, in either case,

$$\left(\sum_{j=1}^N |y_j|^q \right)^{1/q} \leq \|\varphi\|_* \quad \text{for every } N,$$

and therefore $y \in \ell^q$. On every finitely supported x , $\varphi(x) = \sum_j x_j y_j$. Such sequences are dense in ℓ^p for $p < \infty$, so continuity extends the identity to every $x \in \ell^p$. Thus $y \mapsto \varphi_y$ is onto as well as isometric.

In $\mathbb{R}^d$, the dual-norm identity says $\sup_{x \in B_p^d} |y^\top x| = \|y\|_q$, which is exactly $(B_p^d)^\circ = B_q^d$. The endpoint $p = \infty$, $q = 1$ follows by taking $x_j = \text{sgn}(y_j)$; this is a finite-dimensional polar identity, not an assertion that $(\ell^\infty)^* = \ell^1$. Finally, $p = q = 2$ yields self-duality. The Hilbert-space version of the same identification is the Riesz representation theorem proved below. $\square$

Remark A.14 (The ℓ^∞ endpoint). The finite-dimensional identity $(B_\infty^d)^\circ = B_1^d$ does not extend to an identification $(\ell^\infty)^* = \ell^1$. Every $y \in \ell^1$ does define a continuous functional on ℓ^∞ through coordinate pairing, but the full dual $(\ell^\infty)^*$ is strictly larger. Establishing that strict inclusion requires additional functional-analysis machinery and is not used elsewhere in the thesis; it is stated here only to prevent the finite-dimensional drawing from being overinterpreted (Lin 2021).

Theorem A.15 (Cauchy–Schwarz inequality). *For all $f, g \in \mathcal{H}$,*

$$|\langle f, g \rangle| \leq \|f\| \|g\|.$$

Proof. If $g = 0$, the assertion is immediate. Otherwise, non-negativity of the squared norm gives, for every $t \in \mathbb{R}$,

$$0 \leq \|f - tg\|^2 = \|f\|^2 - 2t\langle f, g \rangle + t^2\|g\|^2.$$

Choose $t = \langle f, g \rangle / \|g\|^2$. Substitution yields $0 \leq \|f\|^2 - |\langle f, g \rangle|^2 / \|g\|^2$, which is equivalent to the claimed inequality. $\square$

Theorem A.16 (Completeness and separability of L^2). *For a compact interval $\mathcal{I}$, the space $L^2(\mathcal{I})$ with its usual inner product is a separable Hilbert space.*

Proof. It remains to verify completeness and separability. Let $\{f_n\}$ be Cauchy in $L^2(\mathcal{I})$. Choose a subsequence $\{f_{n_k}\}$ such that

$$\|f_{n_{k+1}} - f_{n_k}\|_2 \leq 2^{-k}.$$

Put $g_k = f_{n_{k+1}} - f_{n_k}$. Since $\mathcal{I}$ has finite measure, Cauchy–Schwarz gives $\|g_k\|_1 \leq |\mathcal{I}|^{1/2} \|g_k\|_2$. Hence $\sum_k \int_{\mathcal{I}} |g_k| < \infty$, so $\sum_k |g_k(t)| < \infty$ for almost every t . Define $f = f_{n_1} + \sum_{k \geq 1} g_k$ at those points. Minkowski’s inequality and Fatou’s lemma imply

$$\|f - f_{n_m}\|_2 \leq \sum_{k=m}^{\infty} \|g_k\|_2 \leq \sum_{k=m}^{\infty} 2^{-k} \longrightarrow 0.$$

Thus the subsequence converges in L^2 ; because the original sequence is Cauchy, the whole sequence converges to the same f . This proves completeness.

For separability, consider step functions taking rational values on finite partitions whose endpoints are rational points of $\mathcal{I}$. This family is countable. Simple functions are dense in L^2 , and measurable sets of finite measure can be approximated in measure by finite unions of intervals; their endpoints and the corresponding function values can then be approximated by rationals. The stated countable family is therefore dense in $L^2(\mathcal{I})$. $\square$

Theorem A.17 (Orthogonal projection). *Let M be a closed linear subspace of $\mathcal{H}$. For every $f \in \mathcal{H}$, there exists a unique $P_M f \in M$ such that $f - P_M f \perp M$. Moreover,*

$$\|f - P_M f\| = \inf_{v \in M} \|f - v\|.$$

If $M = \text{span}\{e_1, \dots, e_m\}$ for orthonormal vectors, then $P_M f = \sum_{j=1}^m \langle f, e_j \rangle e_j$.

Proof. Put $d = \inf_{v \in M} \|f - v\|$ and choose $v_n \in M$ with $\|f - v_n\| \rightarrow d$. The parallelogram identity and $(v_n + v_k)/2 \in M$ imply

$$\|v_n - v_k\|^2 \leq 2\|f - v_n\|^2 + 2\|f - v_k\|^2 - 4d^2,$$

so $\{v_n\}$ is Cauchy. Completeness of $\mathcal{H}$ and closedness of M give a limit $v \in M$ with $\|f - v\| = d$. For $h \in M$, the function $t \mapsto \|f - v - th\|^2$ is minimised at zero; its linear term therefore vanishes, giving $\langle f - v, h \rangle = 0$. If v' has the same property, then $v - v' \in M$ and $v - v' \perp M$, so $v = v'$. This proves existence, orthogonality, and uniqueness. Pythagoras gives $\|f - w\|^2 = \|f - v\|^2 + \|v - w\|^2$ for every $w \in M$, which establishes best approximation. In the finite orthonormal case, direct substitution shows that $f - \sum_{j=1}^m \langle f, e_j \rangle e_j$ is orthogonal to every basis vector and hence to M . $\square$

Corollary A.18 (Basis expansion and Parseval’s identity). *If $\{e_j\}_{j \geq 1}$ is an orthonormal basis of $\mathcal{H}$, then, for every $f \in \mathcal{H}$,*

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j, \quad \|f\|^2 = \sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2,$$

where the series converges in norm.

Proof. Let $P_m f = \sum_{j=1}^m \langle f, e_j \rangle e_j$. Because the linear span of the basis is dense, the best-approximation property in Theorem A.17 implies $\|f - P_m f\| \rightarrow 0$. Orthogonality and Pythagoras give

$$\|f\|^2 = \|f - P_m f\|^2 + \sum_{j=1}^m |\langle f, e_j \rangle|^2.$$

Letting $m \rightarrow \infty$ proves both assertions. $\square$

Theorem A.19 (Riesz representation). *Every continuous linear functional $\ell : \mathcal{H} \rightarrow \mathbb{R}$ has a unique representative $b \in \mathcal{H}$ such that $\ell(f) = \langle f, b \rangle$ for all $f \in \mathcal{H}$.*

Proof. If $\ell = 0$, take $b = 0$. Otherwise, $M = \text{Null}(\ell)$ is a closed subspace. Select $z \notin M$ and put $u = z - P_M z$. Then $u \neq 0$ and $u \perp M$. For arbitrary f , the element

$$f - \frac{\ell(f)}{\ell(u)} u$$

belongs to M . Consequently, $f = m + \{\ell(f)/\ell(u)\}u$ for some $m \in M$. With $b = \ell(u)u/\|u\|^2$, orthogonality gives $\langle f, b \rangle = \ell(f)$. If both b and b' represent ℓ , then $\langle f, b - b' \rangle = 0$ for every f ; taking $f = b - b'$ proves $b = b'$. $\square$

A.3 Random functions and covariance operators

Proposition A.20 (Existence of the Hilbert-space mean). *If X is a random element of $\mathcal{H}$ with $\mathbb{E}\|X\|^2 < \infty$, there is a unique $\mu_X \in \mathcal{H}$ satisfying*

$$\langle \mu_X, f \rangle = \mathbb{E}\langle X, f \rangle \quad \text{for every } f \in \mathcal{H}.$$

Moreover, $\|\mu_X\| \leq (\mathbb{E}\|X\|^2)^{1/2}$.

Proof. The map $\ell(f) = \mathbb{E}\langle X, f \rangle$ is linear. By Theorem A.15 and Cauchy–Schwarz for scalar random variables,

$$|\ell(f)| \leq \mathbb{E}(\|X\| \|f\|) \leq (\mathbb{E}\|X\|^2)^{1/2} \|f\|.$$

It is therefore continuous. Theorem A.19 gives the unique representative μ_X , and the displayed bound gives the norm inequality. $\square$

Theorem A.21 (Properties of a covariance operator). *Let $Z = X - \mu_X$ and suppose $\mathbb{E}\|Z\|^2 < \infty$. The operator*

$$Kf = \mathbb{E}\{\langle Z, f \rangle Z\}$$

is bounded, self-adjoint, non-negative, trace class, and compact. In particular,

$$\|K\|_{\text{op}} \leq \mathbb{E}\|Z\|^2, \quad \text{tr}(K) = \mathbb{E}\|Z\|^2.$$

Proof. For $f, g \in \mathcal{H}$,

$$\langle Kf, g \rangle = \mathbb{E}\{\langle Z, f \rangle \langle Z, g \rangle\}.$$

The expression is symmetric in f and g , proving self-adjointness. With $g = f$ it is non-negative. Moreover,

$$|\langle Kf, g \rangle| \leq \mathbb{E}(\|Z\|^2) \|f\| \|g\|,$$

which proves boundedness and the operator-norm bound.

For any orthonormal basis $\{e_j\}$, Tonelli's theorem and Corollary A.18 yield

$$\sum_{j=1}^{\infty} \langle Ke_j, e_j \rangle = \mathbb{E} \sum_{j=1}^{\infty} |\langle Z, e_j \rangle|^2 = \mathbb{E}\|Z\|^2 < \infty.$$

Thus, K is trace class and has the asserted trace.

For completeness, compactness can also be seen directly. Let P_m project onto

$$M_m = \text{span}\{e_1, \dots, e_m\},$$

and set $K_m = P_m K P_m$, which has finite rank. For unit vectors f, g , Cauchy–Schwarz gives

$$|\langle (I - P_m)Kf, g \rangle| \leq \{\mathbb{E}\|Z\|^2\}^{1/2} \{\mathbb{E}\|(I - P_m)Z\|^2\}^{1/2}.$$

The final expectation tends to zero by Parseval and dominated convergence. The same bound applies to $P_m K (I - P_m)$, so $\|K - K_m\|_{\text{op}} \rightarrow 0$. An operator-norm limit of finite-rank operators is compact. $\square$

Proposition A.22 (Covariance-kernel representation). *Suppose that $Z = X - \mu_X$ has pointwise second moments and that the integrability conditions for Fubini's theorem hold. If $c(s, t) = \mathbb{E}\{Z(s)Z(t)\}$, then, for almost every s ,*

$$(Kf)(s) = \int_{\mathcal{I}} c(s, t) f(t) dt.$$

Proof. Using the definition of K and then Fubini's theorem,

$$\begin{aligned} (Kf)(s) &= \mathbb{E} \left[Z(s) \int_{\mathcal{I}} Z(t) f(t) dt \right] \\ &= \int_{\mathcal{I}} \mathbb{E}\{Z(s)Z(t)\} f(t) dt = \int_{\mathcal{I}} c(s, t) f(t) dt. \end{aligned}$$

The stated integrability assumptions justify interchanging expectation and integration. $\square$

Theorem A.23 (Spectral representation). *Let K be compact, self-adjoint, and non-negative. Its positive eigenvalues can be indexed as $\{\lambda_j : j \in \mathcal{J}_K\}$ in non-increasing order, where $\mathcal{J}_K = \{1, \dots, q\}$ if K has finite rank q and $\mathcal{J}_K = \mathbb{N}$ if it has infinite rank (with $\mathcal{J}_K = \emptyset$ when $K = 0$). There are corresponding orthonormal eigenfunctions $\{v_j : j \in \mathcal{J}_K\}$ spanning $\overline{\text{Range}(K)}$ such that*

$$Kf = \sum_{j \in \mathcal{J}_K} \lambda_j \langle f, v_j \rangle v_j.$$

If K has infinite rank, then $\lambda_j \rightarrow 0$.

Proof. If $K = 0$, the result is immediate. Otherwise, define the maximal Rayleigh quotient $\lambda_1 = \sup_{\|f\|=1} \langle Kf, f \rangle > 0$. Take a maximising sequence. The unit ball is weakly sequentially compact, and compactness of K converts a weakly convergent subsequence $f_n \rightharpoonup v_1$ into strong convergence $Kf_n \rightarrow Kv_1$. Hence $\langle Kf_n, f_n \rangle \rightarrow \langle Kv_1, v_1 \rangle = \lambda_1$. Since $\langle Kv_1, v_1 \rangle \leq \lambda_1 \|v_1\|^2$ and $\|v_1\| \leq 1$, it follows that $\|v_1\| = 1$. Considering the Rayleigh quotient along $v_1 + th$ for $h \perp v_1$ shows that $\langle Kv_1, h \rangle = 0$; therefore $Kv_1 = \lambda_1 v_1$.

Self-adjointness makes $v_1^\perp$ invariant under K . Repeating the same argument on successive orthogonal complements produces orthonormal eigenvectors with non-increasing eigenvalues until the restriction of K is zero. In the infinite-rank case, if infinitely many eigenvalues were bounded below by some $c > 0$, then the sequence $Kv_j = \lambda_j v_j$ would have pairwise distances at least $\sqrt{2}c$ and hence no convergent subsequence, contradicting compactness. Thus $\lambda_j \rightarrow 0$ in that case.

Finally, if a vector orthogonal to every constructed v_j had non-zero image under K , the Rayleigh-quotient argument on that invariant orthogonal complement would produce a unit eigenvector with a positive eigenvalue ρ . In a finite construction this contradicts termination. In an infinite construction that vector belongs to every preceding orthogonal complement, so maximality of each Rayleigh quotient gives $\lambda_j \geq \rho$ for every j , contradicting $\lambda_j \rightarrow 0$. Hence K is zero on the remaining complement. This gives the expansion, and the eigenvectors with positive eigenvalues span $\overline{\text{Range}(K)}$. $\square$

Proposition A.24 (Karhunen–Loève expansion and optimality). *Under the assumptions of Theorem A.21, let (λ_j, v_j) be the positive eigenpairs of K and set $\xi_j = \langle X - \mu_X, v_j \rangle$. Then*

$$X = \mu_X + \sum_{j \in \mathcal{J}_K} \xi_j v_j \quad \text{in mean square,}$$

$\mathbb{E}\xi_j = 0$, and $\mathbb{E}(\xi_j\xi_k) = \lambda_j$ for $j = k$ and zero otherwise. For each $m \leq \text{rank}(K)$, among all m -dimensional subspaces, $\text{span}\{v_1, \dots, v_m\}$ minimises the expected squared reconstruction error. If K has finite rank, the full expansion is the case $m = \text{rank}(K)$.

Proof. Centredness gives $\mathbb{E}\xi_j = 0$, while

$$\mathbb{E}(\xi_j\xi_k) = \langle Kv_j, v_k \rangle = \lambda_j \langle v_j, v_k \rangle.$$

Every component of $Z = X - \mu_X$ in $\text{Null}(K)$ has zero second moment because, for $h \in \text{Null}(K)$,

$$\mathbb{E}\langle Z, h \rangle^2 = \langle Kh, h \rangle = 0.$$

For each $m \leq \text{rank}(K)$, the trace identity therefore implies

$$\mathbb{E} \left\| Z - \sum_{j=1}^m \xi_j v_j \right\|^2 = \text{tr}(K) - \sum_{j=1}^m \lambda_j.$$

If K has infinite rank, the right-hand side tends to zero as $m \rightarrow \infty$; if it has finite rank q , it equals zero at $m = q$. This proves the asserted mean-square expansion.

For any orthonormal $u_1, \dots, u_m$, the expected reconstruction error is

$$\text{tr}(K) - \sum_{\ell=1}^m \langle Ku_\ell, u_\ell \rangle.$$

Writing each u_ℓ in the eigenbasis gives $\sum_\ell \langle Ku_\ell, u_\ell \rangle = \sum_j \lambda_j a_j$, where $0 \leq a_j \leq 1$ and $\sum_j a_j \leq m$. Since the eigenvalues are ordered, this sum cannot exceed $\sum_{j=1}^m \lambda_j$. Equality is attained by $u_\ell = v_\ell$, proving optimality. $\square$

Proposition A.25 (Rank of the empirical covariance). *For observations $X_1, \dots, X_n$ and a fixed population mean μ_X , the population-centred empirical covariance operator*

$$\widehat{K}_\mu f = \frac{1}{n} \sum_{i=1}^n \langle X_i - \mu_X, f \rangle (X_i - \mu_X)$$

has rank at most n . The sample-centred empirical covariance operator

$$\widehat{K}_{\overline{X}} f = \frac{1}{n} \sum_{i=1}^n \langle X_i - \overline{X}, f \rangle (X_i - \overline{X})$$

has rank at most $n - 1$.

Proof. The range of $\widehat{K}_\mu$ is contained in the span of the n curves $X_i - \mu_X$, so its rank is at most n . The range of $\widehat{K}_{\overline{X}}$ is contained in the span of the n curves $X_i - \overline{X}$. These curves

sum to zero, so their span has dimension at most $n - 1$. The rank of the corresponding operator cannot exceed that dimension. $\square$

A.4 The functional normal equation and inverse problem

Proposition A.26 (Centring and the normal equation). *Suppose*

$$Y = \alpha + \langle X, \beta \rangle + \varepsilon, \quad \mathbb{E}(\varepsilon \mid X) = 0,$$

with finite second moments. Then $\alpha = \mu_Y - \langle \mu_X, \beta \rangle$, and the centred variables satisfy $Y^c = \langle X^c, \beta \rangle + \varepsilon$. If $r = \mathbb{E}(Y^c X^c)$ and K is the covariance operator of X , then $K\beta = r$.

Proof. Taking expectations and using $\mathbb{E}\varepsilon = \mathbb{E}\{\mathbb{E}(\varepsilon \mid X)\} = 0$ gives the formula for α and therefore the centred model. For arbitrary $f \in \mathcal{H}$,

$$\begin{aligned} \langle r, f \rangle &= \mathbb{E}\{Y^c \langle X^c, f \rangle\} \\ &= \mathbb{E}\{\langle X^c, \beta \rangle \langle X^c, f \rangle\} + \mathbb{E}\{\varepsilon \langle X^c, f \rangle\}. \end{aligned}$$

The last expectation is zero by iterated expectations. The first equals $\langle K\beta, f \rangle$. Since the equality holds for every f , $r = K\beta$. $\square$

Proposition A.27 (Excess prediction risk). *For $R(b) = \mathbb{E}\{Y^c - \langle X^c, b \rangle\}^2$ under the correctly specified model,*

$$R(b) - R(\beta) = \langle K(b - \beta), b - \beta \rangle = \|K^{1/2}(b - \beta)\|^2.$$

Proof. Let $h = b - \beta$. The prediction residual at b is $\varepsilon - \langle X^c, h \rangle$. Expanding the square gives

$$R(b) = \mathbb{E}\varepsilon^2 - 2\mathbb{E}\{\varepsilon \langle X^c, h \rangle\} + \mathbb{E}\langle X^c, h \rangle^2.$$

The middle term is zero by the conditional mean restriction, and the last term is $\langle Kh, h \rangle$. Since $R(\beta) = \mathbb{E}\varepsilon^2$, the first equality follows. The spectral representation defines $K^{1/2}f = \sum_j \lambda_j^{1/2} \langle f, v_j \rangle v_j$ and gives $\langle Kh, h \rangle = \|K^{1/2}h\|^2$. $\square$

Proposition A.28 (Variance identities for the discrete simulation). *Let*

$$X = \sum_{j=1}^J \sqrt{\lambda_j} Z_j \phi_j, \quad Y = \langle X, \beta \rangle_T + \varepsilon,$$

where the Z_j are independent with mean zero and variance one, ε is independent of them with mean zero and variance σ^2 , and $\langle \cdot, \cdot \rangle_T$ is the discrete inner product. Then

$$\begin{aligned}\mathbb{E}\|X\|_T^2 &= \sum_{j=1}^J \lambda_j \|\phi_j\|_T^2, \\ \text{Var}\{\langle X, \beta \rangle_T\} &= \sum_{j=1}^J \lambda_j \langle \phi_j, \beta \rangle_T^2, \\ \text{Var}(Y) &= \sum_{j=1}^J \lambda_j \langle \phi_j, \beta \rangle_T^2 + \sigma^2.\end{aligned}$$

Proof. Write $c_j = \langle \phi_j, \beta \rangle_T$. Bilinearity gives

$$\langle X, \beta \rangle_T = \sum_{j=1}^J \sqrt{\lambda_j} Z_j c_j.$$

All cross-products have expectation zero because the scores are independent and centred, while $\mathbb{E}Z_j^2 = 1$. Therefore the variance of this sum is $\sum_j \lambda_j c_j^2$. Similarly,

$$\mathbb{E}\|X\|_T^2 = \sum_{j,k} \sqrt{\lambda_j \lambda_k} \mathbb{E}(Z_j Z_k) \langle \phi_j, \phi_k \rangle_T = \sum_j \lambda_j \|\phi_j\|_T^2.$$

Finally, independence of ε and the scores makes the covariance between the noiseless response and the error zero, so their variances add. No discrete orthogonality assumption on the ϕ_j is required. $\square$

Proposition A.29 (Empirical bias–variance identity). *For arbitrary $b_1, \dots, b_R, \beta$ in a real Hilbert space, let $\bar{b} = R^{-1} \sum_{r=1}^R b_r$. Then*

$$\frac{1}{R} \sum_{r=1}^R \|b_r - \beta\|^2 = \|\bar{b} - \beta\|^2 + \frac{1}{R} \sum_{r=1}^R \|b_r - \bar{b}\|^2.$$

Proof. For every r ,

$$b_r - \beta = (b_r - \bar{b}) + (\bar{b} - \beta).$$

Expanding the squared norm and averaging yields the two displayed squared terms plus

$$\frac{2}{R} \sum_{r=1}^R \langle b_r - \bar{b}, \bar{b} - \beta \rangle.$$

The cross-term is zero because $\sum_r (b_r - \bar{b}) = 0$. The argument applies in particular to the grid inner product used in the simulation and explains why the empirical variance uses divisor R when an exact decomposition of the Monte Carlo mean squared error is

desired. □

Proposition A.30 (Identification and the minimum-norm target). *Two slopes β and $\tilde{\beta}$ generate the same centred conditional mean if and only if $\tilde{\beta} - \beta \in \text{Null}(K)$. In the uncentred model, if $h = \tilde{\beta} - \beta \in \text{Null}(K)$, the parameter pairs (α, β) and $(\alpha - \langle \mu_X, h \rangle, \tilde{\beta})$ generate the same conditional mean. Hence the slope is fully identified exactly when K is injective. Otherwise, every identified equivalence class has a unique minimum-norm representative in*

$$\text{Null}(K)^\perp = \overline{\text{Range}(K)}.$$

Proof. For $h = \tilde{\beta} - \beta$,

$$\mathbb{E}\langle X^c, h \rangle^2 = \langle Kh, h \rangle.$$

If $Kh = 0$, the non-negative random variable on the left has expectation zero, so $\langle X^c, h \rangle = 0$ almost surely. Conversely, this almost sure equality implies

$$Kh = \mathbb{E}\{\langle X^c, h \rangle X^c\} = 0.$$

This proves the centred equivalence and the injectivity statement. If $h \in \text{Null}(K)$, then $\langle X, h \rangle = \langle \mu_X, h \rangle$ almost surely. Consequently,

$$\alpha - \langle \mu_X, h \rangle + \langle X, \beta + h \rangle = \alpha + \langle X, \beta \rangle \quad \text{almost surely,}$$

which proves the uncentred assertion.

Every $b \in \mathcal{H}$ has the orthogonal decomposition $b = b_0 + h$ with $b_0 \in \text{Null}(K)^\perp$ and $h \in \text{Null}(K)$. All elements $b_0 + h$ give the same conditional mean, while $\|b_0 + h\|^2 = \|b_0\|^2 + \|h\|^2$ is uniquely minimised by $h = 0$. Finally, for every bounded operator, $\text{Range}(K)^\perp = \text{Null}(K^*)$; self-adjointness and taking orthogonal complements give $\overline{\text{Range}(K)} = \text{Null}(K)^\perp$. □

Theorem A.31 (Picard condition and instability of inversion). *Let K be compact, self-adjoint, and non-negative, with positive eigenpairs (λ_j, v_j) . The equation $K\beta = r$ has a solution if and only if $r \perp \text{Null}(K)$ and*

$$\sum_{j: \lambda_j > 0} \frac{|\langle r, v_j \rangle|^2}{\lambda_j^2} < \infty.$$

When these conditions hold, the unique minimum-norm solution is

$$\beta^\dagger = \sum_{j: \lambda_j > 0} \frac{\langle r, v_j \rangle}{\lambda_j} v_j.$$

If K has infinite rank, the restriction of the minimum-norm inverse to $\text{Range}(K)$, $K^\dagger|_{\text{Range}(K)} : \text{Range}(K) \rightarrow \text{Null}(K)^\perp$, is unbounded in the $\mathcal{H}$ norm.

Proof. If $K\beta = r$, then r lies in $\overline{\text{Range}(K)} = \text{Null}(K)^\perp$ and $\langle r, v_j \rangle = \lambda_j \langle \beta, v_j \rangle$. Parseval therefore implies the summability condition. Conversely, the displayed condition makes the proposed series for $\beta^\dagger$ square summable. Applying K term by term yields the component of r in $\text{Null}(K)^\perp$, which equals r by the first condition. All other solutions differ from $\beta^\dagger$ by an element of $\text{Null}(K)$, so Proposition A.30 gives the minimum-norm claim.

For the final statement, $v_j \in \text{Range}(K)$ because $v_j = K(v_j/\lambda_j)$. Yet

$$\frac{\|K^\dagger v_j\|}{\|v_j\|} = \frac{1}{\lambda_j} \rightarrow \infty$$

as $\lambda_j \rightarrow 0$. Thus no finite operator-norm bound for $K^\dagger$ exists on $\text{Range}(K)$. $\square$

A.5 Regularised and Krylov solutions

Proposition A.32 (Tikhonov solution). *For $\rho > 0$, the criterion $R(b) + \rho\|b\|^2$ has the unique minimiser*

$$\beta_\rho = (K + \rho I)^{-1}r.$$

If $r \perp \text{Null}(K)$, its spectral form is

$$\beta_\rho = \sum_{j:\lambda_j>0} \frac{\langle r, v_j \rangle}{\lambda_j + \rho} v_j.$$

Proof. Apart from a constant independent of b , the population risk equals $\langle Kb, b \rangle - 2\langle r, b \rangle$. The spectral representation shows directly that $K + \rho I$ is invertible and that $\|(K + \rho I)^{-1}\|_{\text{op}} \leq 1/\rho$. The directional derivative of the penalised criterion at b in direction h is

$$2\langle (K + \rho I)b - r, h \rangle.$$

It vanishes for every h exactly when $(K + \rho I)b = r$. Since $\langle (K + \rho I)h, h \rangle \geq \rho\|h\|^2$, the criterion is strictly convex and the solution is unique. Expansion in the eigenbasis produces the stated filter. $\square$

Proposition A.33 (FPCR restriction). *Let $V_m = \text{span}\{v_1, \dots, v_m\}$ with $\lambda_1, \dots, \lambda_m > 0$.*

The minimiser of $R(b)$ over V_m is

$$\beta_m^{\text{FPCR}} = \sum_{j=1}^m \frac{\langle r, v_j \rangle}{\lambda_j} v_j.$$

The empirical formula follows after replacing (K, r) by $(\hat{K}, \hat{r})$.

Proof. Write $b = \sum_{j=1}^m b_j v_j$. Up to an additive constant,

$$R(b) = \sum_{j=1}^m \{\lambda_j b_j^2 - 2b_j \langle r, v_j \rangle\}.$$

Each term is a strictly convex quadratic whose minimiser is $b_j = \langle r, v_j \rangle / \lambda_j$. Combining the coefficients proves the result. $\square$

Proposition A.34 (Polynomial form of Krylov estimators). *Every $\beta_m \in \mathcal{K}_m(K, r)$ can be written as $\beta_m = p_{m-1}(K)r$ for a polynomial of degree at most $m-1$. Its moment residual satisfies*

$$r - K\beta_m = \pi_m(K)r, \quad \pi_m(\lambda) = 1 - \lambda p_{m-1}(\lambda).$$

Proof. Membership in $\mathcal{K}_m(K, r)$ means that there are coefficients $a_0, \dots, a_{m-1}$ with $\beta_m = \sum_{j=0}^{m-1} a_j K^j r$. Taking $p_{m-1}(\lambda) = \sum_{j=0}^{m-1} a_j \lambda^j$ gives the first assertion. Then $r - K\beta_m = \{I - Kp_{m-1}(K)\}r = \pi_m(K)r$ by direct algebra. $\square$

Proposition A.35 (Raw and Arnoldi Krylov spaces). *In exact arithmetic, if Arnoldi orthogonalisation does not break down before step m , its vectors $q_1, \dots, q_m$ form an orthonormal basis of $\mathcal{K}_m(K, r)$. Consequently, a uniquely defined restricted least-squares solution is invariant to whether the raw power basis or the Arnoldi basis is used.*

Proof. The first Arnoldi vector is $q_1 = r/\|r\|$ and hence spans $\mathcal{K}_1(K, r)$. Suppose inductively that $\text{span}\{q_1, \dots, q_j\} = \mathcal{K}_j(K, r)$. Arnoldi forms Kq_j and removes its projections onto the existing vectors. The remainder therefore belongs to $\mathcal{K}_{j+1}(K, r)$ and is orthogonal to $\mathcal{K}_j(K, r)$. Absence of breakdown means that this remainder is non-zero, so its normalisation adds one new direction and spans $\mathcal{K}_{j+1}(K, r)$. Induction proves the first claim. The optimisation problem depends on the subspace, not on its coordinates; uniqueness therefore makes its solution basis-invariant. Finite-precision discrepancies arise because computed raw powers may lose linear independence, not because the exact subspaces differ. $\square$

Proposition A.36 (Response-space Krylov characterisation). *Let $T : \mathcal{H} \rightarrow \mathbb{R}^n$ be bounded, let $K = T^*T$ and $r = T^*y$, and put $M_m = \mathcal{K}_m(K, r)$. If T is injective on M_m ,*

the restricted response least-squares problem

$$\min_{b \in M_m} \|y - Tb\|_n^2$$

has a unique solution β_m^Y . Its moment residual $e_m^Y = r - K\beta_m^Y$ satisfies

$$e_m^Y \perp M_m.$$

Equivalently, β_m^Y uniquely minimises $\Phi(b) = \frac{1}{2}\langle Kb, b \rangle - \langle r, b \rangle$ over M_m .

Proof. Expanding the response criterion and using $K = T^*T$ and $r = T^*y$ gives

$$\frac{1}{2}\|y - Tb\|_n^2 = \frac{1}{2}\|y\|_n^2 + \frac{1}{2}\langle Kb, b \rangle - \langle r, b \rangle.$$

The first term is constant in b . For $h \in M_m$, the directional derivative of the remaining expression is

$$\left. \frac{d}{dt} \Phi(\beta_m^Y + th) \right|_{t=0} = \langle K\beta_m^Y - r, h \rangle.$$

It vanishes for every $h \in M_m$ precisely when $e_m^Y \perp M_m$. Injectivity of T on the finite-dimensional space gives $\langle Kh, h \rangle = \|Th\|_n^2 > 0$ for every non-zero $h \in M_m$. The criterion is therefore strictly convex on M_m , so this stationary point exists uniquely and is the claimed minimiser. $\square$

Proposition A.37 (NIPALS–APLS equivalence). *Let $T : \mathcal{H} \rightarrow \mathbb{R}^n$ be bounded, set $K = T^*T$ and $r = T^*y$, and consider scalar-response PLS with common centring and inner products. Suppose that the first m score directions generated by the NIPALS-type score-and-deflation construction are non-zero and that T is injective on $M_m = \mathcal{K}_m(K, r)$. Then the NIPALS score space is TM_m , its fitted response is the orthogonal projection of y onto TM_m , and its reconstructed slope equals the unique APLS solution in Proposition A.36.*

Proof sketch. At the first step, the NIPALS weight is proportional to $T^*y = r$, so its score lies in TM_1 . Predictor deflation at each subsequent step replaces the current sampling operator T_{h-1} by $(I - P_{t_h})T_{h-1}$, where P_{t_h} is the response-space orthogonal projection onto the current score. Response deflation applies the same projection to the current response. Expanding these projections and repeatedly using $K = T^*T$ gives the standard PLS1 polynomial induction: the unnormalised h th weight has the form

$$c_h = \pi_{h-1}(K)r,$$

where π_{h-1} has degree $h - 1$. Its leading coefficient is non-zero whenever the new score

is non-zero. Consequently,

$$\text{span}\{c_1, \dots, c_h\} = \text{span}\{r, Kr, \dots, K^{h-1}r\} = M_h,$$

because the change-of-basis matrix between the two displayed sequences is triangular with non-zero diagonal. Applying T gives equality of the score space and TM_h .

The predictor deflation makes successive scores mutually orthogonal. At each step, the NIPALS response loading is the least-squares coefficient of the current residual response on the new score. Hence the sum of the first m fitted score contributions is the orthogonal projection of y onto their span, namely TM_m . Minimising $\|y - Tb\|_n^2$ over $b \in M_m$ produces the same projection. Injectivity on M_m makes the restricted slope unique, so the reconstructed NIPALS slope and the APLS slope coincide. This operator argument is the standard fixed-component equivalence established for functional PLS by Delaigle and Hall (2012). $\square$

Proposition A.38 (Response and moment Krylov objectives). *Let K be self-adjoint and non-negative, $r \in \mathcal{H}$, and $M_m = \mathcal{K}_m(K, r)$. Assume that $\dim(M_m) = m$ and that K is injective on M_m . Then both restricted quadratic criteria below are strictly convex. Define*

$$\beta_m^Y = \arg \min_{b \in M_m} \left\{ \frac{1}{2} \langle Kb, b \rangle - \langle r, b \rangle \right\}, \quad \beta_m^M = \arg \min_{b \in M_m} \frac{1}{2} \|r - Kb\|^2.$$

These response-space and moment-residual solutions are characterised by

$$\begin{aligned} r - K\beta_m^Y &\perp M_m, \\ r - K\beta_m^M &\perp KM_m. \end{aligned}$$

With $g_j = K^{j-1}r$ and $\beta_m = \sum_{j=1}^m a_j g_j$, their coefficient systems are respectively

$$\begin{aligned} \sum_{k=1}^m \langle r, K^{j+k-1}r \rangle a_k &= \langle r, K^{j-1}r \rangle, \\ \sum_{k=1}^m \langle r, K^{j+k}r \rangle a_k &= \langle r, K^j r \rangle, \quad j = 1, \dots, m. \end{aligned}$$

Thus, the two methods use the same Krylov space but do not generally have the same finite- m solution.

Proof. For the response criterion

$$\Phi(b) = \frac{1}{2} \langle Kb, b \rangle - \langle r, b \rangle,$$

the directional derivative at b in direction $h \in M_m$ is

$$\left. \frac{d}{dt} \Phi(b + th) \right|_{t=0} = -\langle r - Kb, h \rangle.$$

It vanishes for every $h \in M_m$ if and only if $r - Kb \perp M_m$. For the moment criterion

$$\Psi(b) = \frac{1}{2} \|r - Kb\|^2,$$

the directional derivative at b in direction $h \in M_m$ is

$$\left. \frac{d}{dt} \Psi(b + th) \right|_{t=0} = -\langle r - Kb, Kh \rangle.$$

It vanishes for every $h \in M_m$ if and only if $r - Kb \perp KM_m$. For non-zero $h \in M_m$, injectivity and non-negativity of K imply $\langle Kh, h \rangle > 0$ and $\|Kh\|^2 > 0$. These are the quadratic forms governing the response and moment criteria, respectively, so both restrictions are strictly convex and the two stationarity conditions are sufficient and uniquely determine their minimisers.

For the response solution, take the inner product of $r - K \sum_k a_k g_k$ with g_j . Self-adjointness and $g_j = K^{j-1}r$ give

$$\langle Kg_k, g_j \rangle = \langle K^k r, K^{j-1} r \rangle = \langle r, K^{j+k-1} r \rangle,$$

which yields the first coefficient system. For the moment solution, take the inner product of the same residual with $Kg_j = K^j r$. Then

$$\langle Kg_k, Kg_j \rangle = \langle K^k r, K^j r \rangle = \langle r, K^{j+k} r \rangle,$$

which yields the second system. The one-power shift proves that equality is not implied merely by using the same M_m . $\square$

Proposition A.39 (Finite termination with conjugate directions). *Let $A \in \mathbb{R}^{d \times d}$ be symmetric positive definite and let $p_0, \dots, p_{d-1}$ be non-zero, pairwise A -conjugate directions. Starting from any x_0 , successively minimise $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ along these directions without changing the coefficients already selected. After at most d directions, the iterate is the unique minimiser $A^{-1}b$. Standard conjugate gradient constructs such directions in exact arithmetic.*

Proof. If $\sum_j c_j p_j = 0$, multiplication by $p_k^\top A$ gives $c_k p_k^\top A p_k = 0$. Positive definiteness implies $c_k = 0$ for every k , so the d directions are linearly independent and form a basis

of $\mathbb{R}^d$. Write an iterate as $x_0 + \sum_j \eta_j p_j$. Then

$$f\left(x_0 + \sum_j \eta_j p_j\right) = f(x_0) + \sum_j \eta_j p_j^\top (Ax_0 - b) + \frac{1}{2} \sum_j \eta_j^2 p_j^\top A p_j,$$

because all cross terms vanish by conjugacy. The minimising coefficient for one direction is therefore independent of every other coefficient. After all basis directions have been minimised, every directional derivative is zero; hence $Ax - b = 0$ and $x = A^{-1}b$. The standard CG recurrence produces pairwise A -conjugate search directions by induction from the residual orthogonality relations, so the conclusion applies to its exact-arithmetic iterates. This result concerns the standard finite-dimensional CG geometry used in Figure 3.2; the released functional recurrence is defined separately in (3.34)–(3.38). $\square$

A.6 Late stopping and inference

The asymptotic arguments in this section follow the inference construction of Babii et al. (2026), with the identification qualification made explicit.

The sampling operator retains the notation T_n used throughout the thesis. To avoid confusing that operator with the scalar test statistic, the latter is denoted by $\mathcal{T}_n$. If K is not injective, the statistically identifiable null is $H_0 : K\beta = Kb$; this becomes the literal slope null $H_0 : \beta = b$ when β and b lie in $\mathcal{H}_{\text{id}}$. Chapter 6 writes $\mathcal{T}_{n,m}$ when the finite iteration must be visible and abbreviates $\mathcal{T}_{n,m_n}$ to $\mathcal{T}_n$ for an inference-specific index sequence.

Proposition A.40 (Finite-iteration decomposition). *For any candidate slope b and any iterate $\hat{\beta}_m$, define*

$$u_n(b) = \hat{r} - \hat{K}b, \quad e_m = \hat{r} - \hat{K}\hat{\beta}_m.$$

Then

$$\hat{K}(\hat{\beta}_m - b) = u_n(b) - e_m.$$

Proof. Add and subtract $\hat{r}$:

$$\hat{K}(\hat{\beta}_m - b) = \hat{K}\hat{\beta}_m - \hat{K}b = (\hat{r} - \hat{K}b) - (\hat{r} - \hat{K}\hat{\beta}_m).$$

$\square$

Theorem A.41 (Late-stopping equivalence and null limit). *Suppose that $(X_i, Y_i)_{i=1}^n$*

are independent and identically distributed and satisfy

$$Y_i^c = \langle X_i^c, \beta \rangle + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid X_i) = 0.$$

Assume $\mathbb{E}\|X^c\|^4 < \infty$ and $\mathbb{E}(\varepsilon^2 \mid X) \leq \sigma_{\max}^2 < \infty$ almost surely, and let the identifiable null $H_0 : K\beta = Kb$ hold. Equivalently, if $\beta, b \in \mathcal{H}_{\text{id}}$, assume $H_0 : \beta = b$. If, for an index $m = m_n$ that may depend on n ,

$$\sqrt{n} \|e_m\| = o_{\mathbb{P}}(1),$$

then

$$\mathcal{T}_n(b) = n \|\widehat{K}(\widehat{\beta}_m - b)\|^2 = n \|u_n(b)\|^2 + o_{\mathbb{P}}(1).$$

Let

$$V = \mathbb{E}\{\varepsilon^2(X^c \otimes X^c)\}$$

be the covariance operator of εX^c . If its positive eigenvalues are $\{\omega_j : j \in \mathcal{J}_V\}$, where $\mathcal{J}_V$ is finite or countably infinite, then

$$\mathcal{T}_n(b) \rightsquigarrow \sum_{j \in \mathcal{J}_V} \omega_j Z_j^2, \quad Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

The sum is finite when V has finite rank and converges almost surely and in L^1 otherwise.

Proof. Under $H_0 : K\beta = Kb$, Proposition A.30 implies $\langle X^c, \beta - b \rangle = 0$ almost surely. With population centring,

$$u_n(b) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i X_i^c.$$

If sample means are used instead, the exact expression is

$$u_n(b) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i X_i^c - \bar{\varepsilon} \bar{X}^c.$$

The two sample means are $O_{\mathbb{P}}(n^{-1/2})$ under the stated second moments, so their product is $O_{\mathbb{P}}(n^{-1})$ and is negligible after multiplication by $\sqrt{n}$.

The random element εX^c is mean zero because $\mathbb{E}(\varepsilon \mid X) = 0$, and it has a finite second moment by the conditional variance bound and $\mathbb{E}\|X^c\|^2 < \infty$. The central limit theorem for independent, identically distributed random elements in a separable Hilbert space therefore gives $\sqrt{n} u_n(b) \rightsquigarrow G$, where G is a mean-zero Gaussian element with covariance operator V . In particular, $\sqrt{n} \|u_n(b)\| = O_{\mathbb{P}}(1)$.

By Proposition A.40,

$$\begin{aligned} |\mathcal{T}_n(b) - n\|u_n(b)\|^2| &\leq 2\sqrt{n}\|u_n(b)\|\sqrt{n}\|e_m\| + n\|e_m\|^2 \\ &= o_{\mathbb{P}}(1). \end{aligned}$$

The continuous mapping theorem now gives $n\|u_n(b)\|^2 \rightsquigarrow \|G\|^2$. Applying the covariance spectral representation to G yields $G = \sum_j \sqrt{\omega_j} Z_j w_j$ in mean square for orthonormal eigenfunctions w_j . Parseval's identity therefore gives $\|G\|^2 = \sum_j \omega_j Z_j^2$. Since $\sum_j \omega_j = \text{tr}(V) = \mathbb{E}\|\varepsilon X^c\|^2 < \infty$, the non-negative series converges almost surely and in L^1 , completing the proof. $\square$

Proposition A.42 (Homoskedastic form of the variance operator). *If $\mathbb{E}(\varepsilon^2 | X) = \sigma^2$ almost surely, then the variance operator in Theorem A.41 satisfies*

$$V = \sigma^2 K.$$

Consequently, if $Kv_j = \lambda_j v_j$, then $Vv_j = \sigma^2 \lambda_j v_j$ and the null limit is $\sum_j \sigma^2 \lambda_j Z_j^2$ over the positive spectral indices.

Proof. For arbitrary $g, h \in \mathcal{H}$, iterated expectation gives

$$\begin{aligned} \langle Vh, g \rangle &= \mathbb{E}[\varepsilon^2 \langle X^c, h \rangle \langle X^c, g \rangle] \\ &= \mathbb{E}[\mathbb{E}(\varepsilon^2 | X) \langle X^c, h \rangle \langle X^c, g \rangle] \\ &= \sigma^2 \langle Kh, g \rangle. \end{aligned}$$

Equality of these bilinear forms for every g and h proves $V = \sigma^2 K$. The spectral conclusion follows immediately. $\square$

Corollary A.43 (Asymptotic size). *Suppose the conditions of Theorem A.41 hold and V has at least one positive eigenvalue. Let $q_{1-\alpha}$ be a continuity-point $(1 - \alpha)$ quantile of $Q = \sum_j \omega_j Z_j^2$. Then the rejection rule*

$$\varphi_n(b) = \mathbf{1}\{\mathcal{T}_n(b) > q_{1-\alpha}\}$$

has limiting null rejection probability α .

Proof. Theorem A.41 gives $\mathcal{T}_n(b) \rightsquigarrow Q$. Since at least one weight is positive, Q has a continuous distribution: condition on all terms except one positive weighted chi-squared term and integrate its continuous distribution function. The portmanteau theorem at $q_{1-\alpha}$ therefore yields

$$\mathbb{P}\{\mathcal{T}_n(b) > q_{1-\alpha}\} \longrightarrow \mathbb{P}\{Q > q_{1-\alpha}\} = \alpha.$$

□

Proposition A.44 (Inference under fixed and local alternatives). *Assume the conditions needed for the Hilbert-space central limit theorem in Theorem A.41, $\|\widehat{K} - K\|_{\text{op}} = o_{\mathbb{P}}(1)$, and $\sqrt{n}\|e_{m_n}\| = o_{\mathbb{P}}(1)$.*

For a fixed alternative satisfying $K(\beta - b) \neq 0$,

$$n^{-1}\mathcal{T}_n(b) \xrightarrow{\mathbb{P}} \|K(\beta - b)\|^2,$$

so $\mathcal{T}_n(b) \rightarrow \infty$ in probability. For the triangular sequence $\beta_n = b + n^{-1/2}\Delta$ with fixed $\Delta \in \mathcal{H}$,

$$\mathcal{T}_n(b) \rightsquigarrow \|G + K\Delta\|^2.$$

If $V\phi_j = \omega_j\phi_j$, the latter variable equals

$$\sum_j \omega_j Z_j^2 + 2 \sum_j \sqrt{\omega_j} Z_j \langle \phi_j, K\Delta \rangle + \|K\Delta\|^2.$$

Proof. Under a fixed alternative, the empirical null moment decomposes as

$$u_n(b) = \frac{1}{n} \sum_{i=1}^n \varepsilon_i X_i^c + \widehat{K}(\beta - b),$$

apart from the lower-order sample-centring term treated in Theorem A.41. The first term is $O_{\mathbb{P}}(n^{-1/2})$, while operator convergence gives $\widehat{K}(\beta - b) \rightarrow_{\mathbb{P}} K(\beta - b)$. Since $e_{m_n} = o_{\mathbb{P}}(n^{-1/2})$, Proposition A.40 implies

$$\widehat{K}(\widehat{\beta}_{m_n} - b) \xrightarrow{\mathbb{P}} K(\beta - b).$$

Squaring the norm proves the first claim.

Under $\beta_n = b + n^{-1/2}\Delta$, multiply the analogous decomposition by $\sqrt{n}$ to obtain

$$\sqrt{n} \widehat{K}(\widehat{\beta}_{m_n} - b) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i X_i^c + \widehat{K}\Delta - \sqrt{n}e_{m_n} + o_{\mathbb{P}}(1).$$

The three displayed terms converge jointly to G , $K\Delta$, and zero, respectively. Slutsky's theorem and the continuous mapping theorem give the local limit. Finally write $G = \sum_j \sqrt{\omega_j} Z_j \phi_j$ and expand $\|G + K\Delta\|^2$; Parseval's identity justifies the displayed series. □

Corollary A.45 (Confidence sets by test inversion). *Let $q_{1-\alpha}$ be a critical value for which the test that rejects when $\mathcal{T}_n(b) > q_{1-\alpha}$ has asymptotic size α on an identified*

parameter space $\Theta \subseteq \mathcal{H}_{\text{id}}$. Then

$$\mathcal{C}_{1-\alpha} = \{b \in \Theta : \mathcal{T}_n(b) \leq q_{1-\alpha}\}$$

is an asymptotic $(1 - \alpha)$ confidence set for every $\beta \in \Theta$.

Proof. By construction,

$$\mathbb{P}_\beta\{\beta \in \mathcal{C}_{1-\alpha}\} = \mathbb{P}_\beta\{\mathcal{T}_n(\beta) \leq q_{1-\alpha}\} = 1 - \mathbb{P}_\beta\{\mathcal{T}_n(\beta) > q_{1-\alpha}\}.$$

The final rejection probability converges to α , so the coverage probability converges to $1 - \alpha$. $\square$

Proposition A.46 (Finite-dimensional confidence ellipsoid and envelope). *Let $H \in \mathbb{R}^{T \times p}$ contain the values of p candidate basis functions on an equally weighted grid, set $A = \widehat{K}H$, and let $f = \widehat{K}\widehat{\beta}_m$. For a critical value $q > 0$, define*

$$C_q = \left\{c \in \mathbb{R}^p : \frac{n}{T} \|f - Ac\|_2^2 \leq q\right\}.$$

If A has full column rank, put

$$\widehat{c} = (A^\top A)^{-1} A^\top f, \quad \rho^2 = \frac{Tq}{n} - \|f - A\widehat{c}\|_2^2.$$

Then C_q is empty when $\rho^2 < 0$ and, when $\rho^2 \geq 0$,

$$C_q = \{c : (c - \widehat{c})^\top A^\top A(c - \widehat{c}) \leq \rho^2\}.$$

For an evaluation vector $h(s) = (h_1(s), \dots, h_p(s))^\top$, the extrema of $h(s)^\top c$ over a non-empty C_q are

$$h(s)^\top \widehat{c} \pm \sqrt{\rho^2 h(s)^\top (A^\top A)^{-1} h(s)}.$$

If A is rank deficient and C_q is non-empty, its coefficient set is unbounded along $\text{Null}(A)$; a pointwise functional envelope is unbounded wherever evaluation does not annihilate those directions.

Proof. The least-squares residual $u = f - A\widehat{c}$ is orthogonal to $\text{Range}(A)$. Therefore, for every c ,

$$\begin{aligned} \|f - Ac\|_2^2 &= \|u + A(\widehat{c} - c)\|_2^2 \\ &= \|u\|_2^2 + (c - \widehat{c})^\top A^\top A(c - \widehat{c}). \end{aligned}$$

Substitution into the defining inequality proves the empty-set and ellipsoid claims.

When A has full column rank, write $d = (A^\top A)^{1/2}(c - \hat{c})$. The constraint becomes $\|d\|_2 \leq \rho$, and

$$h(s)^\top (c - \hat{c}) = \{(A^\top A)^{-1/2} h(s)\}^\top d.$$

Cauchy–Schwarz gives the displayed upper and lower values, attained when d is parallel or antiparallel to $(A^\top A)^{-1/2} h(s)$.

If A is rank deficient, take non-zero $v \in \text{Null}(A)$. For any accepted c , every $c + tv$ has the same residual for all $t \in \mathbb{R}$. Thus the coefficient set is unbounded, and so is the value $h(s)^\top (c + tv)$ whenever $h(s)^\top v \neq 0$. $\square$

Late stopping removes iteration bias. Identifying a unique slope still requires restriction to $\text{Null}(K)^\perp$.

Appendix B

Supplementary Monte Carlo Diagnostics

This appendix collects the basis-coefficient and pointwise distribution plots supporting Section 5.7. Each page uses the same frozen $R = 5,000$ archive as Chapter 5. The horizontal ranges of the histograms are based on the central 99% of the pooled distribution so that the centre remains readable; nonzero omitted tail counts are printed inside the panels as “out: N ”. Each histogram is normalised over its displayed observations. All replay-based raw-FPLS panels—the basis-coefficient histograms, pointwise slope histograms, and quantile plots—use the archived fits not flagged by the raw stability diagnostic, while all replications are retained for CG-FPLS, Arnoldi FPLS, and FPCR.

In these plots, “CG”, “Raw”, and “Arnoldi” denote the released implementation of CG-FPLS, raw FPLS, and Arnoldi FPLS. Model identity is given in the caption; method columns retain the same order throughout. The full numerical distribution summaries remain in the frozen archive.

B.1 Leading basis coefficients

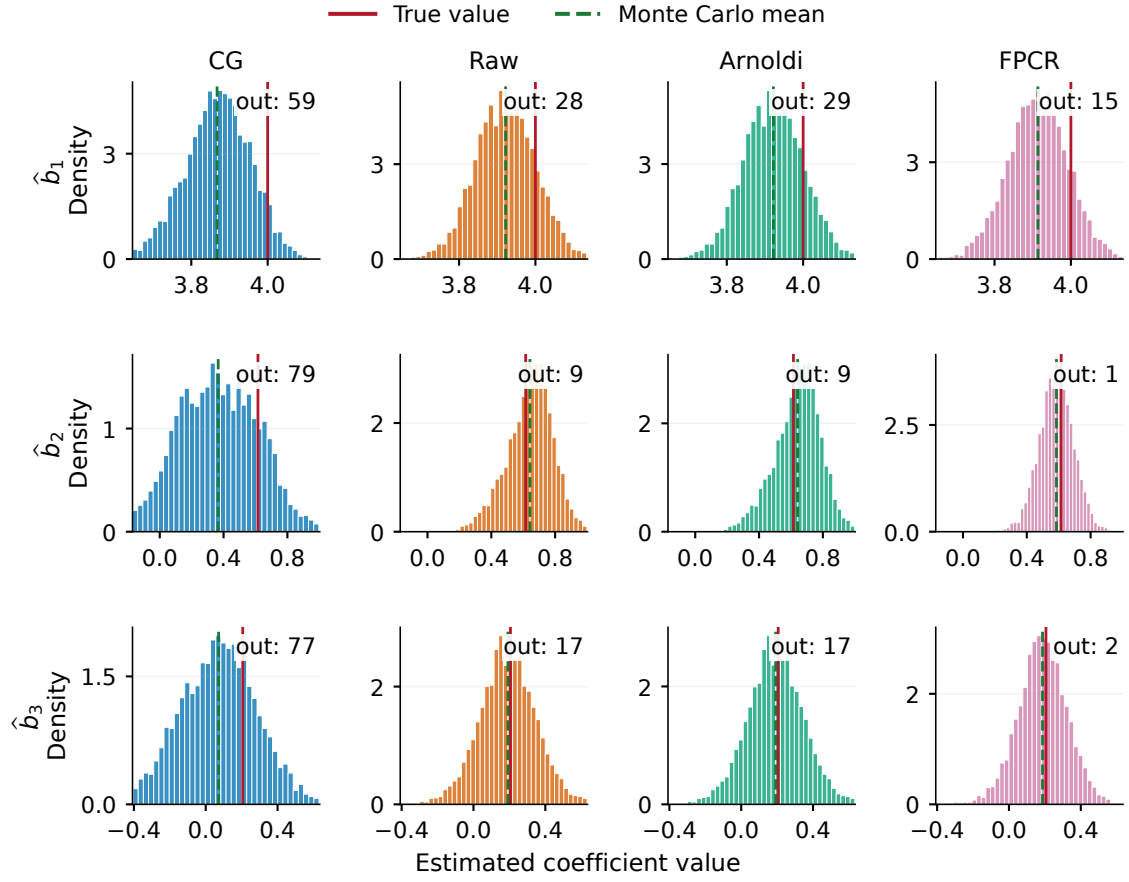


Figure B.1: Empirical distributions of the first three basis coefficients in Model 1. Solid vertical lines mark the true coefficients and dashed lines mark Monte Carlo means.

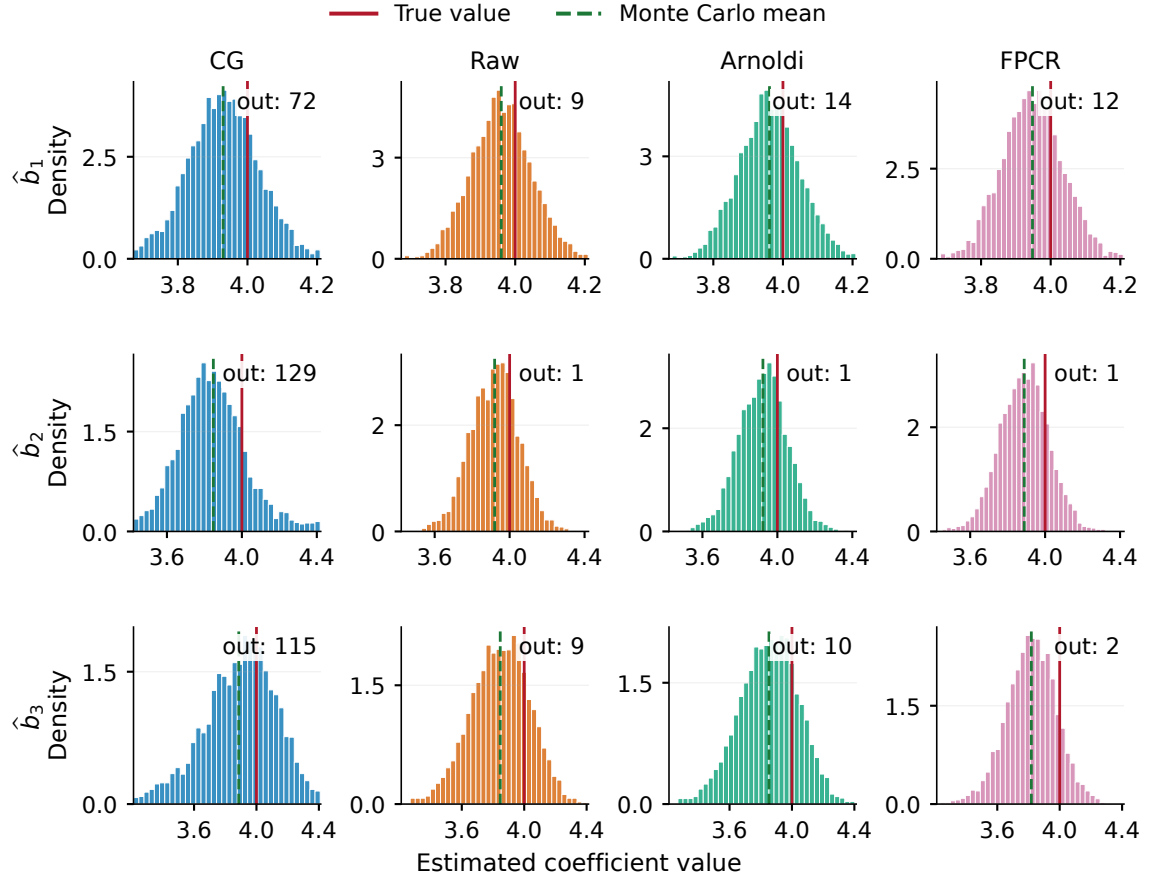


Figure B.2: Empirical distributions of the first three basis coefficients in Model 2.

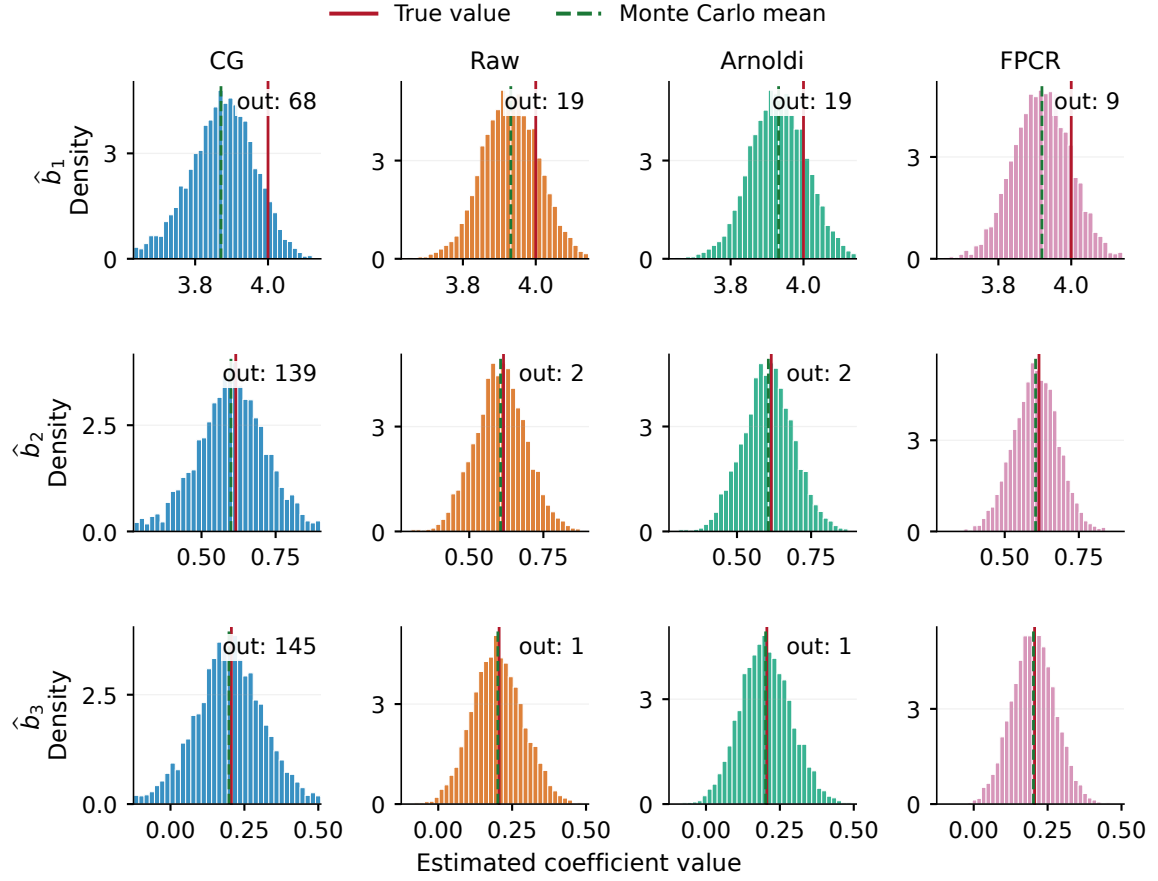


Figure B.3: Empirical distributions of the first three basis coefficients in Model 3.

B.2 Pointwise slope distributions

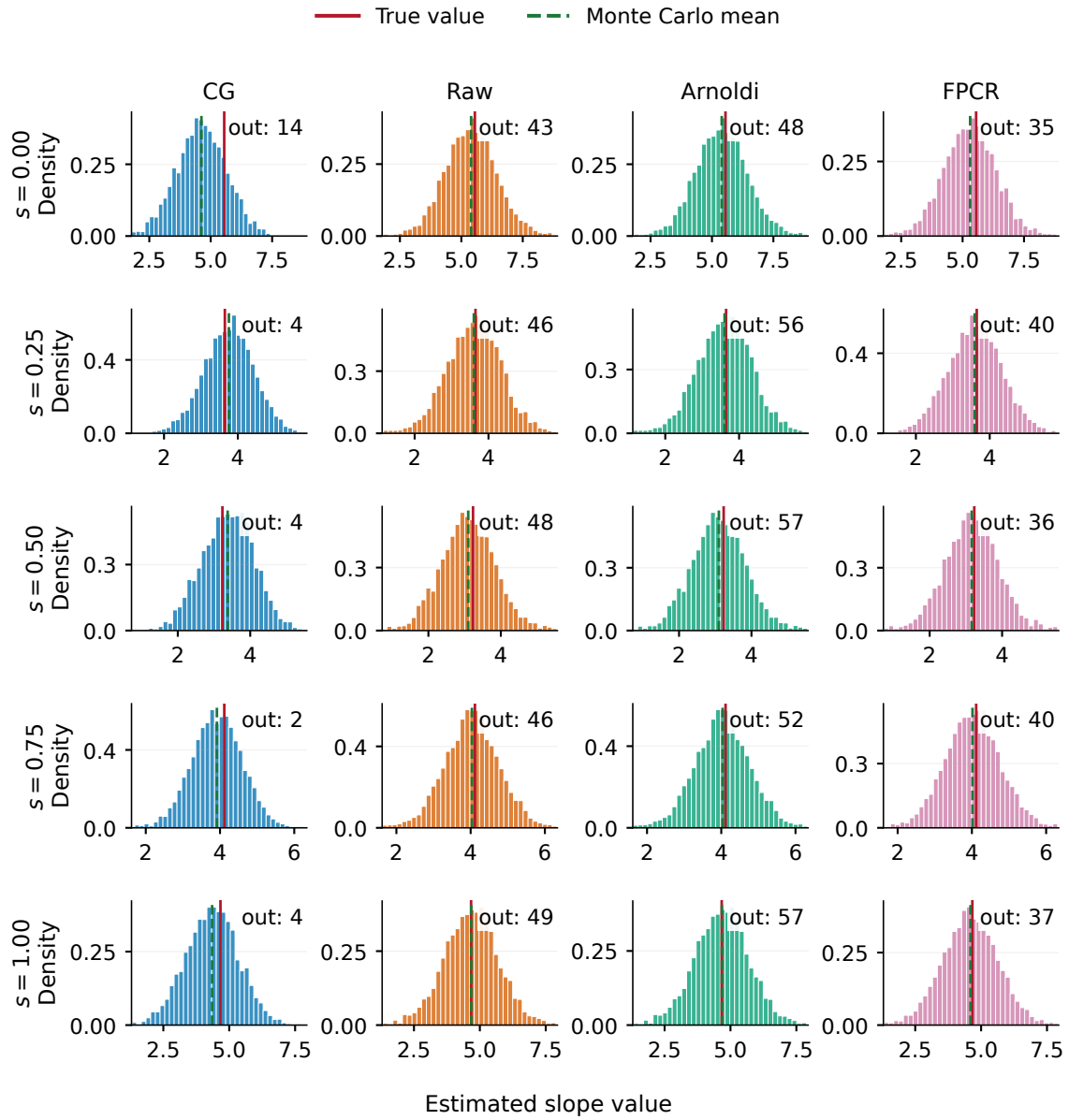


Figure B.4: Pointwise distributions of the selected slope estimator at five locations in Model 1.

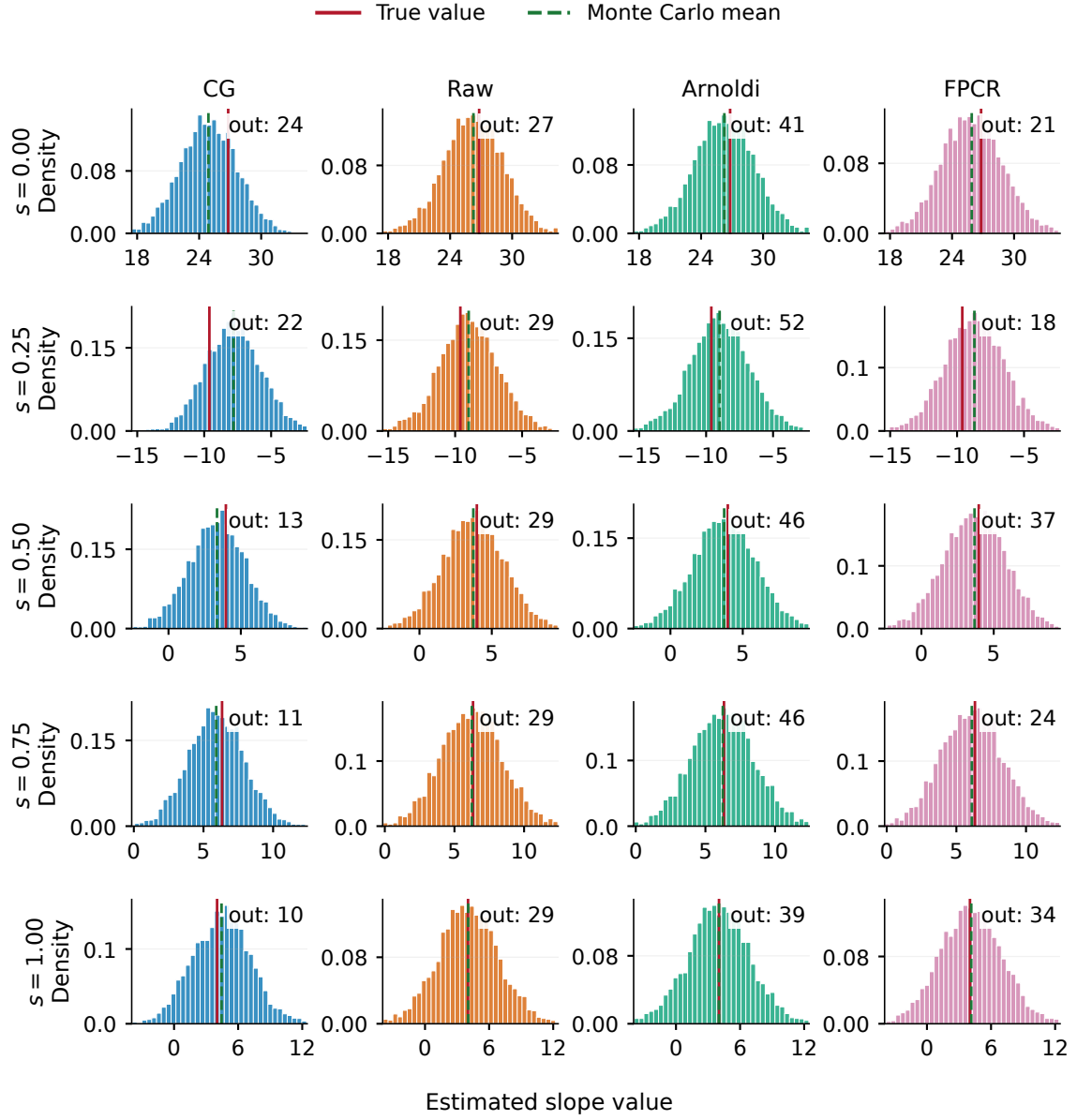


Figure B.5: Pointwise distributions of the selected slope estimator at five locations in Model 2.

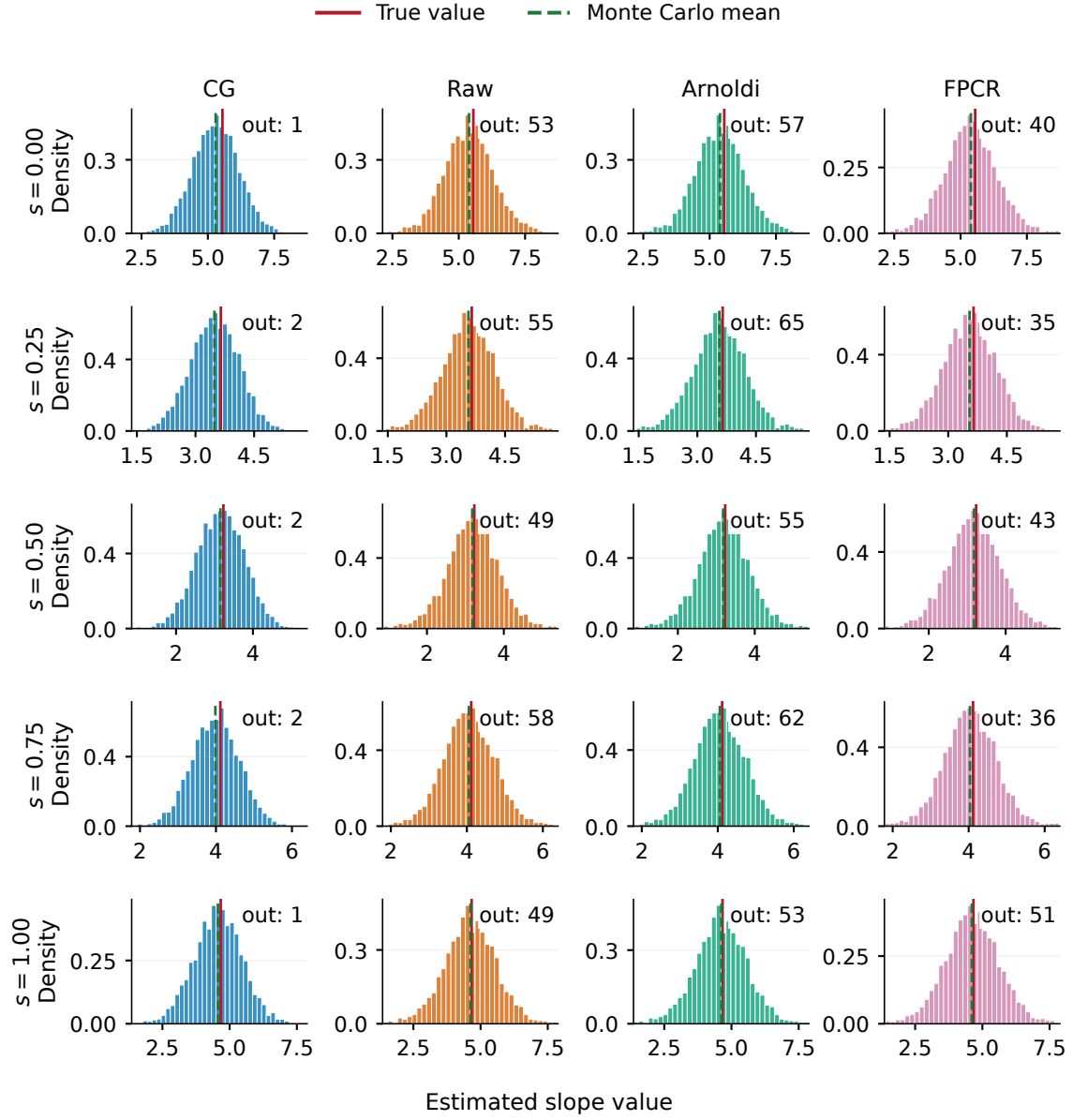


Figure B.6: Pointwise distributions of the selected slope estimator at five locations in Model 3.

B.3 Normal quantile–quantile diagnostics

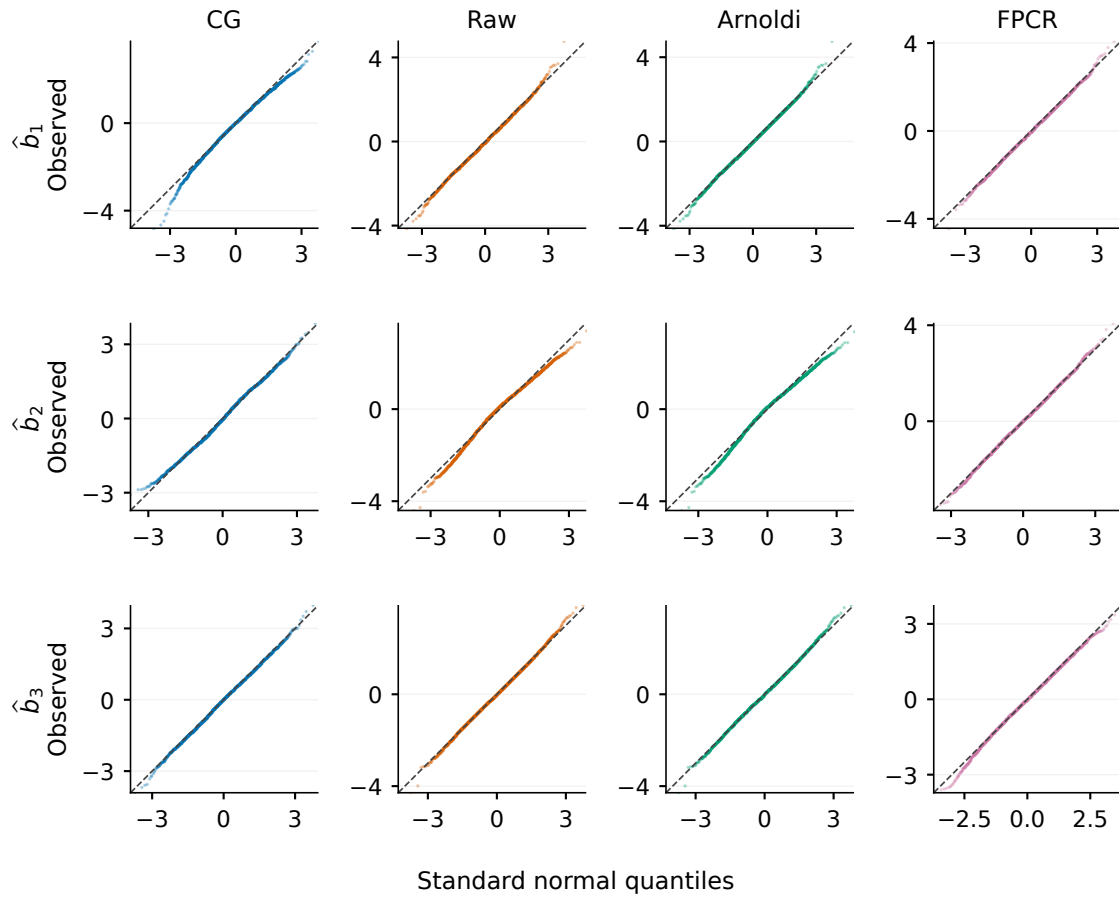


Figure B.7: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 1. Each coefficient sample is standardised by its empirical mean and standard deviation; the dashed line is the 45-degree identity line.

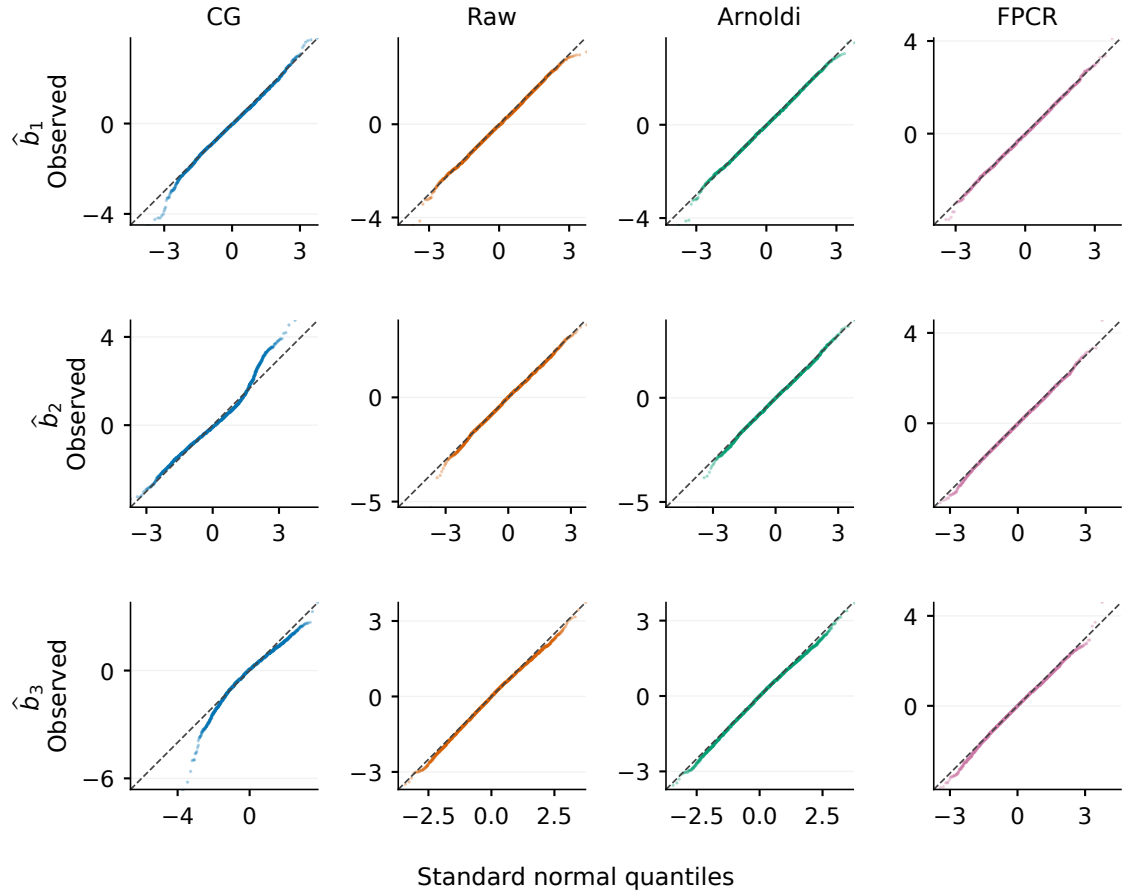


Figure B.8: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 2, using the same empirical standardisation and identity reference line as in Model 1.

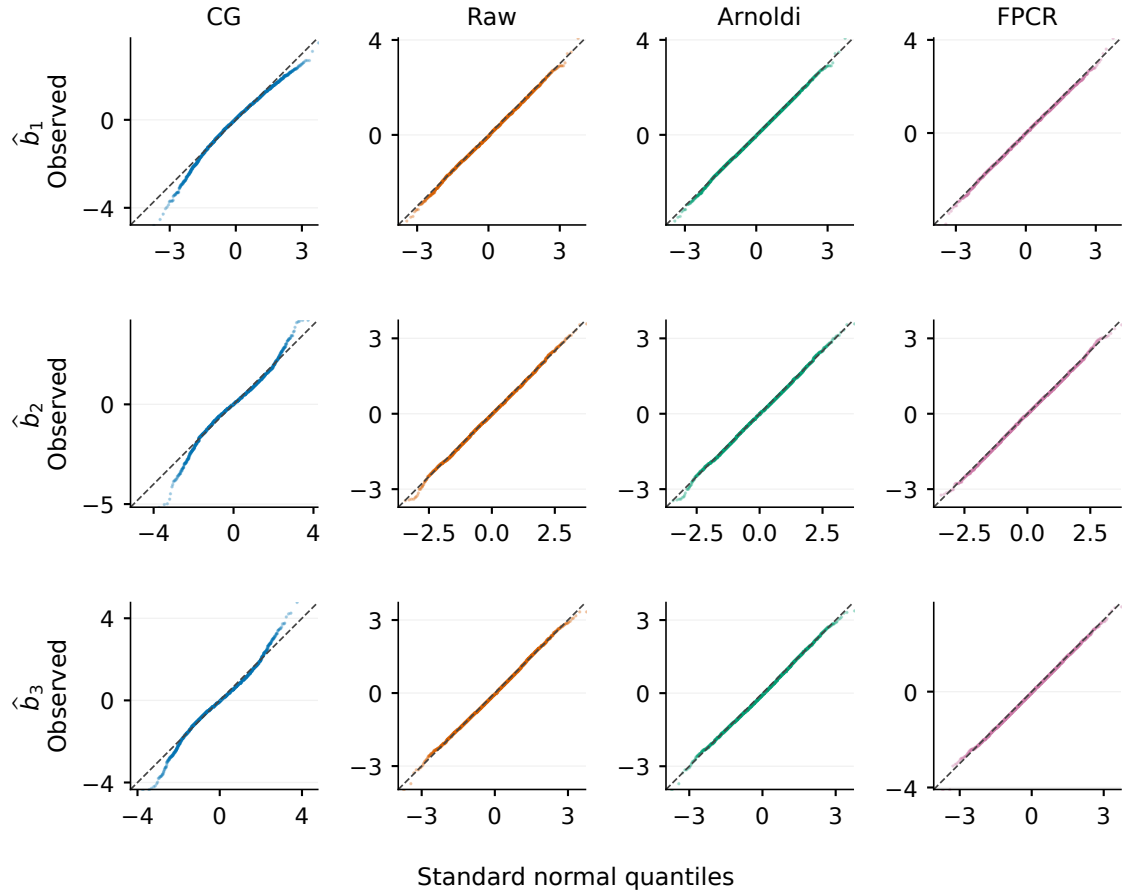


Figure B.9: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 3, using the same empirical standardisation and identity reference line as in Model 1.

Appendix C

Supplementary Inference Diagnostics

Chapter 7 reports the exact rejection frequencies at the 1%, 5%, and 10% levels in Table 7.2. The additional figure below isolates the principal 5% comparison and displays its conditional Monte Carlo uncertainty. It is retained for visual completeness but is placed here because the numerical table communicates the same evidence more compactly.

C.1 Oracle-calibrated size at the 5% level

All three point estimates lie modestly above 5%, matching the heavier finite-sample upper tails documented in Figure 7.2. The figure should therefore be read as a finite-sample size assessment of the oracle reference, not as evidence that oracle calibration is exact or available in an empirical application.

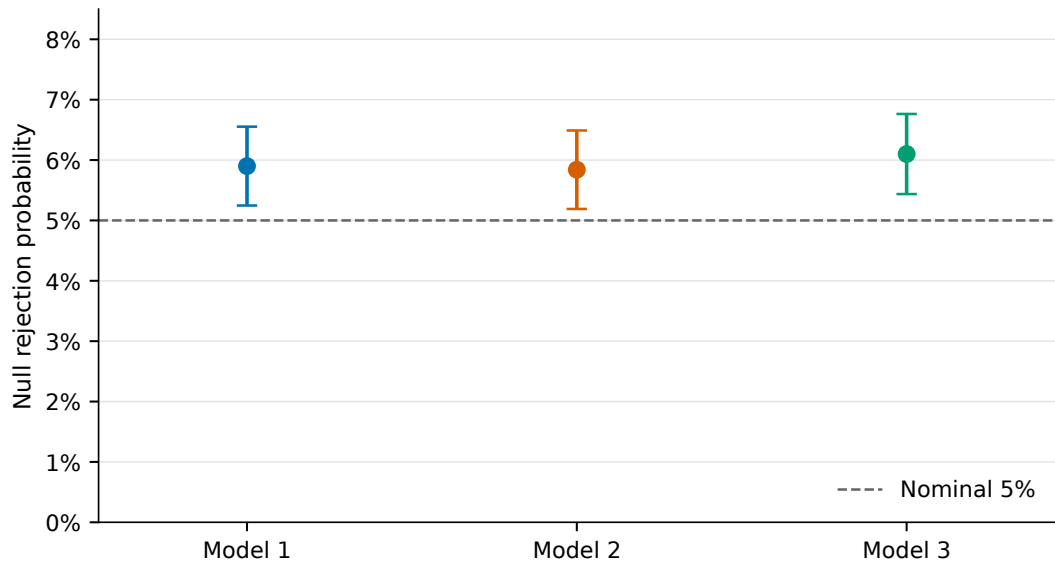


Figure C.1: Empirical null rejection probability of $\mathcal{T}_{n,70}$ at $n = 100$, using the oracle weighted chi-square 5% critical value. Each point is based on 5,000 finite-sample replications; vertical bars are normal-approximation intervals equal to the estimate plus or minus 1.96 binomial Monte Carlo standard errors. They condition on the realised critical values and do not include uncertainty from the 50,000 oracle draws. Since only oracle calibration was selected, the figure compares empirical size with the nominal line rather than comparing alternative feasible calibration procedures.

Appendix D

Supplementary Spectroscopy Analysis and Diagnostics

This appendix is the audit trail for Chapter 8. It records the dataset identity, every preprocessing and validation rule, the complete development and sensitivity summaries, and the numerical diagnostics needed to distinguish finite prediction from reliable Krylov coordinates. The chapter and appendix are derived from one frozen archive; the material below does not redefine the primary analysis after observing the benchmark results.

D.1 Data provenance and structural audit

The source is the 2002 pharmaceutical-tablet NIR Shoot-out (Hopkins 2003; Eigenvector Research 2026). The archive was downloaded from the Eigenvector Research data page on 8 September 2026. Its identifiers are:

- ZIP SHA-256:

04c03a4314ab8e8f63e0ecc788b657b5d1b5450791ce465ad41ad90cb5d52125;

- contained MAT-file SHA-256:

129a32ec9e194e568cbd96a200c88aabeb347156081b5dfbb0120372fdd6c22a.

The distributed file contains 155 calibration, 40 validation, and 460 test rows, giving 655 rather than the catalogue’s stated 654 tablets. All rows and all spectral entries are finite, and there are no exact duplicate spectra within either instrument. Every complete row is retained. Because the file does not supply global tablet identifiers, immutable analysis identifiers join the source split to one-based row order, for example

`calibrate:001`. Corresponding instrument-1 and instrument-2 rows are treated as paired observations.

Table D.1 quantifies the response imbalance that is visible in Figure 8.1. The validation response has sample standard deviation 2.393 and range 188.2–199.5, whereas the calibration and benchmark distributions are much wider. In fact, 267 of the 460 benchmark assay values fall outside the validation range. Validation-set R^2 therefore has a small denominator and is not comparable with benchmark R^2 .

Table D.1: Assay response by supplied archive split. Standard deviations use denominator $n - 1$.

Split	n	Mean	Standard deviation	Minimum	Maximum
Calibration	155	192.885	21.980	151.6	239.1
Validation	40	194.795	2.393	188.2	199.5
Benchmark	460	188.400	15.763	154.3	237.7

The wavelength units and domain are supplied by the external description; they are not embedded as ordinary numeric metadata in the MAT object. The archive and catalogue provide no explicit open-data licence. The frozen empirical archive therefore contains checksums and a reproducible preparation workflow, but does not redistribute the source MAT file.

D.2 Functional representation and preprocessing

Each spectrum is represented by its $T = 650$ evaluations on the uniform 600–1898 nm grid. The discrete geometry is

$$\langle f, g \rangle_T = \frac{1}{T} \sum_{j=1}^T f_j g_j, \quad T = 650. \quad (\text{D.1})$$

For a training sample $\mathcal{A}$, wavelength and response means are computed using only rows in $\mathcal{A}$. The centred estimator has intercept

$$\hat{\alpha}_{\mathcal{A}} = \bar{Y}_{\mathcal{A}} - \frac{1}{T} \bar{X}_{\mathcal{A}}^{\top} \hat{\beta}_{\mathcal{A}}, \quad (\text{D.2})$$

which is algebraically equivalent to the prediction form in (8.2). The training means are applied unchanged to held-out observations. The source code also archives the resulting original-input coefficient vector and intercept for every fitted model, so its stored prediction can be reconstructed directly.

No primary operation other than this fold-local centring is applied. In particular,

there is no variance scaling, smoothing, derivative construction, wavelength selection, standard-normal-variate transformation, multiplicative- scatter correction, or baseline correction. These choices matter because every such operation can add regularisation or alter the covariance geometry.

The primary final fit has $650 > 195$ and numerical covariance rank 194 after centring. At least 456 grid-coordinate null directions therefore remain, so a unique unrestricted physical slope is not identified. Coefficient vectors are archived for replay and numerical comparison, not interpreted as recovered chemical effects.

Three alternatives were fixed before benchmark fitting:

1. instrument 1 restricted to 600–1798 nm inclusive, retaining 600 wavelengths;
2. instrument 1 smoothed row by row with a 15-point cubic Savitzky–Golay filter, derivative order zero and interpolation boundary mode; and
3. the complete 650-wavelength instrument-2 spectrum for the same tablets.

Each transformed dataset is centred within the relevant training sample. The fixed smoothing operator itself does not estimate parameters from held-out rows; all sample-dependent centring remains fold local.

D.3 Evaluation design and leakage safeguards

Table D.2 summarises the complete evaluation design. Random folds are formed separately within the supplied calibration and validation sources. Within each source, observations are ordered by assay in groups of five and assigned one of the five fold labels at random. Every development outer fold thus contains exactly 31 calibration and 8 validation tablets. This preserves source composition while avoiding folds dominated by one part of the response range.

Across five repeats, $195 \times 5 = 975$ out-of-fold predictions are stored for each method. These are repeated predictions of the same 195 tablets, not 975 independent observations. Correspondingly, repeat summaries and fold distributions are descriptive diagnostics rather than a basis for standard independent-sample tests. The development completion record additionally states `benchmark_variables_loaded=false`: the development stage did not load the test variables. Only after its checks passed was the full stage allowed to fit the 195-row models and evaluate the benchmark.

The benchmark bootstrap resamples tablet indices in common across methods. This preserves pairing and estimates conditional variation over the supplied benchmark

Table D.2: Frozen spectroscopy evaluation design. Exact assignments and resampling matrices, rather than only seeds, are included in the archive.

Stage	Frozen rule
Repeated development	Five repeats of five source-specific, assay-balanced outer folds; one complete out-of-fold metric per repeat.
Inner selection	New five-fold balanced partitions inside each outer training set; Raw and Arnoldi share indices but choose independently.
Order sensitivity	Five source-specific contiguous outer folds, each holding out 31 calibration and 8 validation rows.
Original split	Fit and tune on 155 calibration tablets; evaluate once on the 40 validation tablets.
Final development fit	Fit on all 195 development tablets; Raw and Arnoldi use one frozen common balanced inner partition.
Designated benchmark	Predict the 460 supplied test tablets once; no benchmark outcome enters centring, tuning, repair, or variant selection.
Conditional bootstrap	10,000 paired iid draws and 10,000 circular moving-block draws at each block length 5, 10, and 20; models are held fixed.
Randomness	Master seed 20260908 with separate deterministic streams for outer folds, inner folds, final tuning, iid bootstrap, and block bootstrap.

empirical distribution. It does not refit centring, tuning, or the models and therefore excludes development-sample and selection uncertainty. The circular block calculations preserve local row-order sequences, but no batch or acquisition-order metadata are available to determine whether archive order has a physical interpretation.

D.4 Complete procedures and tuning rules

The estimator formulas are given in Chapter 3; Table D.3 records the application-specific tuning and diagnostics. The requested path maximum is 70, further limited by grid size, centred training-sample size, numerical rank, and recurrence breakdown.

Table D.3: Estimator-specific regularisation rules in the frozen application. The procedures retain different finite- m objectives and stopping rules.

Method	Regularisation choice	Principal diagnostics
CG-FPLS	Released-code conjugate-residual recurrence and first discrepancy crossing; $\tau = 1.01$, $\delta = 0.10$, with an internal moment-GCV FPCR variance pilot.	Threshold, crossing status, moment-residual path, pilot dimension, and variance estimate.
Raw FPLS	Minimum five-fold inner prediction MSE over the unscaled raw Krylov power sequence, fitted by response-space normal equations.	Candidate validity, raw-basis and response-design condition numbers, and Krylov column norms.
Arnoldi FPLS	Independent minimum on the same inner folds, using twice reorthogonalised Gram-Schmidt and response-space QR.	Effective dimension, rank breakdown, response-design condition, and basis orthogonality defect.
FPCR	Response-space GCV over covariance-eigenvalue cut-offs.	Numerical rank, retained eigenvalue, and retained spectral condition number.

Invalid or non-finite raw candidates are excluded from selection. They are not silently repaired by a pseudoinverse, ridge penalty, or substitute method. A complete method failure would remain in the diagnostics and create missing predictions rather than a dropped fold. In the realised study, all 140 stored fits and all predictions are finite. These safeguards make clear why selected component counts are method-specific indices rather than a common complexity scale.

D.5 Development evidence

Table D.4 gives the repeat-level development results. Arnoldi has the smallest mean repeated-CV RMSE, while Raw has the second-smallest mean but a much larger standard deviation. Raw’s five repeat RMSEs are 4.850, 4.960, 5.997, 4.844, and 4.760; the single 5.997 repeat creates its adverse tail. The contiguous-fold check reverses the top pair, with Raw at 4.913 and Arnoldi at 5.120. Hence development prediction supports Raw and Arnoldi as the leading pair but does not establish a stable ordering between them.

Table D.4: Development RMSE. The first three columns summarise five complete repeated nested-CV out-of-fold estimates. The final two columns are single prespecified sensitivities.

Method	Mean (SD)	Median	Range	Contiguous folds	Original validation
CG-FPLS	5.924 (0.083)	5.940	5.784–5.990	6.149	5.998
Raw FPLS	5.082 (0.517)	4.850	4.760–5.997	4.913	4.132
Arnoldi FPLS	4.917 (0.131)	4.954	4.741–5.084	5.120	4.150
FPCR	5.364 (0.219)	5.396	5.074–5.624	5.349	3.978

The original 155-to-40 split gives the metrics in Table D.5. All biases are positive, but the small validation set and its restricted assay support make this a weak basis for general calibration claims. In particular, every R^2 is negative because its SSE is compared with the unusually small validation SST. This is compatible with RMSE around four for three methods and does not contradict the broader benchmark signal.

Table D.5: Original supplied-split sensitivity: fitting on the 155 calibration tablets and evaluating on the 40 validation tablets. Bias is prediction minus observation.

Method	RMSE	MAE	Bias	R^2
CG-FPLS	5.998	4.820	2.109	-5.442
Raw FPLS	4.132	3.566	1.460	-2.057
Arnoldi FPLS	4.150	3.558	1.406	-2.085
FPCR	3.978	3.431	1.512	-1.834

Selection is more revealing than the small differences among the leading development prediction errors. Table D.6 records every outer-fit range. CG stops at three throughout.

Arnoldi remains tightly concentrated between four and six. Raw spans almost the entire requested path, reaches the cap in two fits, and triggers the fixed instability rule in 24 of 25. FPCR also varies widely, although its spectral truncation index is not directly comparable with a Krylov iteration.

Table D.6: Selection and numerical stability over 25 repeated-development outer fits. Quartiles use the stored selected indices.

Method	Minimum	Q1	Median	Q3	Maximum	Cap / flags
CG-FPLS	3	3	3	3	3	0 / 0
Raw FPLS	4	34	44	63	70	2 / 24
Arnoldi FPLS	4	5	5	6	6	0 / 0
FPCR	7	12	21	39	57	0 / n/a

D.6 Benchmark uncertainty and residuals

Chapter 8 gives all four primary point estimates and marginal iid intervals. Table D.7 supplies the complete prespecified RMSE sensitivity for the three declared paired contrasts. The focal Raw-minus-Arnoldi interval remains below zero for every block length, as do the larger CG-minus-Arnoldi and FPCR-minus-Arnoldi differences. This order robustness applies to conditional RMSE contrasts. Bias conclusions are less stable: the longer-block bias intervals for the secondary contrasts cross zero.

Table D.7: Declared benchmark RMSE contrasts and paired 95% percentile intervals. Circular blocks follow the supplied archive order.

Contrast	Estimate	Iid	Block 5	Block 10	Block 20
Raw – Arnoldi	-0.0363	[-0.0565, -0.0169]	[-0.0595, -0.0158]	[-0.0622, -0.0141]	[-0.0649, -0.0117]
CG – Arnoldi	0.6500	[0.3973, 0.9208]	[0.3286, 0.9783]	[0.2565, 1.0169]	[0.1664, 1.0687]
FPCR – Arnoldi	0.5420	[0.2855, 0.8442]	[0.2403, 0.8839]	[0.2015, 0.9218]	[0.1886, 0.9384]

Figure D.1 shows prediction-minus-observation residuals against observed assay. The methods share a few large errors, including one high-assay tablet predicted near the centre of the range and a low-assay tablet overpredicted by all procedures. These rows remain in every reported metric and in the bootstrap resampling population. The common extremes widen marginal RMSE intervals while largely cancelling in paired method differences. Their presence does not establish data error or a batch effect in the absence of supporting metadata.

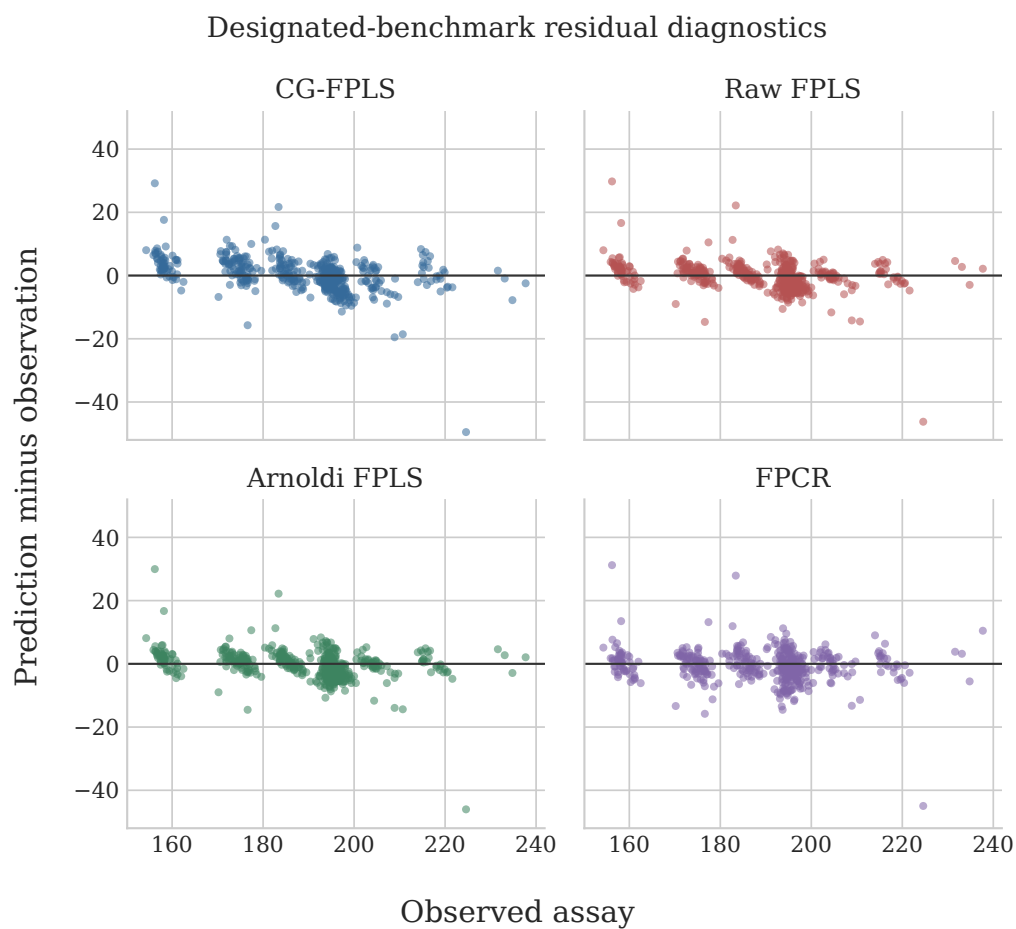


Figure D.1: Designated-benchmark residuals, defined as prediction minus observation, plotted against observed assay. All 460 tablets are retained. Shared large residuals are descriptive; the archive contains no batch or acquisition-order metadata with which to assign a physical cause.

D.7 Final-fit numerical diagnostics

Table D.8 contrasts final training fit with numerical representation. Raw and Arnoldi achieve nearly the same training RMSE and coefficient norm, but through sharply different coordinate systems. Raw’s selected condition numbers are at the reporting ceiling, whereas Arnoldi retains an orthonormal basis to machine precision. FPCR has the smallest training RMSE and largest coefficient norm; these in-sample quantities do not override the benchmark ordering.

Table D.8: Final fit on all 195 development tablets. Coefficient norms use the archived original-input vector under the T^{-1} prediction convention.

Method	m	Training RMSE	$\ \hat{\beta}\ _2$	Principal diagnostic
CG-FPLS	3	5.544	18,997	Pilot $m = 70$, $\hat{\sigma}^2 = 10.467$; discrepancy threshold 0.191 was crossed.
Raw FPLS	55	4.399	21,494	Basis and response-design conditions both recorded at 4.50×10^{15} ; instability flag set.
Arnoldi FPLS	5	4.395	21,573	Basis condition 1; response-design condition 11.95; orthogonality defect 7.40×10^{-16} .
FPCR	53	3.400	45,334	Retained eigenvalue 7.02×10^{-7} ; retained spectral condition 35,919.

Figure D.2 gives the complete relevant paths. For Raw, the basis condition is already 5.74×10^8 at $m = 6$, above the fixed $1/\sqrt{\epsilon_{\text{mach}}} = 6.71 \times 10^7$ threshold; the response design is still worse. The raw Krylov column norm falls by more than 80 orders of magnitude by the selected $m = 55$, and the aggregate inner-CV criterion contains invalid candidate dimensions. Its finite selected prediction must therefore not be equated with stable continuation of the intended power basis.

Arnoldi’s basis condition remains one along the requested path, and its orthogonality defect stays near machine precision. The defect is exactly zero at $m = 1$, so that point is absent on the logarithmic vertical scale. At the selected $m = 5$, the response-space least-squares problem is also well conditioned. Raw and Arnoldi nevertheless make almost identical benchmark predictions: correlation 0.999934, RMS paired difference 0.174, and maximum absolute difference 0.445 assay units. The empirical role of orthogonalisation is thus reliable representation of essentially the same predictive relation, not guaranteed improvement of every realised loss.

D.8 Preprocessing and instrument sensitivities

Table D.9 and Figure D.3 report every prespecified benchmark variant. The full instrument-1 analysis remains the sole primary result. Trimming the high-wavelength tail improves CG-FPLS, Arnoldi, and FPCR and compresses the four-method RMSE range, but slightly worsens Raw. Smoothing changes both errors and selected indices without

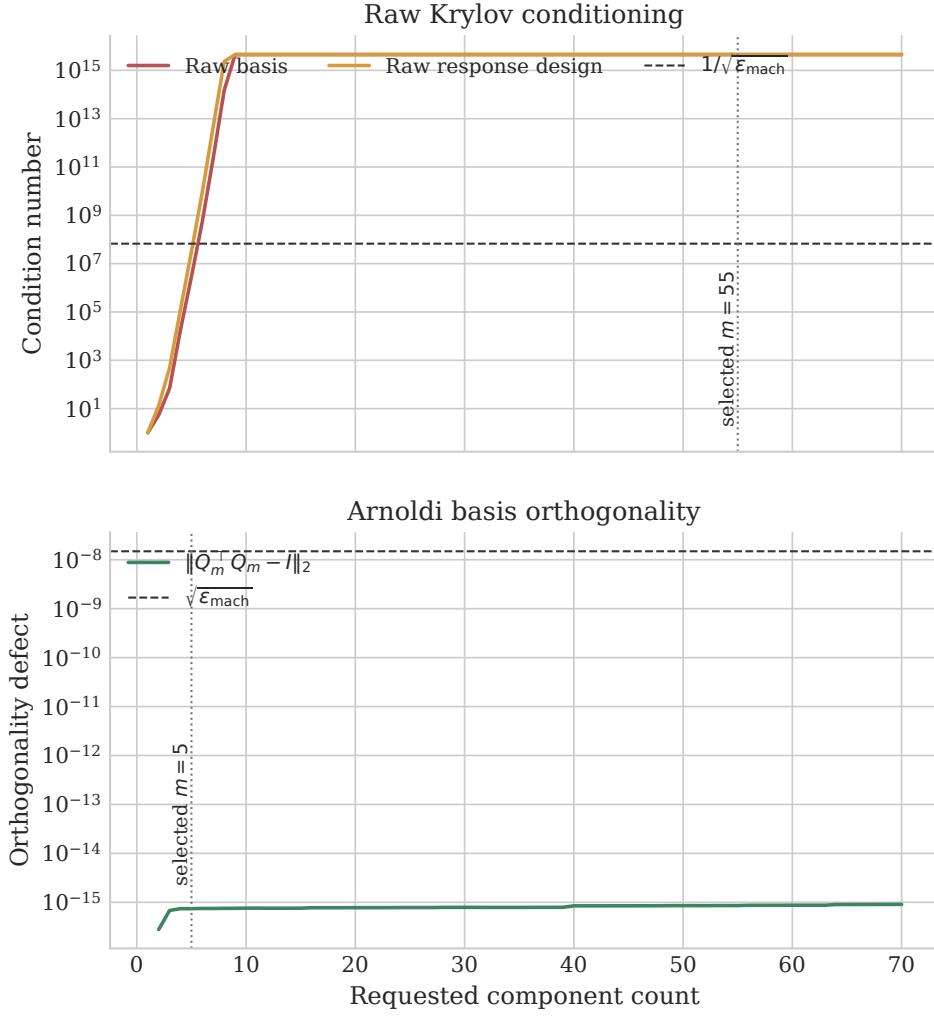


Figure D.2: Final raw and Arnoldi Krylov diagnostics. The upper panel shows raw-basis and response-design condition numbers against the fixed $1/\sqrt{\epsilon_{\text{mach}}}$ flag threshold. The lower panel shows Arnoldi's spectral-norm orthogonality defect; its exactly zero $m = 1$ value is not displayed on the logarithmic scale. Vertical markers identify the selected dimensions 55 and 5.

uniformly improving them. Instrument 2 raises every RMSE; because it measures the same tablets, this is measurement-system sensitivity rather than independent replication.

Table D.9: Complete prespecified benchmark sensitivities. Parenthesised values in the RMSE column are selected component or stopping indices. Sensitivity rows are descriptive point estimates without variant-specific uncertainty intervals.

Variant	Method	RMSE (<i>m</i>)	MAE	Bias	R^2
Instrument 1, full	CG-FPLS	5.293 (3)	3.673	0.177	0.887
	Raw FPLS	4.607 (55)	3.036	-0.303	0.914
	Arnoldi FPLS	4.643 (5)	3.086	-0.301	0.913
	FPCR	5.185 (53)	3.547	-0.628	0.892
Instrument 1, 600–1798 nm	CG-FPLS	4.669 (3)	3.007	-0.456	0.912
	Raw FPLS	4.767 (10)	3.105	-0.476	0.908
	Arnoldi FPLS	4.632 (8)	2.846	-0.513	0.913
	FPCR	4.719 (14)	2.877	-0.575	0.910
Instrument 1, SG smooth	CG-FPLS	5.341 (3)	3.717	0.013	0.885
	Raw FPLS	4.663 (18)	3.056	-0.497	0.912
	Arnoldi FPLS	4.953 (9)	3.042	-0.491	0.901
	FPCR	4.823 (27)	3.053	-0.444	0.906
Instrument 2, full	CG-FPLS	6.764 (3)	5.100	1.022	0.815
	Raw FPLS	5.042 (53)	3.242	-0.601	0.897
	Arnoldi FPLS	5.095 (5)	3.279	-0.562	0.895
	FPCR	6.395 (37)	4.150	-0.921	0.835

The Raw–Arnoldi point ordering changes across variants, reinforcing that the primary 0.036-unit difference is not a universal ranking. Trimming improves all methods except Raw, but this association does not prove that visible high-wavelength spikes caused the primary residuals. Similarly, poorer instrument-2 results do not identify a transfer mechanism without a dedicated calibration-transfer design.

D.9 Reproducibility manifest

The frozen directory `tablet_nir_study_FINAL` contains 59 payload files plus its checksum list and manifest. All listed SHA-256 checks pass. Independent replay reconstructs benchmark metrics from 7,360 prediction rows to within 1.8×10^{-15} , development metrics from 4,840 prediction rows to within 1.6×10^{-14} , and predictions from archived coefficients and intercepts to within 6.3×10^{-13} . All outer, inner, and bootstrap indices regenerate exactly from the declared streams.

The recorded environment is Python 3.12.13, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3, Matplotlib 3.10.8, and OpenBLAS 0.3.30 with the numerical thread limit set to one. The invocation and Git metadata are archived, including commit `75b3f069`, but the worktree was not clean. Reproduction must therefore use the exact source snapshots and hashes stored under `provenance/source/`, rather than assuming that the commit alone identifies the executed program.

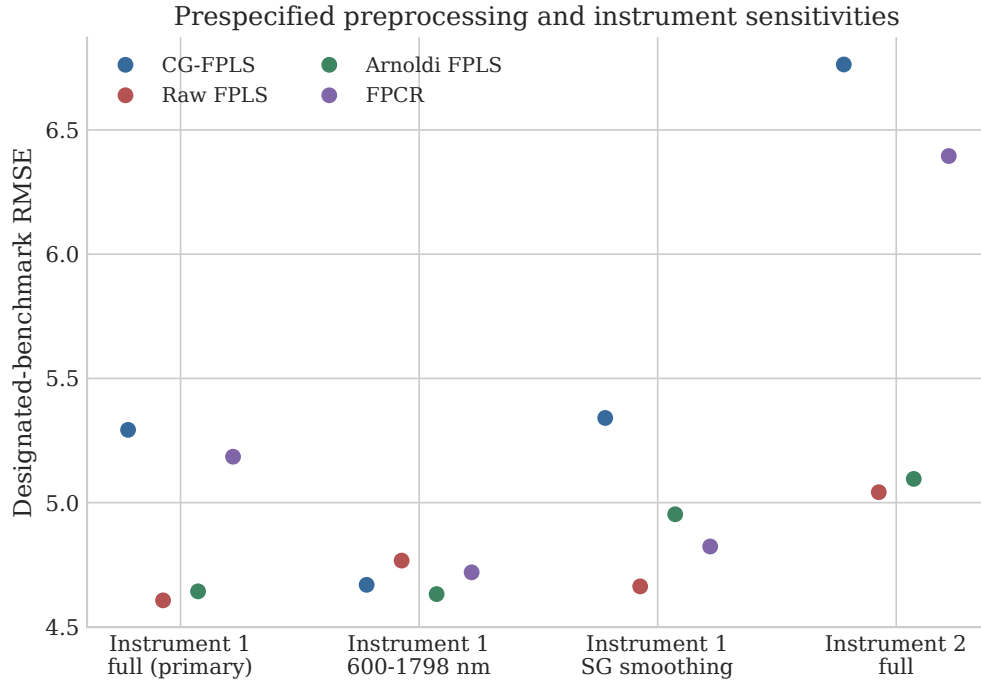


Figure D.3: RMSE across the primary analysis and three prespecified sensitivities. Points are descriptive benchmark estimates without uncertainty intervals. Horizontal comparisons change the spectral representation or instrument and do not define a post hoc replacement for the primary full-spectrum instrument-1 result.

Table D.10: Frozen empirical archive map.

Location	Contents
provenance/	Frozen protocol, resolved configuration, exact source snapshots, run invocation, code hashes, software, platform, and Git state.
resampling/	Exact development outer and inner assignments, final inner assignment, and compressed iid and block bootstrap index matrices.
development/	Observation-level predictions, fold and repeat metrics, selected coefficients, complete paths, and fit diagnostics.
benchmark/	Predictions for all variants, primary bootstrap summaries and draws, metrics, final coefficients, paths, and diagnostics.
tables/, figures/	Deterministic reduced tables and scientific displays generated from the preceding files.
Manifest and checksums	File inventory, sizes, hashes, and whole-archive integrity check.